

Sta301 Mid term

STA301 Midterm Past Papers

Objective Questions

Question # 1

Which of the following is called measure of central tendency?

- Harmonic mean
- Geometric mean
- Mean
- All of these

Question # 2

There are -----types of measure of dispersion.

- Two
- Three
- Four
- Five

Correct Ans BY AR

Question # 3

In a drawer there are 5 black socks and 3 green socks. Two socks are picked randomly one after the other without replacement. What is the possibility that both the socks are black?

- $5/14$
- $5/16$
- $3/8$
-

Correct Ans BY AR BY AR BY AR

5/8

Question # 4

Which of the following pairs of A and B events are mutually exclusive?

- A: the numbers above 100;
B: the numbers less than -200
Correct Ans BY AR
- A: the even numbers;
B: the numbers greater than 10
- A: the odd numbers;
B: the number 5
- A: the numbers less than 5;
B: all negative numbers

Question # 5

Statistics deals with

- Observations
- Aggregates of facts
- Isolated items
- Individuals

Question # 6

Which of the display of the data is divided into two parts?

- Component Bar Chart
- Frequency Distribution
- Simple Bar Chart
- Stem and leaf display

Question # 7

If the mean of two observations is 15, then the median of these two observations

will be:

- 10.5
- 15
- 9.5
- 8.5

Correct Ans BY AR

Question # 8

Total number of words formed by 2 vowels and 3 consonants taken from 4 vowels and 5 consonants is equal to

- 60
- 720
- 120
- 7200

Correct Ans BY AR

Question # 9

Which of the following technique is not used to represent the bivariate qualitative data?

- Pie Chart
- Component Bar Chart
- Multiple Bar Chart
- Line Chart

Correct Ans BY AR

Question # 10

In statistics, conducting a survey means

- drawing graphs and pictures

- making mathematical calculations
- collecting information from elements

Correct Ans BY AR

- None of the above

Question # 11

In how many ways, a team of 11 players can be chosen from a total of 16 players?

- 5400
- 2680
- 2426
- 4368

Correct Ans BY AR

Question # 12

Which of the following comes first to make frequency distribution:

- Class interval
- Tally marks
- Range

Correct Ans BY AR

- No. of Groups

Question # 13

In a left skewed distribution:

- $X_m - Q_3$ less and equal to $Q_1 - X_0$
- $X_m - Q_3$ greater and equal to $Q_1 - X_0$
- $X_m - Q_3$ greater than $Q_1 - X_0$

Correct Ans BY AR

- Xm-Q3 less than Q1-X0

Question # 14

The standard deviation is always _____ than mean deviation.

- Greater
Correct Ans BY AR
- Less
- Equal
- None

Question # 15

Harmonic mean is the reciprocal of

- Arithmetic mean
Correct Ans BY AR
- Median
- Mode
- Geometric mean

Question # 16

The difference between the sample value expected and the estimates value of the parameter is called as?

- Difference
- Bias
Correct Ans BY AR
- Contradiction
- Error

Question # 17

The median is _____:

- The highest number
- The average
- The middle point
Correct Ans BY AR
- Affected by extreme scores

Question # 18

Symbolically, a conditional probability is:

- $P(A \cup B)$
- $P(A/B)$
Correct Ans BY AR
- $P(AB)$
- $P(A)$

Question # 19

Fourth moment about mean provides information about the _____ of the distribution.

- Dispersion
- Kurtosis
- Symmetry
- Centre
Correct Ans BY AR

Question # 20

In right-skewed distributions:

- median > mid-quartile range < midrange
- median midrange

- median > mid-quartile range > midrange
- median < mid-quartile range < midrange

Correct Ans BY AR

Question # 21

Which of the measure of dispersion is used to compare variation between two series?

- C.V
- S.D
- Q.D
- M.D

Correct Ans BY AR

Question # 22

The function abbreviated to d.f. is also called the.....

- Probability density function
- Discrete function
- Probability distribution function
- Commutative distribution function

Correct Ans BY AR

Question # 23

A group consists of three persons, A, B, & C. How many ways are there of selecting a group of two persons out of these three?

- 2
- 6
- 1
-

Question # 24

If one event is not affected by the outcome of the other event, the two events are said to be:

- Dependent
- Not Mutually Exclusive
- Mutually Exclusive
- Independent

Question # 25

Measure of central tendency is used to measure:

- Average
- Location
- Both Average and Location
- Variability

Question # 26

The number of classes in a frequency distribution generally should be:

- Less than five
- Between five and twenty
- More than five
- Between ten and twenty

Question # 27

From the following table, what is lower class boundry when mode is calculated?

Class Boundaries - Frequency

29.95 - 395 - - - -8

395 - 35.95 - - - -87

35.95 - 38.95 - - - -110

38.95 - 495 - - - -104

495 - 495 - - - -101

- 35.95
- 395
- 38.95
- 29.95

Question # 28

For a certain distribution, A.M=136.75, Median=148.37 and Mode=1580, then the distribution will be:

- Negatively Skewed
- Extremely negative J shaped
- Symmetrical
- Positively skewed

Question # 29

If $\mu = 925$ and S.D = 2, then C.V(X) will be:

- 236%
- 14%
- 20.4%
- 15.3%

Question # 30

In absolute measures of dispersion our answer is expressed in the units as our original data set.

- Different
- equal

- Unitless
- Same

1 For any two estimators T1 and T2, if $\text{VAR}(T1) < \text{VAR}(T2)$, then T1 is:

- Consistent
- Unbiased
- Sufficient
- Efficient**

2 In a set of 10 values all the values are 5, what will be the P50?

- 2
- 5**
- 10
- 20

3 What is the graphical shape of the chi-square distribution?

- Positively skewed**
- Negatively skewed
- Uniformly distributed
- Normmally distributed

4 Which of the following is a characteristic of the normal distribution:

- It is a skewed distribution
- It is bell-shaped**
- It is not asymptotic
- It is leptokurtic

5 Please choose one

A five number summery consist of :

- ▶ **0 1 3 , , , (97) m X Q** Median Q andX Page
- ▶ 1 3 0 , , , m X Q Mean Q and X
- ▶ 1 3 0 , , , m X Q Mode Q and X
- ▶ 0 1 2 , , , m X Q Median Q andX

6 Please choose one

in a multiplication theorem $P(A \cap B)$ equals:

- ▶ $P(A) P(B)$
- ▶ $P(A) + P(B)$
- ▶ **$P(A) * P(B | A)$ (Page 159)**
- ▶ $P(B \setminus A) * P(B)$

7 The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is

- ▶ 0
- ▶ **3/15**
- ▶ 1/12
- ▶ 1/2

8 If A and B are mutually exclusive events then $P(A \text{ or } B)$ equals:

- ▶ $P(A) + P(B) - P(A \text{ and } B)$

- ▶ $P(A) * P(B)$
- ▶ $P(A) + P(B)$ (Page 155)
- ▶ $P(A|B) + P(B|A)$

9 The simultaneous occurrence of two events is called

- ▶ Prior probability
- ▶ Subjective probability not confirmed
- ▶ Conditional probability
- ▶ Joint probability

10 First moment about origin is always equals to:

- ▶ Mean (Page 119)
- ▶ Variance
- ▶ Standard Deviation
- ▶ Zero

11 Which one of the following measurement does not divide a set of observations into equal parts?

- ▶ Quartiles
- ▶ Deciles
- ▶ Percentiles
- ▶ Standard deviations

12 if first and third quartile are 21.16 and 56.36 respectively, then the quartile deviation is:

- ▶ 17.1 (Page 84)
- ▶ 34.2
- ▶ 51.3
- ▶ 50.5

13 The height of student is 60 inch. This is an example of.....?

- ▶ Continues data
- ▶ Qualitative data
- ▶ Categorical data
- ▶ Discrete data

14 Which of the statement is true regarding a sample?

- ▶ It is a part of population (Page 13)
- ▶ It must contain at least five observations
- ▶ It refers to descriptive statistics
- ▶ It produces true value

15 Which one of the following graphs is used for a time series data?

- ▶ Histogram
- ▶ Historigram
- ▶ Frequency curve
- ▶ Frequency polygon

16 Which of the following comes first to make frequency distribution

- ▶ No. of groups
- ▶ Class intervals
- ▶ Rang (Page 28)

► Tally marks

17 The average which is defines as reciprocal of arithmetic mean of the reciprocals of the values is called

► Geometric Mean

► Harmonic mean (Page 77)

► Mode

► Median

18 If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

► Weighted mean (Not sure)

► Harmonic mean

► Arithmetic mean

► Geometric mean

19 The probability of drawing a 'jack card 'from 52 playing cards is:

► 1/52

► 4/52

► 13/52

► 26/52

20 If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

► $0 \leq r \leq 1$

► $r \geq 0$

► $r = +1$

► $r = -1$

21 If a curve has a longer tail to the right, it is called:

► Positively skewed (Page 39)

► Negatively skewed

► J-shaped

► Symmetric

22 Which one of the following is not included in measures of central tendency?

► Quartile deviation (Page 82)

► Harmonic mean

► Geometric mean

► Arithmetic mean

23 Which of the following is not based on all the observations?

► Arithmetic Mean

► Geometric Mean

► Harmonic mean

► Mode

24 What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

► 49

► 1

► 0 (Standard deviation will always be zero if all the values in data are same)

► 7

25 $10! = \dots\dots\dots$

► 362880

► 3628800

► 362280

► 362800

26 If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

► $1/52$

► $13/52$

► $4/52$

► $26/52$

27 Which dispersion is used to compare variation of two series?

► C.V. (Page 93)

► Q.D.

► M.D.

► S.D.

28 In a regression line $Y = a + bX$, the value of the correlation coefficient will be zero if:

► ► Intercept $a = 0$

► Intercept $a \neq 0$

► ► Slope $b = 0$

► ► Slope $b \neq 0$

29 When two coins are tossed the probability of at least one head is:

► $1/4$

► $3/4$ (HH, HT, TH, TT)

► $2/4$

► $4/4$

30 In the model $Y = mX + a$, Y is also known as the:

► Predictor variable

► Independent variable

► Predicted (dependent) variable (Page 121)

► Explanatory variable

31 According to empirical rule approximately 95% of the measurements will fall under which interval?

► $X \pm S$

► $X \pm 2S$ correct

► $X \pm 3S$

► $X \pm 4S$

32 Which one of the following is written at the top of the table?

► Source note

► Foot note

► Prefatory note

► Title Click here for detail

33 Which one of the following is the class frequency?

- ▶ The number of observations in each class
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

34 If X is a discrete random variable, then the function $f(x)$ is

- ▶ A probability function
- ▶ A probability density function
- ▶ A density function
- ▶ A distribution function (Page 173)

35 If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ▶ Ogive
- ▶ Frequency polygon
- ▶ Frequency curve (Page 38)
- ▶ Histogram

36 If mean of the two observations is 10.5, then median of these two observations will be:

- ▶ 7.5
- ▶ 8.5
- ▶ 9.5
- ▶ 10.5 (Page 68)

37 Which one is the formula of mid range?

$$\frac{x_0 + x_m}{2}$$
 (Page 80)

38 For the given data 2, 3, 7, 0, -8 G. M will be:

- ▶ Negative
- ▶ Positive
- ▶ Zero (Page 75)
- ▶ Undefined

39 Which one is equal to explained variation divided by total variation?

- ▶ Sum of square due to regression
- ▶ Coefficient of determinant
- ▶ Standard error estimate
- ▶ Coefficient of correlation

40 In the given series 1,2,1,1,2,2,2,3,4,5,3,2,3,1,4,2,3 mode of given is

- ▶ 4
- ▶ 3
- ▶ 1

► 2

- 41 True for the population,
- it must be large number of values
 - It must refer to people
 - It is collection of individual objects or measurement (Page 12)
 - It is the small part of whole
- 42 Data arrangement in ascending or descending order
- Array data (Page 47)
 - Group data
 - Ungroup data
 - Raw data
- 43 What is the main objective of Descriptive statistics?
- To test population properties
 - To describe the data we collected
 - To infer something about the population
 - Making estimate
- 44 Which measure of central tendency?
- Variation of distribution
 - Average of distribution (Page 51)
 - Scattering of distribution
 - Dispersion of distribution
- 45 If $a=4$ $b=2$ estimate line (i.e $y=a+bx$) and independent variable has value 3 the value of dependent variable
- 6
 - 9
 - 10 $4+2(3)=10$
 - 11
- 46 The number of ways in which 4 books can be arranged
- 4
 - 6
 - 12
 - 24 (Page 142)
- 47 If we plot paired observed $(x,y)=1.....n$ on graph is called,
- Polygon
 - Freasito diagram
 - Scatter diagram
 - Cumulative frequency diagram
- 48 The simultaneous occurrence of two events is called
- Descriptive probability
 - Subjective probability
 - Conditional probability
 - Joint probability

49 Which one is the not measure of dispersion.

- ▶ The range
- ▶ 50th percentile
- ▶ Inter quartile range
- ▶ Variance

50 In positively skew cure which relation is

- ▶ The mean, median and mode are equal
- ▶ Mean is greater then median
- ▶ Median is greater then mean
- ▶ Standard deviation must be greater then mean or median

51 When coin tossed we get only

- ▶ 1 outcome
- ▶ 2 outcomes
- ▶ 3 outcomes
- ▶ 4 outcomes

52 When mean is 25 and S.D is 5 then CV is

- ▶ 100%
- ▶ 25%
- ▶ 20%
- ▶ 10%

53 For a positively skewed distribution m3 will be:

- ▶ Positive (Page 119)
- ▶ Negative
- ▶ Zero
- ▶ 1

54 When data is labeled to identify an attribute of element, the measurement scale is:

- ▶ Ordinal (Page 9)
- ▶ Interval
- ▶ Nominal
- ▶ Ratio

55 Suppose the estimated equatio $\hat{Y} = 5 - 2$ has been calculated for a set of data. What is slop of the line:

- ▶ 0
- ▶ 2
- ▶ -2 (Page 121)
- ▶ 5

56 If $P(B|A) = 0.25$ and $P(A \cap B) = 0.20$ then $P(A)$ is:

- ▶ 0.05
 - ▶ 0.80 (Page 159)
 - ▶ 0.95
 - ▶ 0.75
- $$P(B|A) = \frac{P(A \cap B)}{P(A)}$$
- $$= P(A) = \frac{P(A \cap B)}{P(B|A)}$$
- $$= \frac{0.20}{0.25} = 0.80$$

57 Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ▶ Advance statistics

- ▶ Probability statistics
- ▶ **Descriptive statistics (Page 61)**
- ▶ Inferential statistics

58 In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors."

This report is an example of :

- ▶ A sample
- ▶ A population
- ▶ Statistical inference
- ▶ **Descriptive statistics**

59 A set that contains all possible outcomes of a system is known as

- ▶ Finite Set
- ▶ Infinite Set
- ▶ **Universal Set (Page 134)**
- ▶ No of these

60 If X and Y are independent, then $\text{Var}(X-Y)$ is equal to:

- $\text{Var}(X) - \text{Var}(Y)$**
- $\text{Var}(X) + \text{Var}(Y)$
- $\text{Var}(X + Y)$
- Zero

61 Which of the following is the class frequency

- ▶ The number of observations in each class
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

66 How to construct the class interval:

- ▶ Divide the class frequencies in half
- ▶ Divide the class frequency by the number of observations
- ▶ **Find the difference between consecutive lower class limits**
- ▶ Count the number of observations in the class

67 Data in the Population Census Report is:

- ▶ **Ungrouped data (Page 11)**
- ▶ Secondary data
- ▶ Primary data
- ▶ Arrayed data

68 What is the range of -2,-3,-5,-10 :

- ▶ **-12 (Page 28)**
- ▶ 8
- ▶ -8
- ▶ 2

69 The algebraic sum of deviations from mean is:

- ▶ Maximum

- ▶ Minimum
- ▶ Zero (Page 86)
- ▶ Undefined

70 The sum of squares of deviations from mean is:

- ▶ Undefined
- ▶ Zero
- ▶ Maximum
- ▶ Minimum

71 Statistic is a numerical quantity, which is calculated from:

- ▶ Population
- ▶ Sample (Page 7)
- ▶ Data
- ▶ Observations

72 If any value in the data is zero, then it is not possible to have:

- ▶ Harmonic Mean
- ▶ Arithmetic Mean
- ▶ Median
- ▶ Mode

73 For a symmetrical distribution...is equidistance from median:

- ▶ X_0 and X_m
- ▶ Q_1 and Q_3 (Page 97)
- ▶ X_0 and Q_1
- ▶ X_m and Q_3

74 Which one of the following measure is not used in 'five number summery':

- ☐ X_0
- ☐ Q_3
- ☐ $\bar{X}$ (Page 97)
- ☐ Q_1

75 If Mean = 25 & S.D is 5 then C.V is

- ▶ 100%
- ▶ 25%
- ▶ 20% (Page 93)
- ▶ 10%

76 A coin is tossed 3 times then, the number of sample points in the sample space is:

- ▶ 3
- ▶ 8
- ▶ 6 (Page 145)
- ▶ 4

77 What is the difference between a permutation and a combination:

- ▶ In a permutation order is important and in a combination it is not
- ▶ In a permutation order is not important and in a combination it is important

- ▶ A combination is based on the classical definition of probability
- ▶ A permutation is based on the classical definition of probability

78 What we consider in simple correlation analysis:

- ▶ Several independent variables
- ▶ Strength of the association between two variables (Page 128)
- ▶ Intercept with the X-axis
- ▶ Intercept with the Y-axis

79 If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

- ▶ $0 < r < 1$
- ▶ $r = 0$
- ▶ $r = +1$ (rep)
- ▶ $r = -1$

80 Which of the following measure of dispersion is least affected by extreme values of observations in a data:

- ▶ Range (Page 82)
- ▶ Quartile deviation
- ▶ Mean absolute deviation
- ▶ Standard deviation

81 The approximate number of observations in a set of data covered by the interval, $\text{Median} \pm Q.D$ are:

- ▶ 50 per cent (Page 84)
- ▶ 68.5 per cent
- ▶ 95.4 per cent
- ▶ 99 per cent

82 The dispersion which is calculated from all observed values is:

- ▶ Standard deviation (Page 87)
- ▶ Quartile deviation
- ▶ Rang
- ▶ Coefficient of Rang

83 In a set of 20 values all the values are 2, what is the value of Geometric Mean:

- ▶ 2 (Page 75)
- ▶ 5
- ▶ 10
- ▶ 20

84 In a set of 10 values all the values are 5, what is the value of 5th Decile:

- ▶ 2
- ▶ 5
- ▶ 10
- ▶ 20

85 If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

- ▶ W.M
- ▶ H.M

- ▶ A.M
- ▶ G.M

86 In stem and leaf plot, data is measured on:

- ▶ Ratio Scale
- ▶ Interval Scale
- ▶ Ordinal scale
- ▶ Nominal Scale

87 In statistics, we deal with:

- ▶ Individuals
- ▶ Isolated items
- ▶ Aggregates of facts (Page 7)
- ▶ Qualitative data

88 Which of the following is a systematic arrangement of data into rows and columns:

- ▶ Component bar chart
- ▶ Classification
- ▶ Tabulation
- ▶ Bar chart

89 For any number $k \geq 1$, at least $1 - 1/k^2$ of the data-values fall within k standard deviations of the mean:

- ▶ Greater than 1 (Page 97)
- ▶ Less than 1
- ▶ Greater or equal to 1
- ▶ Less or equal to 1

90 When E is an impossible event, then $P(E)$ is:

- ▶ 0 (Page 146)
- ▶ 0.5
- ▶ 1
- ▶ 2

91 The data for an ogive is found in which distribution:

- ▶ A cumulative frequency distribution (Page 43)
- ▶ A relative frequency distribution
- ▶ A frequency distribution
- ▶ A joint frequency distribution

92 Which of the following statements is true regarding a sample:

- ▶ It is a part of population (Page 13)
- ▶ It must contain at least five observations
- ▶ It refers to descriptive statistics
- ▶ It produces True value

93 Which level of measurement is required for the median:

- ▶ Nominal
- ▶ Ordinal
- ▶ Interval

► Ratio

94 In a the regression line $Y = a + bX$ € the variable which is non-random is:

► X (Page 122)

► Y

► Both X and Y

► Neither X nor Y

95 The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

► Scatter variable

► Independent variable

► Dependent variable

► Correlation variable

96 Which is the formula of range:

► $x_m - x_0$ (Page 28)

► $x_0 - x_m$

► $\frac{x_m - x_0}{2}$

► $\frac{x_m + x_0}{2}$

97 The descriptive measure of dispersion that is based on the concept of a deviation about the mean is:

► Range

► Inter quartile range

► Quartile deviation

► Standard deviation

98 Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

► Advance statistics

► Probability statistics

► Descriptive statistics

► Inferential statistics

99 A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

► Infinite population

► Finite population

► Concrete population

► Hypothetical population (Page 12)

100 When two dice are rolled, the numbers of possible sample points are:

► 6

► 12

► 24

► 36 (6*6=36)

101 The correlation of coefficient lies between :

► 0 to 1

► 0 to ∞

► - 1 to +1 (Page 128)

► +1 to ∞

102 Median can be found only when:

- Data is Discrete
- Data is Attributed Data is continuous
- Data is continuous Data is attributed

► Data is arranged (Page 52)

103 From the following observations 2,3,4,5,4,6,4, the mode is:

- 2
- 3
- 4
- 5

104 How to construct the class interval:

- Divide the class frequencies in half
- Divide the class frequency by the number of observations
- Find the difference between consecutive lower class limits
- Count the number of observations in the class

105 How many elements are in the sample space of rolling one die:

- 6
- 12
- 24
- 36

106 When two coins are tossed the probability of at most one head is:

- 1/4
- 2/4
- 3/4
- 4/4

107 In scatter diagram the variable plotted along Y-axis is:

- Independent variable
- Dependent variable
- Any one
- Undefined

108 Positive square root of variance of a distribution is:

- Rang
- Quartile deviation
- Standard deviation (Page 91)
- only (a) &(c)

109 When more values are lying at the start of the distribution, it is a:

- Symmetrical distribution
- Positively skewed
- Negatively skewed
- U shape figure

- 110 What is 'm f' in the formula of mode:
- ▶ First frequency
 - ▶ Last frequency
 - ▶ **Maximum frequency** (Page 54)
 - ▶ Minimum frequency
- 111 If median = 7 and Mean = 5, what is the value of 2 Q :
- ▶ 1
 - ▶ 3
 - ▶ 5
 - ▶ **7** (because median = Q2)
- 112 The probability of drawing a king of spade from a pack of 52 cards is:
- ▶ 1/4
 - ▶ 1/13
 - ▶ 1/26
 - ▶ **1/52**
- 113 When referring to a curve whose longer tail is to the left, you would call it:
- ▶ U shape
 - ▶ **Skewed to the left** (Page 39)
 - ▶ Skewed to the right
 - ▶ Symmetrical
- 114 In statistics, we deal with:
- ▶ Individuals
 - ▶ Isolated items
 - ▶ **Aggregates of facts** (Page 7)
 - ▶ Qualitative data
- 115 The distribution is mesokurtic if the Moment Coefficient of kurtosis b_2 is:
- ▶ Equal to 0
 - ▶ **Equal to 3**
 - ▶ Less than 3
 - ▶ Greater than 3
- 116 A histogram consists of a set of adjacent rectangles whose bases are marked off by:
- ▶ **Class boundaries** (Page 32)
 - ▶ Class limits
 - ▶ Class marks
 - ▶ Class frequency
- 117 Stem and leaf was introduced by:
- ▶ Francis Galton
 - ▶ Friedman
 - ▶ John Tukey (Page 47)
 - ▶ Pearson

118 For positively skewed distribution

Mean.....Median.....Mode:

- ▶ =
- ▶ <
- ▶ >
- ▶ ≠

119 For a positively skewed distribution:

- ▶ Mean<Mode<Median
- ▶ Median<Mode<Mean
- ▶ Mode>Mean>Median
- ▶ Mean>Median>Mode

120 5C_5 5 C Equals :

- ▶ 1
- ▶ 5
- ▶ 10
- ▶ 25

121 If a curve has a longer tail to the right, it is called :

- ▶ Positively skewed (Page 39)
- ▶ negatively skewed
- ▶ J-shaped
- ▶ Symmetric

122 In measures of relative dispersion unit of measurement is:

- ▶ Changed
- ▶ Vanish (Page 82)
- ▶ Does not changed
- ▶ Dependent

123 Data used by an agency which is originally collected by them is :

- ▶ Primary data (Page 11)
- ▶ Raw data
- ▶ Secondary data
- ▶ Grouped data

124 When we toss a coin , we get only:

- ▶ 1 outcome
- ▶ 2 outcomes
- ▶ 3 outcomes
- ▶ 4 outcomes

125 Which of the following is the class frequency

- ▶ The number of observations in each class
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

126 Standard deviation is affected by the change of _____:

- ▶ Origin & scale
- ▶ Origin only
- ▶ Scale only [Click here for detail](#)
- ▶ Not origin & scale

127 For a moderately skewed distributions, the approximate percentage of case included between

$\bar{X} - 2S$ and $\bar{X} + 2S$ is :

- ▶ 99.7%
- ▶ 68%
- ▶ **95%** (Page 95)
- ▶ 50%

128 A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- ▶ 1/2
- ▶ 1/3
- ▶ **2/3** (greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)
- ▶ 5/6

129 The probability of drawing any one spade card is:

- ▶ 1/52
- ▶ 4/52
- ▶ **13/52**
- ▶ 52/52

130 Which of the following is not the type of frequency curves?

- ▶ The symmetrical frequency curve
- ▶ The extremely skewed frequency curve
- ▶ The U-shaped frequency curve
- ▶ **Frequency polygon** (Page 38)

131 If the third moment about mean is zero 3 ($m_3 = 0$) , then the distribution is:

- ▶ **Symmetrical**
- ▶ Negatively skewed
- ▶ Positively skewed
- ▶ Mesokurtic

A histogram is consists of a set of a set of adjacent rectangle whose bases are marked off by:

Class boundaries (Lecture no 4 Page no 32 Handouts)

Class limits

Class frequency

Class marks

The middle value of an ordered array of numbers is the

Mean

Median (Lecture no 7 Page no 59 Handouts)

Mode

Midpoint

If Mean =25 & S.D is 5 then C.V is:

100%

20% (Lecture no 11 Example Page no 88 Handouts)

10%

25%

Solution: C.V = (S.D/X) x 100 = (5/25) x 100 = 20% Answer

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, you will get:

Ogive

Frequency polygon

Frequency curve

Historigram

The conditional probability $P(A/B)$ is:

$P(A \cap B)/P(B)$ (Lecture no 20 Page no 154 Handouts)

$P(A \cap B)/P(A)$

$P(A \cup B)/P(B)$

$P(A \cup B)/P(A)$

Serious disadvantage of using range as a measure of dispersion is that it is based on only:

Minimum Values

Maximum Values

Both Minimum and Maximum values

None of the above

Frequency of a variable is always in:

Fraction form
Percentage form
Less than form
Integer form

5C_5 is equal to

- a. 5
- c. 10

- b. 1**
- d. 24

Chebychev's inequality does not hold for $k = ?$

- 3
- 2
- 1

0 (Lecture no 12 Page no 94 Read CHEBYCHEV'S THEOREM)

If $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$ then $A - B$ will be:

- $\{1,2\}$**
- $\{3,4\}$
- $\{3,2,1\}$
- $\{1,2,3,4,5,6\}$

If $P(B|A) = 0.25$ and $P(A \text{ and } B) = 0.20$, then $P(A)$ is

- 0.05
- 0.80**
- 0.95
- 0.75

Solution: $P(B|A) = [P(A \text{ and } B)] / P(A)$

$$P(A) = [0.20 / P(B|A)]$$

$$P(A) = [0.20 / 0.25]$$

$$\mathbf{P(A) = 0.80}$$

According to Empirical rule, approximately 68% of the measurements will fall within:

- (Mean - S.D, Mean + S.D) (Lecture no 12 Page no 90 Handouts)**
- (Mean - 2S.D, Mean + 2S.D)
- (Mean - 3S.D, Mean + 3S.D)
- None of these

For Platykurtic distribution, b_2 (moment ratio) will be:

- Greater than 3
- Less than 3 (Lecture no 14 Page no 114 Handouts)**
- Equal to 3

Equal to zero

There are two broad categories of data, which are:

Weighted and Un-weighted

Grouped and Un-grouped

Qualitative and Quantitative (Lecture no 3 Page no 16 Handouts)

Primary and Secondary

In a linear regression, $Y = a + bX$, the variable "Y" will always:

A random variable (Lecture no 15 Page no 117 Handouts)

A non-random variable

Qualitative variable

Quantitative variable

In regression line $Y = a + bX$, X is called:

Dependent variable

Independent variable (Lecture no 15 Page no 116 Handouts)

Explained variable

Regressand

Difference between the largest and the smallest data values is called

variance

interquartile range

range

coefficient of variation

The extremely positively skewed curve is also known as:

Frequency curve

U-shaped curve

J-shaped curve

Reverse J-shaped curve (Lecture no 5 Page no 35 Handouts)

A list of 7 pulse rates is: 70, 64, 80, 74, 92, 96, 98. What is the median for this list?

a. 70

b. 80

c. 92

d. 98

Solution: Median is the middle value. First of all we will arrange them in ascending order 64, 70, 74, **80**, 92, 96, 98

20. The value of $10C_9$:

a. 45

b. 35

c. 35

d. 10

21. The most frequent value in the data is called

a. Mean

b. Median

c. Mode

d. Harmonic mean

An event that contains more than one sample points is called:

Mutually exclusive event

Not mutually exclusive event

Hyper event

Compound event (Lecture no 17 Page no 140 Handouts)

Calculate range for the following data: 22, 22, 30, 32, 37, 48, 60, 88, 90.

a. 22

b. 90

c. 37

d. 68

Solution: Largest value – Smallest value = Range

90 – 22 = **68 Answer.**

24. What is the median of this set of numbers: 4, 6, 7, 9, 2000000?

a. 9

b. 6

c. 7.5

d. 7

25. The value of the middle term in a ranked (ordered) data set is called the:

a. Mode

b. Mean

c. Median

d. Harmonic mean

26. The median is _____.

a. The highest number

b. The middle point

c. The average

d. Affected by extreme scores

27. The Mode of 8, 5, 7, 10, 15, 21, 5, 7, 2, 5 is

a. 8

b. 5

c. 7

d. 21

Note: Mode is the most repeated value so in this case 5 is the most repeated value.

When all the values falling in a class are equal to the mid point of the class interval is called?

Random error

Unbiased Error

Biased Error

Grouping Error (Lecture no 7 Page no 56 handouts)

For a Leptokurtic distribution, b_2 (moment ratio) will be:

Greater than 3 (Lecture no 14 Page no 114 Handouts)

Less than 3
Equal to 3
Equal to zero

Let $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$ Then $A \cap B$:

$\{3,4\}$
 $\{1,4\}$
 $\{3,5\}$
 $\{3,6\}$

The range of the scores 29, 3, 143, 27, 99 is:

- a. **140** b. 143
c. 146 d. 70

Solution: $(143 - 3) = 140$ Answer

In how many ways, a team of 11 players can be chosen from a total of 16 players?

- a. **4368** b. 2426
c. 5400 d. 2680

Direct personal investigation is _____ when the area to be covered is vast. **(Doubtful) Note: It can be both read lecture no 2 Page no 6 Handouts.**

Costly (Lecture no 2 Page no 6 Handouts)
Time-consuming (Lecture no 2 Page no 6 Handouts)
Localized inquires
Inaccurate

What is the mean of this set of numbers: 4, 6, 7, 9, 200000?

- a. 7.5 c. 7
b. **400,005.2** d. 4

Solution: Mean = $(4+6+7+9+200000)/5 = 400,005.2$ **Answer**

According to this empirical rule, approximately how much values will fall within (Mean – 3S.D, Mean + 3S.D)?

- a. **100% (Lecture no 12 Page no 90 Handouts)** b. 95%
c. 75% d. 68%

In a linear regression, $Y = a + bX$, the variable "X" will always: **a.** A random variable

A non-random variable (Lecture no 15 Page no 117 Handouts)
Qualitative variable
Quantitative variable

Variance is expressed in _____ units as the units of data set.

Squared (Lesson no 11 Page no 85 Handouts)

Cube
Single
Same

In a linear regression, best fitted line is obtained through: **Page 117**

- a. Method of moment
- c. Method of least square**
- b. Method of likelihood
- d. Method of semi average

39. In case of frequency distribution, the median is given by the formula:

- a. $l + \frac{h}{f} \left(\frac{n}{2} - 2c \right)$
- b. $l + \frac{h}{f} \left(\frac{n}{2} - c \right)$**
- c. $l + \frac{f}{h} \left(\frac{n}{2} - c \right)$
- d. $l + \frac{f}{h} \left(\frac{n}{4} - c \right)$

40. The sum of squared deviations from mean is: **page 119**

- a. Maximum
- b. Minimum**
- c. Zero
- d. Undefined

41. In a week the prices of a bag of rice were 350,280,340,290,320,310,300.

- a. 320
- b. 315**
- c. 300
- d. 420

Which scale will you use to measure the temperature?

Nominal scale

Interval scale (Lecture no 1 Page no 4 Handouts)

Ratio scale

Ordinal scale

From the given data, which one of the following chart we can draw: Year Turnover

1965 350000
1966 42000
1967 43500
1968 48000
1969 48500

- a. Simple bar chart**
- b. Multiple bar charts
- c. Component bar chart
- d. Pie chart

In a Venn diagram, the overlap between two circles represents:

The union of two sets

The intersection of two sets

The elements that are in either of two sets

The difference between the number of elements in two sets

Calculate range for the following data: 10, 32, 33, 34, 37, 42, 55, 58, 70

- Mutually exclusive event
- Normal event
- Simple event
- Compound event**

Frequency distribution is considered as negatively skewed if all values of distribution moves to

- a. **lower tail**
- b. median tail
- c. variance tail
- d. upper tail

Which of the following is NOT a common measure of central tendency?

Page 77

- a. Mode
- b. **Range**
- c. Median
- d. Mean

From the table given below, how many students obtained marks between 60 and 69?

Marks f Mid-Points
50-59 5 54.5
60-69 7 64.5
70-79 8 74.5

80-89 5 84.5

- a. 64.5
- b. 12
- c. 60
- d. **7**

If the first and third quartiles are 22, 16 and 56,36 respectively, then the quartile deviation is: **$Q_3 - Q_1 / 2$**

- a. **17.1**
- b. 30.5
- c. 50.5
- d. 51.3

Adding all the squared deviations taken from mean and dividing by the number of observations, we get: **Page 85**

- a. Standard Deviation
- b. **Variance**
- c. Mean Deviation
- d. None of the above

Standard deviation divided by mean is known as: **Page 87**

Co-efficient of standard deviation

Co-efficient of variation

Both

None

5C5 equals to:

- a. 5
- b. **1**
- c. 10
- d. 15

Which of the measure of dispersion is used to compare variation between two series? **Page 88**

- a. **C.V.**
- b. Q.D.
- c. M.D.
- d. S.D.

62. If $Y=3X+5$, then S.D of Y is equal to **(Doubtful)**

- a. $9 s. d(x)$ b. $3 s. d(x)$ c. $s. d(x)+5$ **d. $3 s. d(x) + 5$**

Which of the following technique is not used to represent the bivariate qualitative data?

- a. Component Bar Chart b. Multiple Bar Chart
c. Line Chart d. Pie Chart

When a frequency distribution involves “open-end” classes, then which average is appropriate?

- Mean
 Mode
Median (Lecture no 8 Page no 62 Handouts)
 None of these

Harmonic mean is extremely useful in averaging _____ types of data.

- Ratios
 Rates
Both ratios and rates
 None of the above

According to Empirical rule, approximately how much values will fall within (Mean-3S.D, Mean+3S.D)?

- 100% (Lecture no Page no 90 Handouts)**
 95%
 75%
 68%

What is probability of drawing two clubs from a well shuffled pack of 52 cards?

- a. $13/51$ **b. $1/17$** c. $1/26$ d. $13/17$

Solution: Total Cards = 52 cards

Number of club cards = 13 Probability of 1st card being club card = $13/52$

Probability of 2nd card being club card = $12/51$

∴ Total Probability

= $(13/52) \times (12/51)$

= **$1/17$ Answer**

Which pair of measure cannot be calculated when one of numbers in the series is zero?

- a. G.M. and A.M. b. H.M. and A.M.
c. G.M. and H.M. d. None of these

Which of the following techniques is used to predict the value of one variable on the basis of other variables?

- a. Correlation analysis
- b. Coefficient of correlation
- c. Covariance
- d. Regression analysis**

A bag contains 12 red balls and 12 blue balls. A ball is drawn at random. The probability that ball drawn is red is

- a. $1/2$**
- b. $5/11$
- c. $6/10$
- d. 1

Solution: Total Red Balls: 12

Total blue balls: 12

Total Balls: $12 + 12 = 24$

Probability of ball drawn is a red ball:

$P(\text{Red}) = 12/24 = 1/2$ **Answer**

If the GM of a set of two observations is 10 and its HM is 8, then the AM of the set of observations is

- a. 100 b. 12.5 c. 64 d. 7.5

RELATION BETWEEN AM, GM AND HM - FORMULA

Relation between AM, GM and HM is:

$$AM \times HM = GM^2$$

Solution: GM = 10, HM = 8, $GM^2 = (10)^2 = 100$, $AM \times HM = GM^2$,

AM x 8 = 100, AM = 100/8 = 12.5 Answer

72. If $A = \{H, T\}$ then which of the following is power set of A? **Page 133**

- a. $\{\{\}, \{H, T\}\}$ b. $\{\{H\}, \{T\}, \{H, T\}\}$
c. $\{\{H\}, \{T\}, \{H, T\}, \{T, H\}\}$ d. $\{\{\}, \{H\}, \{T\}, \{H, T\}\}$

What do understand by the term power set?

In set theory, the **power set** (or **power set**) of a **Set A** is defined as the **set** of all subsets of the **Set A** including the **Set** itself and the null or empty **set**. It is denoted by $P(A)$. Basically, this **set** is the combination of all subsets including null **set**, of a given **set**. Simple formula 2^n . n is the number of subsets. $2^2 = 4$.

A = {H, T} = $\{\{\}, \{H\}, \{T\}, \{H, T\}\}$ Answer

73. 2nd Quartile = 5th Decile = 50th Percentile =? **Page 65**

- a. Geometric Mean b. **Median**
c. Mode d. Harmonic Mean

74. The total number of observations, which are below a certain value are known as

- A. class boundaries B. class marks
C. cumulative frequency D. variances

Histogram can be drawn only for:

- A. Discrete frequency distribution **B Continuous frequency distribution**
C Continuous frequency distribution D Relative frequency distribution

76. Classification is the process of arranging data according to:

- A. One characteristic B. Two or more characteristic
C. Similar characteristic D. None of these

77. Which of the following, measures the dispersion around mean?

- a. Mean deviation
- b. Standard deviation
- c. Mean deviation and Standard deviation**
- d. None of these

78. What is mode for the following set of data: 1,1,2,2,5,5,7

- a. 1
- b. 1,2
- c. 1,2,5**
- d. no mode in the data

The mode is the value that appears most frequently in a data set. A set of data may have one mode, more than one mode, or no mode at all. Other popular measures of central tendency include the mean, or the average of a set, and the median, the middle value in a set.

Important: The mode can be the same value as the mean and/or median, but this is usually not the case.

Which of the following averages give information about central value in the distribution?

- a. Mean
- b. Median**
- c. Mode
- d. Harmonic mean

80. In a Pie diagram, the sector of a circle is obtained by:

- a. (component part/total)*100**
- b. (component part/total)*360
- c. (component part/total)*180
- d. (component part/total)*300

81. Relationship among the averages **Page 75**

- a. $HM \geq GM \geq AM$
- b. $AM \geq GM \geq HM$**
- c. $GM \leq HM \leq AM$
- d. $AM \geq HM \leq GM$

82. Which of the scale is best to use for measuring the salary of an employee?

- a. nominal
- b. ordinal
- c. interval
- d. ratio**

The mean of a distribution is 30, the mode is 24 and the standard deviation is 4,
then the coefficient of skewness will be: $\frac{\text{mean} - \text{mode}}{\text{standard deviation}}$

- a. Less than zero
- b. Equal to zero
- c. Greater than zero**
- d. None of the above

Smaller standard error of estimate shows: **[page 123]**

Data points are very far to the line

Data points are close to the line

There is no difference between line and points

Difference is additive

Data arranged in ascending or descending order of magnitude is called: **Page 42**

- a. Ungrouped data
- b. Grouped data
- c. Discrete frequency distribution
- d. Arrayed data**

86. A circle in which sectors represents various quantities is called:

- a. Histogram
- b. Frequency Polygon
- c. Pie Chart**
- d. Component Bar Chart

Sum of the absolute deviations of the values is least when deviations are taken from:

- a. Mean
- b. Median**
- c. Mode
- d. G.M

Chebychev's inequality is valid for the data set

- a. Sample
- b. Entire population
- c. Both sample and entire population**
- d. None of the above

Which of the following terms best describes data that were originally collected at an earlier time by a different person for a different purpose?

- a. Primary data
- b. Secondary data**
- c. Experimental data
- d. Field notes

90. Statistics deals with

- a. Individuals
- b. Isolated items
- c. Observations**
- d. Aggregates of facts

If a box contains six red, three blue and five pink ties then probability of blue ties will be equal to:

- a. $1/14$
- b. $3/14$**
- c. $5/14$
- d. $6/14$

Which of the rule is applied to any data set, regardless shape of the frequency distribution? **Page 89**

- a. Chebychev's rule**
- b. Empirical rule
- c. Combination rule
- d. Permutation rule

Which average is used in the situation where the number of floors in the buildings at the center of a city?

- a. Mean
- b. Median**
- c. Mode
- d. Variance

Rankings of the finishes of competitors in a foot race is an example of a(n)_____.

- a. ratio scale
- b. ordinal scale**
- c. nominal scale
- d. interval scale

Among 18 articles, six having minor Defects and three have major defects. Determine the probability that an article selected at random has major defect.

- a. **1/6** b. 1/5 c. 0.25 d. 0.11

A series of data with exclusive classes along with the corresponding frequencies is called:

- a. Discrete frequency distribution **b. Continuous frequency distribution**
c. Percentage frequency distribution d. Cumulative frequency distribution

97. Using the following table, calculate $P(X < 2)$ **Page 171**

X	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

- a. 1/8 b. 3/8 **c. 4/8** d. 7/8

98. If a distribution has two modes then the distribution is called: **Page 53**

- a. Uni-Modal **b. Bi-Modal** c. Tri-Modal d. Multi-Modal

A group consists of three persons, A, B, & C. How many ways are there of selecting a group of two persons out of these three?

- a. 1 b. 2 **c. 3** d. 6

100. If any of the value in data set is zero then it is not possible to compute

- a. Mean b. Mode
c. Geometric Mean **d. Harmonic Mean**

101. By using identity law A U S is equal to: **page 133**

- a. A **b. S** c. A&S d. None of them

102. Which average gives the more weightage to the smaller values? **Google**

- a. Harmonic Mean** b. Geometric Mean
c. Arithmetic Mean d. None of the above

If an observation in the data set is zero, then its geometric mean will be:
Google

- a. Positive b. Negative **c. Zero** d. Indeterminant

In a Box and Whisker plot, a line which divides the box into two equal parts is referred to as: **Page 99**

- a. Mean **b. Median** c. Mode d. Range

105. The Random errors are also called **Page 5**

- a. Systematic Errors b. Biased Errors

d. Unbiased Errors

a. 4 b. 8 c. 12 **d. 16**

a. Mode
b. Median
c. **Mean**
d. All above

a. 4th Decile
b. 2nd Quartile
c. 51th Percentile
d. 50th Percentile

a. 4th Decile
b. 3rd Quartile
c. 51th Percentile
d. 75th Percentile

a. Theoretically concept
c. Scientific concept

b. Mathematical concept
d. Probabilistic concept

a. Histogram
b. Frequency Polygon
c. Ogive
d. Frequency Curve

a. A sample
b. A Parameter
c. A Population
d. Both sample and parameter

X	0	1	2	3
f(x)	1/6	1/3	1/3	1/6

a. 1 **b. 0** c. $1/6$ d. Not possible to find

In a five number summary, which of the following is not used for data summarization? **Page 65**

- a. the smallest value
- c. the mean**
- b. the largest value
- d. the 25th percentile

115. The arrangement of data in rows and columns is called:

- a. Classification
- b. Tabulation**
- c. Frequency distribution
- d. Cumulative frequency distribution

116. Relative measure of dispersion for Quartile deviation is: **Page 80**

- a. Co-efficient
- b. Co-efficient of Quartile Deviation**
- c. Co-efficient of Deviation
- d. None of the above

117. If the mean of 6 numbers is 41 then the sum of these numbers is **[6x41]**

- a. 250
- b. 246**
- c. 134
- d. 456

Which branch of statistics deals with the techniques that are used to organize, summarize and present the data? **Google**

- a. Probability Statistics
- c. Descriptive Statistics**
- b. Inferential Statistics
- d. Bayesian Statistics

119. One of the major type of statistics is

- a. Basic
- b. Descriptive**
- c. Advanced
- d. Probabilistic

120. Using the following table, choose the correct answer:

X	0	1	2	3
f(x)	1/6	1/3	1/3	1/6

$P(1 < X < 3)$ equals to:

- a. 1/3**
- b. 2/3
- c. 5/6
- d. 4/6

A value which has half of the observations above it and half the observations below it, is known as:

- a. Mean
- b. Median**
- c. Mode
- d. Standard deviation

122. When data are classified according to a single characteristic, it is called:

- a. Quantitative classification
- b. Qualitative classification
- c. Area classification
- d. Simple classification**

123. The difference of Mode and Mean is equal to

- a. 3(mean-median)**
- b. 2(mean-median)
- c. 3(mean-mode)
- d. 2(mode-mean)

124. Which of the following comes first to make frequency distribution: **Page**

23 a. No. of Groups b. Class interval c. **Range** d. Tally marks 125. Sum of the deviations taken from the mean is equal to:

- a. Positive number **b. 0**
- c. Negative number d. 1

126. In a frequency distribution the categories the categories must

- a. Be mutually exclusive b. Have at least 5 observations
- c. Be of the same size** d. Be of nominal scale

Which of the following statements about histogram rectangles is correct?

The rectangles are proportional height to the number of items falling in the classes

There are generally five rectangles in every histogram

Heights of all rectangles are equal

None of these

128. Using the following table, calculate $P(X < 1)$

X	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

- a. 0 b. 1 c. 1/4 **d. 1/8**

A coin is tossed 4 times in succession. What is the probability that at least one head occurs? **Page 145**

- a. 15/16** b. 1/16 c. 2/16 d. 16/15

130. When a coin is tossed, then sample space consists of:

- a. 2 outcomes** b. 4 outcomes c. 6 outcomes d. 8 outcomes

If we flip a coin three times, then the possible outcomes of the sample space are: **Page 145**

- a. HHH TTT THT HTH HHT TTH HTH
- b. HHH HTT HTH TTT HTT THH HHT THT
- c. HTT THT HTH HHH TTH TTT
- d. HHH HHT HTH HTT THH THT TTH TTT**

The number of classes in a frequency distribution generally should be: **Page 24**

- a. Less than five b. More than five
- c. Between five and twenty **d. Between ten and twenty**

133. If $P(A) = 0.45$, $P(B) = 0.35$ and $P(B \text{ and } A) = 0.25$, then $P(B | A)$ is:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

- a. **0.555** b. 0.801 c. 0.751 d. 0.750

134. If $\sum \{x P(x)\} = 5.925$ and $S.D = 1.3$, then $C.V(X)$ will be:

- a. 16.7% b. 20.1% c. 14.5% **d. 21.9%**

135. The occurrence of a 6 when a die is thrown is an example of: **Page 141**

- a. Mutually exclusive event b. Random event
c. Simple event d. Compound event

In case of frequency distribution, the median is given by the formula: **Page**

- a. $I + \frac{h}{f} (3n/4 - c)$ **b. $I + \frac{h}{f} (n/2 - c)$**
c. $I + \frac{f}{h} (3n/4 - c)$ d. $I + \frac{3f}{h} (n/4 - c)$

A fair die is rolled. Probability of getting even face given that face is less than 5 is given by:

- a. **1/2** b. 5 c. 2 d. 6

138. The data which has not undergone any statistical treatment is called:

- a. Discrete data b. Secondary data
c. Primary data d. Qualitative data

The occurrence of sum of 11 when a die is thrown is an example of:
Page 141

- a. Mutually exclusive event b. Random event
c. Simple event **d. Compound event**

140. Data used by an agency which is originally collected by them is called:

- a. Primary data** b. Ungroup data c. Secondary data d. Grouped data

141. In regression analysis, the variable that is being predicted is the **Page 121**

- a. Dependent variable** b. Independent variable
c. Intervening variable d. None of these

Two dice are tossed. The probability that the total score is a prime number is: (e.g. 2, 3, 5, 7, 11).

- a. 1/6 **b. 5/12** c. 1/2 d. 7/9

Total number of words formed by 2 vowels and 3 consonants taken from 4 vowels and 5 consonants is equal to **[4C2 x 5C3] = [6x10]**

- a. **60** b. 120 c. 720 d. 520

144. If a curve has longer tail to the right, it is called: **Page 34**

- a. Negatively skewed distribution **b. Positively skewed distribution**
c. Symmetrical distribution d. J-shaped distribution

Which of the following measures of dispersion are based on deviations from the mean? **Google**

- a. Variance b. Standard deviation
c. Mean deviation **d. All of the these**

Bowley's coefficient of skewness for a certain data is ZERO. What is the shape of the distribution of data? **Page 109**

- a. Symmetrical** b. Asymmetrical c. Uniform d. Bi-modal

The probability of drawing 'zero white' ball from a bag containing 4 red, 8 black and 3 white balls is:

- a. 0** b. 12/15 c. 8/15 d. $\frac{1}{2}$

Mean deviation is a measure of dispersion in which deviations are taken around the:

- a. Mean** b. First Quartile
c. Third Quartile d. None of the above

149. In a simple bar chart: **Page 19**

- a. Width of bar is meaningful b. 0.5
b. Height of bar is meaningful
c. Height of bar is not meaningful
d. Height and width both are meaningful

150. A sector diagram is also called:

- a. Bar diagram b. Histogram c. Histogram **d. Pie diagram**

If $P(A) = 0.7$ and $P(B) = 0.2$ then find $P(A \cup B)$ where A and B are mutually exclusive events. **$P(A \cup B) = P(A) + P(B)$**

- a. 0.1 **c. 0.9** c. 0.14

152. Absolute dispersion measures spread in terms of: **Page 77**

a. Same units
c. Free of units

b. Different units
d. Independent of units

A fair coin is tossed three times. What is the probability that at least one head appears?

a. $1/8$ b. $7/8$ c. $4/8$ d. $6/8$

154. Harmonic mean is better than other means if the data are of

a. Speed or rates
c. Binary values like 0 & 1

b. Heights or lengths
d. Ratios or proportions

155. Which of the following is not an example of random sampling? **Page 10**

a. Purposive sampling
c. Cluster sampling

b. Systematic sampling
d. Stratified sampling

If X is a discrete random variable and $f(x)$ is the probability of X, then the expected value of this random variable is equal to: **page 64**

a. $\sum[f(x)+x]$ b. $\sum f(x+3)$ c. $\sum xf(x)$ d. $\sum f(x)$

157. The suitable average for the qualitative data is: **Google**

a. Mean b. **Median** c. Geometric mean d. Weighted mean

158. Relative dispersion is expressed in terms of: **Page 77**

a. Addition
c. **Ratio**

b. Subtraction
d. All of these

In regression line $Y = a + bX$, Y is called: **Page 121**

Dependent variable
Independent variable
Explanatory variable
Regressor

In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?

a. 970 b. **50400** c. 640 d. 720

Explanation:

161. Statistic is a numerical quantity which is calculated from **Page 2**

a. Data b. Observations c. Population d. **Sample**

To find the average speed of a journey which is the appropriate measure of central tendency.

Vowel O is 3 times, so vowels can be arranged = $5!/3!$

= 20

- a. Geometric Mean b. Weighted Mean c. Mode **d. Harmonic Mean**

There are 30 people in a group. If all shake hands with one another, how many handshakes are possible?

- a. 870 b. **435** c. 291 d. 370

Step-by-step explanation:

Given : There are 30 people in a group.

To Find: Now we are given that if all shake hands with one another, how many handshakes are possible?

Solution :

We will use combination to find the no. of handshakes .

Formula of combination : ${}^nC_r = \frac{n!}{r!(n-r)!}$

$$n = 30$$

$$r = 2$$

$${}^{30}C_2 = \frac{30!}{2!(30-2)!}$$

$${}^{30}C_2 = \frac{30!}{2!(28)!}$$

$${}^{30}C_2 = \frac{30 \times 29 \times 28!}{2! \times 28!}$$

$${}^{30}C_2 = \frac{30 \times 29}{2 \times 1}$$

$${}^{30}C_2 = 435$$

OR

Step-by-step explanation:

From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?

- a. 1/15 b. 25/57 c. 1/122 d. **1/221**

which is : $a + (n-1)d$

Let the first term is 0

Two cards can be drawn from a pack of 52 playing cards in ${}^{52}C_2$ ways,

$$\text{i.e., } \frac{52 \times 51}{2} = 1326 \text{ ways}$$

The event that two kings appear in a single draw can appear in 4C_2 ways, i.e., 6 ways,

$$\therefore \text{The probability that the two cards drawn from a pack of 52 cards are kings} = \frac{6}{1326} = \frac{1}{221}$$

$$= 29 \times 15 =$$

$$435$$

OR

Therefore the Probability that **BOTH** cards are Kings is:

$$\text{Prob (2 Kings)} = (4/52) * (3/51)$$

1) β is the probability of

Select the correct option

A) Reject H_0/H_0 is true

B) Reject H_0/H_0 is false

C) Accept H_0/H_0 is false (Correct)

D) Accept H_0/H_0 is true

2) An estimator is said to be efficient if it hasSelect

the correct option

- A)** Smallest Variance
- B)** Unbiased estimate
- C)** Both (A) & (B) (Correct)
- D)** None of these

3) Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

Select the correct option

- A)** σ (sigma)
- B)** β (beta)
- C)** θ (theta)
- D)** α (alpha) (Correct)

4) The end points that bound the confidence interval are called

Select the correct option

- A)** Lower limits
- B)** Bounded its
- C)** Lower and upper limits (Correct)
- D)** Upper limits

5) An estimator is said _____ to be is expected value is equal to true value of its parameter.

Select the correct option

- A)** Efficient
- B)** Consistant
- C)** Based
- D)** Unbiased (Correct)

6) The width of the confidence interval is called _____ of the estimate.

Select the correct option

- A)** Biased
- B)** Accuracy
- C)** Precision (Correct)
- D)** Unbiasedness

7) If we draw a sample of size 'n' from a population then sample is called "large sample" if.

Select the correct option

- A)** $n > 40$
- B)** $n > 30$ (Correct)
- C)** $n > 50$
- D)** $n > 25$

8) The method of Maximum Likelihood (ML) is used to find out Select

the correct option

- A)** Sample values
- B)** Random numbers
- C)** Correlation coefficient
- D)** Point estimates (Correct)

9) The precision of an estimator can be increased by increasing the

Select the correct option

- A)** Size of sample (Correct)
- B)** Number of estimators
- C)** Size of population
- D)** Number of parameters

10) Which of the following is most important and most widely used method in point estimation?

Select the correct option

- A)** The method of moments
- B)** The method of fractional moments
- C)** The method of maximum likelihood (Correct)
- D)** The method of least square

***** (repeat right
14)*****

11) Sample Proportion is a/an _____ estimator of population mean.

Select the correct option

- A)** None of these
- B)** Unbiased and consistent estimator
- C)** Unbiased (Correct)
- D)** Consistent

12) In a random sample of 500 men from Lahore city, 300 are found to be smokers; the proportion of smokers is equal to:

Select the correct option

A) 0.6 (Correct)

B) 0.3

C) 0.5

D) 0.4

13) An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32). This means:

Select the correct option

A) None of these

- B)** We are sure the true mean yield of this new variety is between 2.48 and 3.32.
- C)** We are 95% confident that the true mean yield of this variety is 2.9
- D)** We are 95% confident that the true mean yield of this variety is between 2.48 and 3.32 (D is Correct)

14) A _____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Select the correct option

- A)** Hypothesis (Correct)
- B)** Lower limit
- C)** Confidence interval
- D)** Probability limit

15) To determine a sample size which test we use _____

Select the correct option

A) t

B) F

C) None of these (Correct)

D) z

16) A statistic whose standard error decreases with an increase in the sample size will be

Select the correct option

A) Consistent (Correct)

B) Unbiased

C) Efficient

D) None

17) A deserving player was not selected in the cricket team. It is an example of:

Select the correct option

- A)** Type-III error
- B)** Type-I error (Correct)
- C)** Type-IV error
- D)** Type-II error

18) In a Geometric distribution, Maximum Likelihood Estimator (MLE) for proportion (P) is equal to:

Select the correct option

- A)** Sample variance
- B)** Reciprocal of the mean (Correct)
- C)** Reciprocal of the sample variance
- D)** Sample mean

19) The precision of an estimator can be increased by decreasing the

Select the correct option

- A)** Level of significance (Correct)
- B)** Number of estimators
- C)** Number of parameters
- D)** Size of sample

20) Conventionally, the probability of making a type-II error is denoted by:

Select the correct option

- A)** β (beta) (Correct)
- B)** θ (theta)

- C)** α (alpha)
- D)** σ (sigma)

21) Sample mean is a/an estimate of population mean.

Select the correct option

- A)** Biased
- B)** None of the above
- C)** Unbiased (Correct)
- D)** Either unbiased or biased

22) How can we interpret the 90% confidence interval of the mean of the normal population.

Select the correct option

- A)** There are 100% chances of falling true value of the parameter

B) There are 90% chances of falling true value of the parameter (Correct)

C) All are correct

D) There are 10% chances of falling true value of the parameter

23) In testing $H_0: \mu = 100$ against $H_1: \mu \leq 100$ at the 10% level of significance, H_0 is rejected if:

Select the correct option

A) The p-value is greater than 0.10

B) The p-value is less than 0.10

C) 100 is contained in the 90% confidence interval

D) The value of the test statistics in the acceptance region (Correct)

24) In the method of moments, how many equations are required for finding two unknowns

Select the correct option

A) 2 (Correct)

B) 4

C) 1

D) 3

25) To determine a sample size in estimating population mean, we use the _____ value.

Select the correct option

A) X

B) Z

C) F

D) t (Correct)

26) The method of Maximum Likelihood (ML) is used to findout:

Select the correct option

- A)** Random numbers
- B)** Sample values
- C)** Point estimates (Correct)
- D)** Correlation coefficient

27) Alpha ' α ' is the probability of

Select the correct option

- A)** Making Type 1 error (Correct)
- B)** Accepting H_0 when it is true
- C)** Rejecting H_1 when it is wrong

D) Making Type II error

28) In interval estimation we obtained a ...of values as an estimate of parameter.

Select the correct option

A) Both a and D

B) Range (Correct)

C) Group

D) None of these

29) We can apply method of Maximum Likelihood on: Select

the correct option

A) Qualitative variables only

B) Discrete variables only

- C)** Discrete as well as continuous variables (Correct)
D) Continuous variables only

30) Inferential statistics involves:

select the correct option

- A)** Testing of hypothesis
B) All of these (Correct)
C) Confidence interval
D) Estimation

Reference:

RATIONALE OF THE METHOD OF MAXIMUM LIKELIHOOD (ML)

"To consider every possible value that the parameter might have, and for each value, compute the probability that the given sample would have occurred if that were the true value of the parameter. That value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." An estimate obtained by this method is called the maximum likelihood estimate (MLE). It should be noted that the method of maximum likelihood is applicable to both discrete and continuous random variables.

EXAMPLES OF MLE'S IN CASE OF DISCRETE DISTRIBUTIONS

Example-1:
For the Poisson distribution given by

$$P(X=x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

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31) Ideally, the width of confidence interval should be

select the correct option

A) 98 (Correct)

B) 1

C) 100

D) 0

32) IF $\text{Var}(T1) > \text{Var}(T2)$,where T1 and T2 are two unbiased estimators ,then:

select the correct option

A) T1 is more consistent

B) T2 is more consistent

C) T1 is more efficient

D) T2 is more efficient (Correct)

33) he confidence intervals become wide and less precise byselect the correct option

- A)** None of these
- B)** increasing the sample size
- C)** decreasing the level of significance
- D)** decreasing the sample size (Correct)

Reference:

by increasing the sample size, the confidence interval becomemore precise.

34) From which of the following methods we can obtain pointestimate of the population parameters.

select the correct option

A) All of the above (Correct)

B) Maximum Likelihood Method

C) Method of Moments

D) Method of Least Squares

35) A confidence interval has a specified probability _____ of containing the true value of the parameter. Select

the correct option

A) $1-\alpha$ (Correct)

B) $\alpha/2$

C) α

D) $1+\alpha$

36) _____ are equivalent.

Select the correct option

- A)** Type-II error and Level of confidence
- B)** Type-I error and Level of significance (Correct)
- C)** Type-II error and Level of significance
- D)** Type-I error and Level of confidence

37) The Maximum Likelihood Estimators (MLE) are
and but not necessarily

Select the correct option

- A)** Consistent, efficient, unbiased (Correct)
- B)** Consistent, unbiased, efficient
- C)** Unbiased, efficient, consistent
- D)** Unbiased, consistent, efficient

38) Interval estimation and Confidence interval are :Select

the correct option

A) Different

B) Independent

C) Dependent

D) Same (Correct)

39) "A failed student was passed by the examiner", is an example of:

Select the correct option

A) Type-I error

B) Type-IV error

C) Type-III error

D) Type-II error (Correct)

40) When sample size is to be considered large, population standard deviation can be replaced by

Select the correct option

A) Sample Variance

B) Sample Mean

C) Population variance

D) Sample Standard Deviation (Correct)

41) Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income null the Hypothesis will be.

Select the correct option

A) $\text{MeanA} > \text{MeanB}$

B) $\text{MeanA} \geq \text{MeanB}$

C) $\text{MeanA} < \text{MeanB}$

D) $\text{MeanA} \leq \text{MeanB}$ (Correct)

*****57*****

42) As sample size goes up, what tends to happen to 95% confidence intervals?

Select the correct option

A) They become more narrow (Correct)

B) They become more precise and narrow

C) They become more precise

D) They become wider

43) An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time

of 19.92 minutes, with a population standard deviation equal to 5.73.
assuming normality of assembly times, what is a 95%.

Select the correct option

A) (14.1123, 22.7277)

B) (17.1123, 24.7277)

C) (17.1123, 22.7277) (Correct)

D) (17.1123, 30.7277)

Reference:

$$\begin{aligned}
 n &= 16, \bar{x} = 19.92, \sigma = 5.73 \\
 1 - \alpha &= 95\% = 0.95 \\
 \alpha &= 0.05 \Rightarrow \frac{\alpha}{2} = 0.025 \\
 \text{we know that} & \\
 \bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} & \\
 19.92 \pm z_{0.025} \frac{5.73}{\sqrt{16}} & \\
 19.92 \pm 1.96 \left(\frac{5.73}{4} \right) & \\
 19.92 \pm 2.8077 & \\
 19.92 - 2.8077, 19.92 + 2.8077 & \\
 17.1123, 22.7277 & \\
 (17.1123, 22.7277) &
 \end{aligned}$$

44) For a particular hypothesis test, Alpha=0.09 and Beta=0.03. what is the value of Type I error?

Select the correct option

A) 0.09 (Correct)

B) 0.91

C) 10.03

D) 0.97

45) Pairing is done only in case of..... observations.

Select the correct option

A) Dependent (Correct)

B) Independent

C) Small

D) Large

46) How can we interpret the 90% confidence interval for the mean of the normal population.

Select the correct option

- A)** There are 10% chances of falling true value of the parameter
- B)** There are 90% chances of falling true value of the parameter (Correct)
- C)** There are 100% chances falling true value of the parameter
- D)** All are correct

47) Alpha ' α ' is the probability of:

Select the correct option

- A)** Committing Type I error
- B)** Reject H_0/H_0 is true
- C)** Type I error
- D)** All of these (Correct)

48) Suppose $H_0: p=0$ $H_1: p<0$ and Critical value is $z=-2.33$. If calculate the value of $Z=-2.60$, then what will be your conclusion?

Select the correct option

- A)** Accept H_0
- B)** Reject H_0 (Correct)
- C)** Impossible to decide
- D)** None of the above

49) If a significance level of 5% is used rather than 1%, the null hypothesis is:

Select the correct option

- A)** Less likely to be rejected
- B)** Just as likely to be rejected
- C)** More likely to be rejected (Correct)
- D)** None of these

50) If the NULL hypothesis says that the average height of Pakistani soldiers exceeds the average height of American soldiers by not more than 3 inches. What will be its ALTERNATIVE hypothesis?

Select the correct option

A) $\mu > 3$ (Correct)

B) $\mu < 3$

C) $\mu - 3$

D) none of these

51) Suppose $H_0: \mu_1 < \mu_2$ $H_1: \mu_1 > \mu_2$ and Critical value is $= 1.645$
If calculated value of $Z = 1.88$ then what will be your conclusion?

Select the correct option

- A)** Accept H_0
- B)** Reject H_0 (Correct)
- C)** Impossible to decide
- D)** None of the above

52) The term $1-B$ is called

Select the correct option

- A)** Level of the test
- B)** Power of the test (Correct)
- C)** Size of the test
- D)** Critical region

53) The t-distribution can never become narrower than: Select

the correct option

- A)** F-distribution
- B)** standard normal distribution (Correct)
- C)** Chi-square distribution
- D)** exponential distribution

54) When examining group difference where no direction of the difference is specified, which of the following is used?

Select the correct option

- A)** One-tailed test
- B)** Two-tailed test (Correct)
- C)** Directional hypothesis
- D)** Difference of modes test

55) If we are testing $H_0: \text{mean} = 50$ against $H_1: \text{mean} < 50$ then the rejection region will be

Select the correct option

- A)** in the center
- B)** on left side of mean (Correct)
- C)** on right side of mean
- D)** on both sides

56) The level of significance is also called:

Select the correct option

- A)** Power of test
- B)** Type II error
- C)** Type I error (Correct)
- D)** Acceptance region

57) A random sample of $n=25$ values gives sample mean 83. Can this sample be regarded as drawn from a normal population with $\mu = 80$ and $s=7$? In this question the alternative hypothesis will be:

Select the correct option

- A) $H_1: \mu = 80$
- B) $H_1: \mu \neq 80$ (Correct)
- C) $H_1: \mu > 80$
- D) $H_1: \mu < 80$

58) The t-test for independent samples assumes which of the following?

Select the correct option

- A)** Scores are independent (Correct)
- B)** Scores are dependent
- C)** Groups have equal means
- D)** Groups have equal medians

59) The critical region is computed from the value of:Select

the correct option

A) Alpha (Correct)

B) Beta

C) 1-alpha

D) 1-Beta

60) For a sample $p_1 = 0.2$. $n_1 = 100$ and for second sample $p_2 = 0.5$ and $n_2 = 200$, the pooled estimator p_c is:Select

the correct option

A) 0.30

B) 0.45

C) 0.40 (Correct)

D) 0.35

61) The location of the critical region depends upon: Select

the correct option

- A)** Null hypothesis
- B)** Alternative hypothesis (Correct)
- C)** Value of alpha
- D)** Value of test-statistic

62) To test the hypothesis about the difference of means for small samples, which test statistic will be used?

Select the correct option

- A)** Z-test
- B)** t-test (Correct)
- C)** F-test

D) All of these

63) The level of significance is also called:

Select the correct option

A) Power of test

B) Type II error

C) Type I error (Correct)

D) Acceptance region

64) Critical region is computed with the value of....

Select the correct option

A) α (Correct)

B) β

C) $1-\alpha$

D) $1-\beta$

65) Mean of t-distribution will not exist when d.f(degree of freedom) will equal to

Select the correct option

A) 1 (Correct)

B) 2

C) 3

D) 4

66) Which of the following is not a criterion for a good problem situation?

Select the correct option

A) It must be authentic

B) It must be clearly structured and defined. (Correct)

- C)** It must be appropriate to the student of development
- D)** It must be such that its will be benefited by group effort

67) An alternative hypothesis is generally denoted by:Select the correct option

- A)** H_2
- B)** H_3
- C)** H_0
- D)** H_1 (Correct)

68) When an interval is constructed, what is the probability of occurrence of parameter in it?

Select the correct option

- A)** 1
- B)** 50
- C)** Either 1 or 0 (Correct)
- D)** 0

69) If an estimator is more efficient then the other estimator,its shape of the sampling distribution will be

Select the correct option

- A) Normal
- B) Skewed to right
- C) Highly peaked (Correct) (NOT SURE) D) Flattered

70) In a Poisson distribution, Maximum Likelihood Estimator(MLE) for μ is equal to:

Select the correct option

- A) Reciprocal of the sample variance
- B) Sample variance
- C) Sample mean (Correct) D) Reciprocal of the mean

Question No: 1 (Marks: 1) - Please choose one
 five number summary consist of :

- ▶ $X_0, Q_1, \text{Median}, Q_3, \text{and } X_m$ (Page 97)
- ▶ $X_m, Q_1, \text{Mean}, Q_3, \text{and } X_0$
- ▶ $X_m, Q_1, \text{Mode}, Q_3, \text{and } X_0$
- ▶ $X_0, Q_1, \text{Median}, Q_2, \text{and } X_m$

Question No: 3 (Marks: 1) - Please choose one
 in a multiplication theorem $P(A \cap B)$ equals:

- ▶ $P(A) P(B)$
- ▶ $P(A) + P(B)$
- ▶ $P(A) * P(B|A)$ (Page 159)
- ▶ $P(B|A) * P(B)$

Question No: 4 (Marks: 1) - Please choose one

The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is

- ▶ 0
- ▶ $3/15$
- ▶ $1/12$
- ▶ $1/2$

Question No: 6 (Marks: 1) - Please choose one

If A and B are mutually exclusive events then $P(A \text{ or } B)$ equals:

- ▶ $P(A) + P(B) - P(A \text{ and } B)$
- ▶ $P(A) * P(B)$
- ▶ $P(A) + P(B)$ (Page 155)

$$P(A|B) + P(B|A)$$

Question No: 7 (Marks: 1) - Please choose one

The simultaneous occurrence of two events is called

- ▶ Prior probability
- ▶ Subjective probability not confirmed
- ▶ Conditional probability
- ▶ **Joint probability** [click here for detail](#)

Question No: 8 (Marks: 1) - Please choose one

First moment about origin is always equals to:

- ▶ **Mean (Page 119)**
- ▶ Variance
- ▶ Standard Deviation
- ▶ Zero

Question No: 9 (Marks: 1) - Please choose one

Which one of the following measurement does *not* divide a set of observations into equal parts?

- ▶ Quartiles
- ▶ Deciles
- ▶ Percentiles
- ▶ **Standard deviations**

Question No: 10 (Marks: 1) - Please choose one

if first and third quartile are 21.16 and 56.36 respectively, then the quartile deviation is:

- ▶ **17.1** (Page 84)
- ▶ 34.2
- ▶ 51.3
- ▶ 50.5

Question No: 11 (Marks: 1) - Please choose one

The height of student is 60 inch. This is an example of.....?

- ▶ Continues data
- ▶ Qualitative data
- ▶ Categorical data
- ▶ **Discrete data**

Question No: 12 (Marks: 1) - Please choose one

Which of the statement is true regarding a sample?

- ▶ **It is a part of population (Page 13)**
- ▶ It must contain at least five observations
- ▶ It refers to descriptive statistics
- ▶ It produces true value

Question No: 13 (Marks: 1) - Please choose one

Which one of the following graphs is used for a time series data?

- ▶ Histogram
- ▶ Historigram
- ▶ Frequency curve
- ▶ **Frequency polygon** [Click here for detail](#)

Question No: 14 (Marks: 1) - Please choose one

Which of the following comes first to make frequency distribution

- ▶ No. of groups
- ▶ Class intervals
- ▶ **Rang** (Page 28)
- ▶ Tally marks

Question No: 15 (Marks: 1) - Please choose one

The average which is defines as reciprocal of arithmetic mean of the reciprocals of the values is called

- ▶ Geometric Mean
- ▶ **Harmonic mean** (Page 77)
- ▶ Mode
- ▶ Median

Question No: 16 (Marks: 1) - Please choose one

If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

- ▶ **Weighted mean** (Not sure)
- ▶ Harmonic mean
- ▶ Arithmetic mean
- ▶ Geometric mean

Question No: 1 (Marks: 1) - Please choose one

The probability of drawing a 'jack card' from 52 playing cards is:

▶ $\frac{1}{52}$

▶ $\frac{4}{52}$

▶ $\frac{13}{52}$

▶ $\frac{52}{52}$

▶

▶ $\frac{26}{52}$

▶ $\frac{52}{52}$

▶

Question No: 2 (Marks: 1) - Please choose one

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

▶ $0 \leq r \leq 1$

▶ $r \geq 0$

▶ $r = +1$

▶ $r = -1$

Question No: 3 (Marks: 1) - Please choose one

If a curve has a longer tail to the right, it is called:

▶ **Positively skewed** (Page 39)

▶ Negatively skewed

▶ J-shaped

▶ Symmetric

Question No: 04 (Marks: 1) - Please choose one

Which one of the following is not included in measures of central tendency?

▶ **Quartile deviation** (Page 82)

▶ Harmonic mean

▶ Geometric mean

▶ Arithmetic mean

Question No: 05 (Marks: 1) - Please choose one

Which of the following is not based on all the observations?

- ▶ Arithmetic Mean
- ▶ Geometric Mean
- ▶ Harmonic mean

▶ **Mode**

Question No: 06 (Marks: 1) - Please choose one

What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

▶ 49

▶ **1**

▶ **0**

Standard deviation will always be zero if all the values in data are same

▶ 7

Question No: 1 (Marks: 1) - Please choose one

10! =.....

▶ 362880

▶ **3628800**

▶ 362280

▶ 362800

Question No: 2 (Marks: 1) - Please choose one

If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

▶ 1/52

▶ 13/52

▶ 4/52

▶ **26/52**

Question No: 3 (Marks: 1) - Please choose one

The probability of drawing a 'jack card 'from 52 playing cards is:

- ▶ 1/52
- ▶ 4/52
- ▶ 13/52
- ▶ 26/52

Question No: 4 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series?

- ▶ C.V. (Page 93)
- ▶ Q.D.
- ▶ M.D.
- ▶ S.D.

Question No: 5 (Marks: 1) - Please choose one

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

- ▶ $0 \leq r \leq 1$
- ▶ $r \geq 0$
- ▶ $r = +1$ (Page 129)
- ▶ $r = -1$

Question No: 6 (Marks: 1) - Please choose one

In a regression line $Y = a + bX$, the value of the correlation coefficient will be zero if:

- ▶ Intercept $a = 0$
- ▶ Intercept $a \neq 0$
- ▶ Slope $b = 0$
- ▶ Slope $b \neq 0$

Question No: 7 (Marks: 1) - Please choose one

When two coins are tossed the probability of at least one head is:

- ▶ 1/4
- ▶ 3/4 (HH, HT, TH, TT)
- ▶ 2/4
- ▶ 4/4

Question No: 8 (Marks: 1) - Please choose one

Which one of the following measurement does *not* divide a set of observations into equal parts?

- ▶ Quartiles
- ▶ Deciles
- ▶ Percentiles
- ▶ **Standard deviations rep**

Question No: 9 (Marks: 1) - Please choose one

In the model $Y = mX + a$, Y is also known as the:

- ▶ Predictor variable
- ▶ Independent variable
- ▶ **Predicted (dependent) variable (Page 121)**
- ▶ Explanatory variable

Question No: 10 (Marks: 1) - Please choose one

According to empirical rule approximately 95% of the measurements will fall under which interval?

- ▶ $\bar{X} \pm S$
- ▶ $\bar{X} \pm 2S$ **correct**
- ▶ $\bar{X} \pm 3S$
- ▶ $\bar{X} \pm 4S$

Question No: 11 (Marks: 1) - Please choose one

Which one of the following is written at the top of the table?

- ▶ Source note
- ▶ Foot note
- ▶ Prefatory note
- ▶ **Title** [Click here for detail](#)

Question No: 12 (Marks: 1) - Please choose one

If a curve has a longer tail to the right, it is called:

- ▶ **Positively skewed rep**
- ▶ Negatively skewed
- ▶ J-shaped
- ▶ Symmetric

Question No: 13 (Marks: 1) - Please choose one

Which one of the following is the class frequency?

- ▶ **The number of observations in each class**
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 14 (Marks: 1) - Please choose one

If X is a discrete random variable, then the function $f(x)$ is

- ▶ A probability function
- ▶ A probability density function
- ▶ A density function
- ▶ **A distribution function (Page 173)**

Question No: 15 (Marks: 1) - Please choose one

Which one of the following graphs is used for a time series data?

- ▶ Histogram
- ▶ Historigram
- ▶ Frequency curve
- ▶ **Frequency polygon**

Question No: 16 (Marks: 1) - Please choose one

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ▶ Ogive
- ▶ Frequency polygon
- ▶ **Frequency curve (Page 38)**
- ▶ Historigram

Question No: 17 (Marks: 1) - Please choose one

If mean of the two observations is 10.5, then median of these two observations will be:

- ▶ 7.5
- ▶ .5
- ▶ 9.5
- ▶ **10.5 (Page 68)**

Question No: 18 (Marks: 1) - Please choose one
Which one is the formula of mid range?

▶ $X_m - X_0$

▶ $X_0 - X_m$

▶ $\frac{X_0 - X_m}{2}$

▶ $\frac{X_0 + X_m}{2}$

(Page 80)

Question No: 19 (Marks: 1) - Please choose one
Which one of the following is not included in measures of central tendency?

▶ **Quartile deviation** (page 82)

▶ Harmonic mean

▶ Geometric mean

▶ Arithmetic mean

Question No: 20 (Marks: 1) - Please choose one
For the given data 2, 3, 7, 0, -8 G. M will be:

▶ Negative

▶ Positive

▶ **Zero** (Page 75)

▶ Undefined

MIDTERM EXAMINATION (Spring 2010)

STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one

If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

▶ 1/52

▶ 13/52

▶ 4/52

▶ **26/52** rep

Question No: 2 (Marks: 1) - Please choose one

The probability of drawing a 'jack card 'from 52 playing cards is:

- ▶ 1/52
- ▶ 13/52
- ▶ 4/52 rep
- ▶ 26/52

Question No: 3 (Marks: 1) - Please choose one

In a regression line $Y = a + bX$, the value of the correlation coefficient will be zero if:

- ▶ Intercept $a = 0$ rep
- ▶ Intercept $a \neq 0$
- ▶ Slope $b = 0$
- ▶ Slope $b \neq 0$

Question No: 4 (Marks: 1) - Please choose one

Which one of the following measurement does *not* divide a set of observations into equal parts?

- ▶ Quartiles
- ▶ Deciles
- ▶ Percentiles
- ▶ Standard deviations rep

Question No: 5 (Marks: 1) - Please choose one

Which one of the following graphs is used for a time series data?

- ▶ Histogram
- ▶ Historigram
- ▶ Frequency curve
- ▶ Frequency polygon rep

Question No: 6 (Marks: 1) - Please choose one

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ▶ Ogive
- ▶ Frequency polygon
- ▶ Frequency curve (Page 38) rep
- ▶ Historigram

Question No:7 (Marks: 1) - Please choose one

Which one is equal to explained variation divided by total variation?

- ▶ Sum of square due to regression
- ▶ **Coefficient of determinant** [Click here for detail](#)
- ▶ Standard error estimate
- ▶ Coefficient of correlation

Question No: 8 (Marks: 1) - Please choose one

in the given series 1,2,1,1,2,2,2,3,4,5,3,2,3,1,4,2,3 mode of given is

- ▶ 4
- ▶ 3
- ▶ **1**
- ▶ 2

Question No: 9 (Marks: 1) - Please choose one

True for the population,

- ▶ it must be large number of values
- ▶ It must refer to people
 - ▶ **It is collection of individual objects or measurement (Page 12)**
- ▶ It is the small part of whole

Question No:10 (Marks: 1) - Please choose one

Data arrangement in ascending or descending order

- ▶ **Array data (Page 47)**
- ▶ Group data
- ▶ Ungroup data
- ▶ Raw data

Question No: 11 (Marks: 1) - Please choose one

What is the main objective of Descriptive statistics?

- ▶ To test population properties
- ▶ To describe the data we collected
- ▶ To infer something about the population
 - ▶ **Making estimate** [Click here for detail](#)

Question No: 12 (Marks: 1) - Please choose one

Which measure of central tendency?

- ▶ Variation of distribution
 - ▶ **Average of distribution (Page 51)**
- ▶ Scattering of distribution
- ▶ Dispersion of distribution

Question No: 13 (Marks: 1) - Please choose one

If $a=4$ $b=2$ estimate line (i.e $y=a+bx$) and independent variable has value 3 the value of dependent variable

- ▶ 6
- ▶ 9
- ▶ 10 $4+2(3)=10$
- ▶ 11

Question No: 14 (Marks: 1) - Please choose one

The number of ways in which 4 books can be arranged

- ▶ 4
- ▶ 6
- ▶ 12
- ▶ 24 (Page 142)

Question No: 15 (Marks: 1) - Please choose one

If we plot paired observed $(x,y)=1.....n$ on graph is called,

- ▶ Polygon
- ▶ Freasito diagram
- ▶ Scatter diagram
- ▶ Cumulative frequency diagram

Question No: 16 (Marks: 1) - Please choose one

The simultaneous occurrence of two events is called

- ▶ Descriptive probability
- ▶ Subjective probability
- ▶ Conditional probability
- ▶ Joint probability

Question No: 17 (Marks: 1) - Please choose one

Which one is the not measure of dispersion.

- ▶ The range
- ▶ 50th percentile
- ▶ Inter quartile range
- ▶ Variance

Question No: 18 (Marks: 1) - Please choose one

In positively skew cure which relation is

- ▶ The mean, median and mode are equal
- ▶ Mean is greater then median [Click here for detail](#)
- ▶ Median is greater then mean
- ▶ Standard deviation must be greater then mean or median

Question No: 19 (Marks: 1) - Please choose one

When coin tossed we get only

- ▶ **1 outcome**
- ▶ 2 outcomes
- ▶ 3 outcomes
- ▶ 4 outcomes

Question No: 20 (Marks: 1) - Please choose one

When mean is 25 and S.D is 5 then CV is

- ▶ 100%
- ▶ 25%
- ▶ **20%**
- ▶ 10%

Question No: 1 (Marks: 1) - Please choose one

For a positively skewed distribution m_3 will be:

- ▶ **Positive** (Page 119)
- ▶ Negative
- ▶ Zero
- ▶ 1

Question No: 2 (Marks: 1) - Please choose one

When data is labeled to identify an attribute of element, the measurement scale is:

- ▶ **Ordinal** (Page 9)
- ▶ Interval
- ▶ Nominal
- ▶ Ratio

Question No: 3 (Marks: 1) - Please choose one

Suppose the estimated equation is $\hat{Y} = 5 - 2X$ has been calculated for a set of data. What is slop of the line:

- ▶ 0
- ▶ 2
- ▶ **-2 (Page 121)**
- ▶ 5

Question No: 4 (Marks: 1) - Please choose one

If $P(B|A) = 0.25$ and $P(A \cap B) = 0.20$, then $P(A)$ is:

- ▶ 0.05
- ▶ **0.80 (Page 159)**
- ▶ 0.95
- ▶ 0.75

$$\begin{aligned} P(B|A) &= \frac{P(A \cap B)}{P(A)} \\ &= P(A) = \frac{P(A \cap B)}{P(B|A)} \\ &= \frac{0.20}{0.25} = 0.80 \end{aligned}$$

Question No: 5 (Marks: 1) - Please choose one

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ▶ Advance statistics
- ▶ Probability statistics
- ▶ **Descriptive statistics (Page 61)**
- ▶ Inferential statistics

Question No: 6 (Marks: 1) - Please choose one

In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors."

This report is an example of :

- ▶ A sample
- ▶ A population
- ▶ Statistical inference
- ▶ **Descriptive statistics**

Question No: 7 (Marks: 1) - Please choose one

A set that contains all possible outcomes of a system is known as

- ▶ Finite Set
- ▶ Infinite Set
- ▶ **Universal Set (Page 134)**
- ▶ No of these

Question No: 8 (Marks: 1) - Please choose one

If X and Y are independent, then $\text{Var}(X-Y)$ is equal to:

- ▶ **$\text{Var}(X) - \text{Var}(Y)$**
- ▶ $\text{Var}(X) + \text{Var}(Y)$
- ▶ $\text{Var}(X + Y)$
- ▶ Zero

Question No: 9 (Marks: 1) - Please choose one

Which of the following is the class frequency

- ▶ **The number of observations in each class**
- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one

How to construct the class interval:

- ▶ Divide the class frequencies in half
- ▶ Divide the class frequency by the number of observations
- ▶ **Find the difference between consecutive lower class limits**
- ▶ Count the number of observations in the class

Question No: 11 (Marks: 1) - Please choose one

Data in the Population Census Report is:

- ▶ **Ungrouped data (Page 11)**
- ▶ Secondary data
- ▶ Primary data
- ▶ Arrayed data

Question No: 12 (Marks: 1) - Please choose one
What is the range of -2,-3,-5,-10 :

- ▶ **-12** (Page 28)
- ▶ 8
- ▶ -8
- ▶ 2

Question No: 13 (Marks: 1)

- Please choose one

The algebraic sum of deviations from mean is:

- ▶ Maximum
- ▶ Minimum
- ▶ **Zero** (Page 86)
- ▶ Undefined

Question No: 14 (Marks: 1)

- Please choose one

The sum of squares of deviations from mean is:

- ▶ Undefined
- ▶ Zero
- ▶ **Maximum**
- ▶ Minimum

Question No: 15 (Marks: 1)

- Please choose one

Statistic is a numerical quantity, which is calculated from:

- ▶ Population
- ▶ **Sample** (Page 7)
- ▶ Data
- ▶ Observations

Question No: 16 (Marks: 1) -

Please choose one Which of the following is not based on all the observations?

- ▶ Arithmetic Mean
- ▶ Geometric Mean
- ▶ Harmonic mean

▶ **Mode** (rep)

MIDTERM EXAMINATION

Spring 2009

STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

If any value in the data is zero, then it is not possible to have:

- ▶ Harmonic Mean
- ▶ Arithmetic Mean
- ▶ Median
- ▶ **Mode (rep)**

Question No: 2 (Marks: 1) - Please choose one

For a symmetrical distribution... is equidistance from median:

- ▶ X_0 and X_m
- ▶ **Q_1 and Q_3 (Page 97)**
- ▶ X_0 and Q_1
- ▶ X_m and Q_3

Question No: 3 (Marks: 1) - Please choose one

Which one of the following measure is not used in 'five number

☐ X_0

☐ Q_3

summery':

☒ **X (Page 97)**

☐ Q_1

Question No: 4 (Marks: 1) - Please choose one

If Mean = 25 & S.D is 5 then C.V is

- ▶ 100%
- ▶ 25%
- ▶ **20% (Page 93)**
- ▶ 10%

Question No: 5 (Marks: 1) - Please choose one

A coin is tossed 3 times then, the number of sample points in the sample space is:

- ▶ **3**
- ▶ **8**
- ▶ **4 (Page 145)**

Question No: 6 (Marks: 1) - Please choose one

What is the difference between a permutation and a combination:

- ▶ **In a permutation order is important and in a combination it is not** [Click here for detail](#)
- ▶ In a permutation order is not important and in a combination it is important
- ▶ A combination is based on the classical definition of probability
- ▶ A permutation is based on the classical definition of probability

Question No: 7 (Marks: 1) - Please choose one

What we consider in simple correlation analysis:

- ▶ Several independent variables
 - ▶ **Strength of the association between two variables** (Page 128)
- ▶ Intercept with the X-axis
- ▶ Intercept with the Y-axis

Question No: 8 (Marks: 1) - Please choose one

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

- ▶ $0 < r < 1$
- ▶ $r = 0$
- ▶ **$r = +1$ (rep)**
- ▶ $r = -1$

Question No: 9 (Marks: 1) - Please choose one

Which of the following measure of dispersion is least affected by extreme values of observations in a data:

- ▶ **Range** (Page 82)
- ▶ Quartile deviation
- ▶ Mean absolute deviation
- ▶ Standard deviation

Question No: 10 (Marks: 1) - Please choose one

The approximate number of observations in a set of data covered by the interval, *Median* $\pm$ *Q.D* are:

- ▶ **50 per cent** (Page 84)
- ▶ 68.5 per cent
- ▶ 95.4 per cent
- ▶ 99 per cent

Question No: 11 (Marks: 1) - Please choose one

The dispersion which is calculated from all observed values is:

- ▶ **Standard deviation** (Page 87)
- ▶ Quartile deviation

- ▶ Rang
- ▶ Coefficient of Rang

Question No: 12 (Marks: 1) - Please choose one

In a set of 20 values all the values are 2, what is the value of Geometric Mean:

▶ 2 (Page 75)

▶ 5

▶ 10

▶ 20

Question No: 13 (Marks: 1) - Please choose one

In a set of 10 values all the values are 5, what is the value of 5th Decile:

▶ 2

▶ 5

▶ 10

▶ 20

Question No: 14 (Marks: 1) - Please choose one

If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

▶ W.M

▶ H.M

▶ A.M

▶ G.M

Question No: 15 (Marks: 1) - Please choose one

In stem and leaf plot, data is measured on:

▶ Ratio Scale

▶ Interval Scale

▶ Ordinal scale

▶ Nominal Scale

Question No: 16 (Marks: 1) - Please choose one

In statistics, we deal with:

▶ Individuals

▶ Isolated items

▶ Aggregates of facts (Page 7)

▶ Qualitative data

MIDTERM EXAMINATION

Spring 2009

STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

Which of the following is a systematic arrangement of data into rows and columns:

- ▶ Component bar chart
- ▶ Classification
- ▶ **Tabulation**
- ▶ Bar chart

Question No: 2 (Marks: 1) - Please choose one

For any number k , at least $1 - 1/k^2$ of the data-values fall within k standard

deviations of the mean:

- ▶ **Greater than 1 (Page 97)**
- ▶ Less than 1
- ▶ Greater or equal to 1
- ▶ Less or equal to 1

Question No: 3 (Marks: 1) - Please choose one

If Mean = 25 & S.D is 5 then C.V is

- ▶ 100%
- ▶ 25%
- ▶ **20% (Page 93)rep**
- ▶ 10%

Question No: 4 (Marks: 1) - Please choose one When E is an impossible event, then $P(E)$ is:

- ▶ **0 (Page 146)**
- ▶ 0.5
- ▶ **1**
- ▶ **2**

Question No: 5 (Marks: 1) - Please choose one

The data for an ogive is found in which distribution:

- ▶ **A cumulative frequency distribution (Page 43)**
- ▶ A relative frequency distribution
- ▶ A frequency distribution
- ▶ A joint frequency distribution

Question No: 6 (Marks: 1) - Please choose one

Which of the following statements is true regarding a sample:

▶ **It is a part of population** (Page 13) rep

- ▶ It must contain at least five observations
- ▶ It refers to descriptive statistics
- ▶ It produces True value

Question No: 7 (Marks: 1) - Please choose one

Which level of measurement is required for the median:

- ▶ Nominal
- ▶ **Ordinal**
- ▶ Interval
- ▶ Ratio

Question No: 8 (Marks: 1) - Please choose one

In a the regression line $Y = a + bX$ the variable which is non-random is:

▶ **X** (Page 122)

- ▶ **Y**
- ▶ Both X and Y
- ▶ Neither X nor Y

Question No: 9 (Marks: 1) - Please choose one

The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

▶ **Scatter variable** [Click here for detail](#)

- ▶ Independent variable
- ▶ Dependent variable
- ▶ Correlation variable

Question No: 10 (Marks: 1) - Please choose one

Which is the formula of range:

- ▶ $X_m - X_0$ (Page 28)
- ▶ $X_0 - X_m$
- ▶ $\frac{X_m - X_0}{2}$
- ▶ $\frac{X_m + X_0}{2}$

Question No: 11 (Marks: 1) - Please choose one

The descriptive measure of dispersion that is based on the concept of a deviation about the mean is:

- ▶ Range
- ▶ Inter quartile range
- ▶ Quartile deviation
- ▶ **Standard deviation**

Question No: 12 (Marks: 1) - Please choose one

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ▶ Advance statistics
- ▶ Probability statistics
- ▶ **Descriptive statistics (rep)**
- ▶ Inferential statistics

Question No: 13 (Marks: 1) - Please choose one

A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ▶ Infinite population
- ▶ Finite population
- ▶ Concrete population
- ▶ **Hypothetical population(Page 12)**

Question No: 14 (Marks: 1) - Please choose one

First moment about origin is always equals to:

- ▶ **Mean** [Click here for detail](#)
- ▶ Variance
- ▶ Standard Deviation
- ▶ Zero

Question No: 15 (Marks: 1) - Please choose one

When two dice are rolled, the numbers of possible sample points are:

- ▶ **6**
- ▶ 12
- ▶ 24
- ▶ **36 ($6*6=36$)**

Question No: 16 (Marks: 1) - Please choose one

The correlation of coefficient lies between :

- ▶ 0 to 1
- ▶ 0 to ∞
- ▶ **- 1 to +1 (Page 128)**

► +1 to ∞

MIDTERM EXAMINATION

Spring 2009

STA301- Statistics and Probability (Session - 2)

Question No: 1 (Marks: 1) - Please choose one

Median can be found only when:

- ▶ Data is Discrete
- ▶ Data is Attributed Data is continuous
- ▶ Data is continuous Data is attributed
- ▶ **Data is arranged (Page 52)**

Question No: 2 (Marks: 1) - Please choose one

From the following observations 2,3,4,5,4,6,4, the mode is:

- ▶ 2
- ▶ 3
- ▶ **4**
- ▶ 5

Question No: 3 (Marks: 1) - Please choose one

How to construct the class interval:

- ▶ Divide the class frequencies in half
- ▶ Divide the class frequency by the number of observations
- ▶ **Find the difference between consecutive lower class limits**
- ▶ Count the number of observations in the class

Question No: 4 (Marks: 1) - Please choose one

How many elements are in the sample space of rolling one die:

- ▶ **6**
- ▶ 12
- ▶ 24
- ▶ 36

Question No: 5 (Marks: 1) - Please choose one

When two coins are tossed the probability of at most one head is:

- ▶ 1/4
- ▶ 2/4
- ▶ **3/4**
- ▶ 4/4

If A and B are mutually exclusive events then $P(A \text{ or } B)$ equals:

- Question No: 7 (Marks: 1) - Please choose one**

In scatter diagram the variable plotted along Y-axis is:

- Question No: 8 (Marks: 1) - Please choose one**

Question No. 6 (Marks: 1) Please choose one
Positive square root of variance of a distribution is:

- Question No: 9 (Marks: 1) - Please choose one**

When more values are lying at the start of the distribution, it is a:

- Question No: 10 (Marks: 1) - Please choose one**

**What f_m ' in the formula of mode:
is '**

- Question No: 11 (Marks: 1) - Please choose one**

If median = 7 and Mean = 5, what is the value of

- 1 (because median = Q2) ► 7 (because median = Q2)

Q_2 :

Question No: 12 (Marks: 1) - Please choose one

The probability of drawing a king of spade from a pack of 52 cards is:

- ▶ 1/4
- ▶ 1/13
- ▶ 1/26
- ▶ 1/52

Question No: 13 (Marks: 1) - Please choose one

When referring to a curve whose longer tail is to the left, you would call it:

- ▶ U shape
- ▶ Skewed to the left (Page 39)
- ▶ Skewed to the right
- ▶ Symmetrical

Question No: 14 (Marks: 1) - Please choose one

In statistics, we deal with:

- ▶ Individuals
- ▶ Isolated items
- ▶ Aggregates of facts (Page 7) rep
- ▶ Qualitative data

Question No: 15 (Marks: 1) - Please choose one

When data is labeled to identify an attribute of element, the measurement scale is:

- ▶ Ordinal (Page 9)
- ▶ Interval
- ▶ Nominal
- ▶ Ratio

Question No: 16 (Marks: 1) - Please choose one

The distribution is mesokurtic if the Moment Coefficient of kurtosis

b_2 is:

- ▶ Equal to 0
- ▶ Equal to 3
- ▶ Less than 3

► Greater than 3

MIDTERM EXAMINATION

Spring 2009

STA301- Statistics and Probability (Session - 2)

Question No: 1 (Marks: 1) - Please choose one

A histogram consists of a set of adjacent rectangles whose bases are marked off by:

▶ [Class boundaries\(Page 32\)](#)

- ▶ Class limits
- ▶ Class marks
- ▶ Class frequency

Question No: 2 (Marks: 1) - Please choose one

Stem and leaf was introduced by:

- ▶ Francis Galton
- ▶ Friedman

▶ [John Tukey \(Page 47\)](#)

- ▶ Pearson

Question No: 3 (Marks: 1) - Please choose one

For positively skewed distribution

Mean.....Median.....Mode:

- ▶ =
- ▶ <
- ▶ >
- ▶ ≠

[Click here for detail](#)

Question No: 4 (Marks: 1) - Please choose one

For a positively skewed distribution:

- ▶ Mean<Mode<Median
- ▶ Mdian<Mode>Mean
- ▶ Mode>Mean>Median

▶ [Mean>Median>Mode Click here for detail](#)

Question No: 5 (Marks: 1) - Please choose one

5C_5 Equals :

- ▶ **1**
- ▶ **5**
- ▶ 10
- ▶ 25

Question No: 6 (Marks: 1) - Please choose one

If a curve has a longer tail to the right, it is called :

▶ **Positively skewed** (Page 39)

- ▶ negatively skewed
- ▶ J-shaped
- ▶ Symmetric

Question No: 7 (Marks: 1) - Please choose one

In measures of relative dispersion unit of measurement is:

- ▶ Changed
- ▶ **Vanish** (Page 82)
- ▶ Does not changed
- ▶ Dependent

Question No: 8 (Marks: 1) - Please choose one

Data used by an agency which is originally collected by them is :

▶ **Primary data**(Page 11)

- ▶ Raw data
- ▶ Secondary data
- ▶ Grouped data

Question No: 9 (Marks: 1) - Please choose one

When we toss a coin , we get only:

▶ **1 outcome**

- ▶ 2 outcomes
- ▶ 3 outcomes
- ▶ 4 outcomes

Question No: 10 (Marks: 1) - Please choose one

Which of the following is the class frequency

▶ **The number of observations in each class**

- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 11 (Marks: 1) - Please choose one

Standard deviation is affected by the change of ____:

- ▶ Origin & scale
- ▶ Origin only
- ▶ **Scale only** [Click here for detail](#)
- ▶ Not origin & scale

Question No: 12 (Marks: 1) - Please choose one

For a moderately skewed distributions, the approximate percentage of case included between

$\bar{X} - 2S$ and $\bar{X} + 2S$ is :

- ▶ 99.7%
- ▶ 68%
- ▶ 95% (Page 95)
- ▶ 50%

Question No: 13 (Marks: 1) - Please choose one

A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- ▶ 1/2
- ▶ 1/3
- ▶ 2/3 (greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)
- ▶ 5/6

Question No: 14 (Marks: 1) - Please choose one

The probability of drawing any one spade card is:

- ▶ 1/52
- ▶ 4/52
- ▶ 13/52
- ▶ 52/52

Question No: 15 (Marks: 1) - Please choose one

Which of the following is not the type of frequency curves?

- ▶ The symmetrical frequency curve
- ▶ The extremely skewed frequency curve
- ▶ The U-shaped frequency curve
- ▶ Frequency polygon (Page 38)

Question No: 16 (Marks: 1) - Please choose one

If the third moment $(m_3 = 0)$, then the distribution is:
about mean is zero

- ▶ Symmetrical
- ▶ Negatively skewed
- ▶ Positively skewed
- ▶ Mesokurtic

Question No: 1 (Marks: 1) - Please choose one

The standard deviation of -1, -1, -1, -1 will be

- 1
- -1
- 0
- Does not exist

Question No: 2 (Marks: 1) - Please choose one

In a multiplication theorem $P(A \cap B)$ equals:

- $P(A) P(B)$
- $P(A) + P(B)$
- $P(A) * P(B|A)$
- $P(B \setminus A) * P(B)$

Question No: 3 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series:

- C.V.
- Q.D.
- M.D.
- S.D.

Question No: 4 (Marks: 1) - Please choose one

A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- $1/2$
- $1/3$
- $2/3$
- $5/6$

Question No: 5 (Marks: 1) - Please choose one

When two dice are rolled the number of possible sample points is :

- 6
- 12
- 24
- 36

Question No: 6 (Marks: 1) - Please choose one

If $P(B|A) = 0.25$ and $P(A \cap B) = 0.20$ then $P(A)$ is

- 0.80
- 0.05

, then $P(A)$ is:

- 0.95
- 0.75

Question No: 7 (Marks: 1) - Please choose one

If we plot the paired observation $(X_i, Y_i) i = 1, 2, \dots, n$ on a graph, the resulting set of points is called:

- Frequency polygon
- Frequency histogram
- Scatter diagram
- Cumulative frequency polygon

Question No: 8 (Marks: 1) - Please choose one

For any number k , at least $1 - 1/k^2$ of the data-values fall within k standard deviations of the mean:

- Greater than 1
- Less than 1
- Greater or equal to 1
- Less or equal to 1

Question No: 9 (Marks: 1) - Please choose one

The measure of dispersion which is based on the concept of deviations about the mean is:

- Range
- Interquartile range
- Quartile deviation
- Standard deviation

Question No: 10 (Marks: 1) - Please choose one

According to empirical rule approximately 95% of the measurements will fall under which interval?

- $\bar{X} \pm S$
- $\bar{X} \pm 2S$
- $\bar{X} \pm 3S$
- $\bar{X} \pm 4S$

Question No: 11 (Marks: 1) - Please choose one

The correlation coefficient lies between :

▸ 0 to 1

▸ 0 to ☐

▸ - 1 to +1

▶ +1 to ☐

The Zip-Code of a city is an example of:

- Qualitative data
- Discrete data
- Quantitative data
- Continuous data

Question No: 13 (Marks: 1) - Please choose one

How to construct the class interval:

- Divide the class frequencies in half
- Divide the class frequency by the number of observations
- Find the difference between consecutive lower class limits
- Count the number of observations in the class

Question No: 14 (Marks: 1) - Please choose one

The collection of all elements in a group is called:

- Population
- Sample
- Data
- Registration

Question No: 15 (Marks: 1) - Please choose one

Which one is the formula to calculate the approximate value of class interval?

- Minimum value/RANGE
- Maximum value/No. of classes
- No. of classes/RANGE
- RANGE/No. of classes

Question No: 16 (Marks: 1) - Please choose one

If the random variable X denotes the number of heads when three distinct coins are tossed, then X assumes the value

- 0, 1, 2, 3
- 1, 3, 3, 1
- 1, 2, 3
- 3 only

Question No: 17 (Marks: 1) - Please choose one

Which of the following comes first to make frequency distribution.

- No. of Groups
- Class interval
- Range

- Tally marks

What will be the mode for the data 2,3,4,5,4,6,4?

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 19 (Marks: 1) - Please choose one

Which one of the following is equal to the 2nd quartile:

P_{33}



- ▶ D_3
- ▶ Median
- ▶ Mode

Question No: 20 (Marks: 1) - Please choose one

In a grouped data, the range is the difference between :

- ▶ Two extreme class frequencies
- ▶ Two extreme open-end class limits
- ▶ Two extreme class boundaries
- ▶ Two extreme class marks

**Question No: 1 (Marks: 1) - Please
choose one**

..... 10!

▶ 362880

*▶ 3628800

▶ 362280

▶ 362800

**Question No: 2 (Marks: 1) - Please
choose one**

..... If
a player well shuffles the pack of 52 playing card, then the probability of a black card
from 52 playing cards is: <http://vustudents.ning.com>

$\frac{1}{52}$



$$\frac{13}{52}$$

* ►

$$\frac{4}{52}$$

►

$$\frac{26}{52}$$

►

**Question No: 3 (Marks: 1) - Please
choose one**

The probability of drawing a ‘jack card ‘ from 52 playing cards is:

$$\frac{1}{52}$$

►

$$\frac{4}{52}$$

* ►

$$\frac{13}{52}$$

►

$$\frac{26}{52}$$

►

**Question No: 4 (Marks: 1) - Please
choose one**

Which dispersion is used to compare variation of two series:

- ▶ C.V.
- *▶ Q.D.
- ▶ M.D.
- ▶ S.D.

**Question No: 5 (Marks: 1) - Please
choose one**

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r': <http://vustudents.ning.com>

- ▶ $0 \leq r \leq 1$
- ▶ $r \geq 0$
- *▶ $r = +1$
- ▶ $r = -1$

**Question No: 6 (Marks: 1) -
Please choose one**

In a

regression line $Y = a + bX$, the value of the correlation coefficient will be zero if:

- *▶ Intercept $a = 0$
- ▶ Intercept $a \neq 0$
- ▶ Slope $b = 0$
- ▶ Slope $b \neq 0$

**Question No: 7 (Marks: 1) -
Please choose one**

When two coins are tossed the probability of at least one head is:

- ▶ $1/4$
- *▶ $3/4$

▶ 2/4

▶ 4/4

**Question No: 8 (Marks: 1) - Please
choose one**

**Which one of the following measurement does *not* divide a set of
observations into equal parts?**

▶ quartiles

- ▶ deciles
- ▶ percentiles
- *▶ **standard deviations**

**Question No: 9 (Marks: 1) - Please
choose one**

In the model $Y = mX + a$, Y is also known as the:

- ▶ Predictor variable
- ▶ Independent variable
- ▶ Predicted variable
- ▶ Explanatory variable

**Question No: 10 (Marks: 1) - Please
choose one**

According to empirical rule approximately 95% of the measurements will fall under which interval? <http://vustudents.ning.com>

- ▶ $\bar{X} \pm S$
- *▶ $\bar{X} \pm 2S$
- ▶ $\bar{X} \pm 3S$
- ▶ $\bar{X} \pm 4S$

**Question No: 11 (Marks: 1) - Please
choose one**

Which one of the following is written at the top of the table?

- ▶ Source note
- ▶ Foot note
- ▶ Prefatory note
- *▶ Title

**Question No:
12 (Marks: 1) - Please
choose one**

curve has a longer tail to the right, it is called :

itively skewed

If a

► Negatively skewed

► J-shaped

- ▶ Symmetric

**Question No: 13 (Marks: 1) - Please
choose one**

Which one of the following is the class frequency?

*** ▶ The number of observations in each class**

- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

**Question No: 14 (Marks: 1) - Please
choose one**

is a discrete random variable, then the function $f(x)$ is **If X**

- ▶ A probability function
 - ▶ A probability density function
 - ▶ A density function
- * ▶ A distribution function**

**Question No: 15 (Marks: 1) - Please
choose one**

Which one of the following graphs is used for a time series data?

<http://vustudents.ning.com>

*** ▶ Histogram**

- ▶ Histogram
- ▶ Frequency curve
- ▶ Frequency polygon

**Question No: 16 (Marks: 1) - Please
choose one**

**If
you connect the mid-points of rectangles in a histogram by a series of lines that also touches
the x-axis from both ends, what will you get?**

- ▶ Ogive
- ▶ Frequency polygon *▶ Frequency curve
- ▶ Histogram

**Question No: 17 (Marks: 1) - Please
choose one**

If mean of the two observations is 10.5, then median of these two observations will be:

- ▶ 7.5
- ▶ 8.5
- ▶ 9.5
- *▶ 10.5

Question No: 18 (Marks: 1) - Please choose one

Which one is the formula of mid range:

- ▶ $x_m \square x_0$
- ▶ $x_0 \square x_m$
- ▶ $\frac{x_0 \square x_m}{2}$
- ▶ $\frac{x_0 \square x_m}{2}$
- *▶

Question No: 19 (Marks: 1) - Please choose one

Which one of the following is not included in measures of central tendency:

- *▶ Quartile deviation
- ▶ Harmonic mean
- ▶ Geometric mean
- ▶ Arithmetic mean

Question No: 20 (Marks: 1) - Please choose one

For

the given data 2, 3, 7, 0, -8 G. M will be:

- ▶ Negative
 - ▶ Positive
 - ▶ Zero
- * ▶ **Undefined**

STA301 - Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one

P (A or B) =P (A) +P (B), then A and B are:

- Mutually exclusive events
- Independent events
- Exhaustive events
- Equally likely events

Question No: 2 (Marks: 1) - Please choose one

The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is:

- 4/52
- 2/52
- 13/52
- 26/52

Question No: 3 (Marks: 1) - Please choose one

The probability of an event always lies between:

- 0 and $-\infty$
- -1 and +1
- $-\infty$ and $+\infty$
- 0 and 1

Question No: 4 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series:

- C.V.
- Q.D.
- M.D.
- S.D.

Question No: 5 (Marks: 1) - Please choose one

How many elements are in the sample space of rolling one die:

- 6
- 12
- 24
- 36

Question No: 6 (Marks: 1) - Please choose one

When two coins are tossed simultaneously, P (one head) is:

$$\frac{1}{4}$$

- $\frac{1}{2}$
- $\frac{3}{4}$
- 1

Question No: 7 (Marks: 1) - Please choose one

For any number k, at least $1 - 1/k^2$ of the data-values fall within k standard deviations of the mean:

- Greater than 1
- Less than 1
- Greater or equal to 1
- Less or equal to 1

Question No: 8 (Marks: 1) - Please choose one

If $\bar{X} = 10$, & $Y = 5 + 2X$ then $\bar{Y}$ is :

- 5
- 10
- 15
- 25

Question No: 9 (Marks: 1) - Please choose one

If median = 7 and Mean = 5, what is the value of Q_2 :

- 1
- 3
- 5
- 7

Question No: 10 (Marks: 1) - Please choose one

Which one is the major branch of Statistics?

- Probabilistic Statistics
- Advanced Statistics
- Basic Statistics

▸ Descriptive Statistics

Which of the following value could not represent a coefficient of correlation:

- $r = 0.99$
- $r = 1.09$
- $r = -0.73$
- $r = -1$

Question No: 12 (Marks: 1) - Please choose one

Tabulation is the process of arranging data into:

- Different classes
- Rows
- Columns
- Rows and Columns

Question No: 13 (Marks: 1) - Please choose one

Which one is commonly called a bell shaped distribution?

- Symmetrical
- Bimodal
- Skewed
- U shaped

Question No: 14 (Marks: 1) - Please choose one

Which one of the following is a suitable frequency distribution for an ogive:

- A relative frequency distribution
- A frequency distribution
- A joint frequency distribution
- A cumulative frequency distribution

Question No: 15 (Marks: 1) - Please choose one

Which one of the following is not a type of frequency curve?

- The symmetrical frequency curve
- The extremely skewed frequency curve
- The U-shaped frequency curve
- Frequency polygon

Question No: 16 (Marks: 1) - Please choose one

In a set of 20 values all the values are 5, what will be the mean?

- 4
- 5
- 10

Question No: 17 (Marks: 1) - Please choose one

For the given set of observations 1, 4, 4, 4 and 7, which statement is true:

- Mean > median > mode
- Mean # median # mode
- Mean = median = mode
- Mean < median < mode

Question No: 18 (Marks: 1) - Please choose one

In a set of 10 observations the mean is 20 and the median is 15. There are 2 values that are 6, and all other values are different. What is the value of mode:

- 15
- 6
- 10
- 2

Question No: 19 (Marks: 1) - Please choose one

For a positively skewed distribution:

- Mean < Mode < Median
- Median < Mode > Mean
- Mode > Mean > Median
- Mean > Median > Mode

Question No: 20 (Marks: 1) - Please choose one

Which one of the following is equal to “explained variation divided by total variation”?

- Sum of squares due to regression
- Coefficient of determination
- Standard error of estimate
- Coefficient of correlation

STA301 - Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one

Which of the following can never be taken as the probability of an event?

- ▶ 1
- ▶ 0
- ▶ 0.5
- ▶ -0.5

Question No: 2 (Marks: 1) - Please choose one

When we toss a coin , we get only:

- ▶ 1 outcome
- ▶ 2 outcomes
- ▶ 3 outcomes
- ▶ 4 outcomes

Question No: 3 (Marks: 1) - Please choose one

The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is:

- ▶ $\frac{4}{52}$
- ▶ $\frac{2}{52}$
- ▶ $\frac{13}{52}$
- ▶ $\frac{26}{52}$

Question No: 4 (Marks: 1) - Please choose one

Which dispersion is used to compare variation of two series:

- ▶ C.V.
- ▶ Q.D.
- ▶ M.D.
- ▶ S.D.

Question No: 5 (Marks: 1) - Please choose one

If A and B are mutually exclusive events then $P(A \text{ or } B)$ equals:

- $P(A) + P(B) \leq P(A \text{ and } B)$
- $P(A) \leq P(B)$
- $P(A) + P(B)$
- $P(A|B) + P(B|A)$

Question No: 6 (Marks: 1) - Please choose one

The probability of an event always lies between:

- -1 to +1
- -1 to 0
- 0 to 1

▸ 0 to ☐

Question No: 7 (Marks: 1) - Please choose one

Standard deviation of the data 7, 7, 7, 7, 7, 7, 7 is

- 49
- Sqrt (7)
- 0
- 7

Question No: 8 (Marks: 1) - Please choose one

What is the range of -2,-3,-5,-10 :

- -12
- 8
- -8
- 2

Question No: 9 (Marks: 1) - Please choose one

A coin is tossed 3 times then, the number of sample points in the sample space is:

- 3
- 8
- 6
- 4

Question No: 10 (Marks: 1) - Please choose one

Sum of absolute deviations of the values is least when deviations are taken from:

- Mean
- Median
- Mode
- Geometric mean

Question No: 11 (Marks: 1) - Please choose one

For positively skewed distribution

Mean.....Median.....Mode:

ode:

- ☐
- =

▶
▶

<
<

Question No: 12 (Marks: 1) - Please choose one

$Q_1 = 20$, $Q_2 = 30$ then Q_3 is

In a symmetrical distribution, if :

- ▶ 50
- ▶ 40
- ▶ 30
- ▶ 10

Question No: 13 (Marks: 1) - Please choose one

What is the collection of all possible individuals, objects, or measurements?

- ▶ A sample
- ▶ A ratio measurement
- ▶ An inference
- ▶ A population

Question No: 14 (Marks: 1) - Please choose one

Which one is the major branch of Statistics?

- ▶ Probabilistic Statistics
- ▶ Advanced Statistics
- ▶ Basic Statistics
- ▶ Descriptive Statistics

Question No: 15 (Marks: 1) - Please choose one

What is the main objective of Descriptive Statistics?

- ▶ To test the population properties
- ▶ To describe the data we collected
- ▶ To infer something about the population
- ▶ To make estimates

Question No: 16 (Marks: 1) - Please choose one

Today, the major area of Statistics is concerned with drawing inferences about

- ▶ Complete universe
- ▶ Complete study
- ▶ Population under study
- ▶ Sample

Question No: 17 (Marks: 1) - Please choose one

The following data shows the scores made by Pakistani cricketers against

New Zealand in one – day match. What type of chart would we draw from the following data:

Cricketers	Inzam	Waseem	Shahid	Saeed	Imran	Razzaq
scores	54	47	26	30	25	23

- Multiple bar chart
- Component bar chart
- Simple bar chart
- Pie chart

Question No: 18 (Marks: 1) - Please choose one

If mean of the two observations is 10.5, then median of these two observations will be:

- 7.5
- 8.5
- 9.5
- 10.5

Question No: 19 (Marks: 1) - Please choose one

In a set of 20 values all the values are 5, what will be the mean?

- 4
- 5
- 10
- 20

Question No: 20 (Marks: 1) - Please choose one

The deviation of a distribution from symmetry is called:

- Kurtosis
- Skewness
- Dispersion
- Flatness

STA301 - Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one

The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is:

- $4/52$
- $2/52$
- $13/52$
- $26/52$

Question No: 2 (Marks: 1) - Please choose one

If X and Y are independent, then $\text{Var}(X-Y)$ is equal to:

$\text{Var}(X) \square \text{Var}(Y)$



$\text{Var}(X) \square \text{Var}(Y)$



$\text{Var}(X \square Y)$



- Zero

Question No: 3 (Marks: 1) - Please choose one

A standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:

- $1/5$
- $1/13$
- $5/52$
- $1/52$

Question No: 4 (Marks: 1) - Please choose one

How many elements are in the sample space of rolling one die:

- 6
- 12
- 24
- 36

Question No: 5 (Marks: 1) - Please choose one

Which of the following is true for coefficient of determination:

- Is the square of the coefficient of correlation
- Can be negative
- Is also called Pearson correlation coefficient
- Also called explained variation

We determine the number of combinations,when

- We are really computing a probability
- The order of the outcomes is not important
- The order of the outcomes is important
- We multiple the likelihood of two independent trials

Question No: 7 (Marks: 1) - Please choose one

When two coins are tossed the probability of at least one head is:

- 1/4
- 3/4
- 2/4
- 4/4

Question No: 8 (Marks: 1) - Please choose one

Stem and leaf was introduced by:

- Francis Galton
- Friedman
- John Tukey
- Pearson

Question No: 9 (Marks: 1) - Please choose one

The ages of 15 patients admitted to a certain hospital during a particular week were as follows:

12,48,31,54,71,29,42,62,67,57,74,27,18,71,26

The stem part is:

- 2,8,1,4,9,2,2,7,7,7,4,7,8,71,6
- 1,2,3,4,5,6,7
- 2,2,3,4,5,7,8
- 9,2,4,7,9,8,7,5,

Question No: 10 (Marks: 1) - Please choose one

If '2' is a leading digit in 24335, then what are the trailing digits in the observation to display a 'Stem-and –Leaf display'?

- 4
- 335
- 4335
- 43

Which of the following measures of central location is affected most by extreme values:

- Geometric Mean
- Median
- Mean
- Mode

Question No: 12 (Marks: 1) - Please choose one

The correlation coefficient lies between :

- 0 to 1
- 0 to ☐
- - 1 to +1
- +1 to ☐

Question No: 13 (Marks: 1) - Please choose one

Which one of the following statements is true regarding a sample?

- It is a part of population
- It must contain at least five observations
- It refers to descriptive statistics
- It produces True value

Question No: 14 (Marks: 1) - Please choose one

How to find the class midpoint?

- Half the sum of upper class limit and lower class limit
- Find the difference between consecutive lower limits
- Count the number of observations in the class
- Divide the class frequency by the number of observations

Question No: 15 (Marks: 1) - Please choose one

A discrete probability distribution may be represented by

- A table
- A graph
- A mathematical equation
- All of these

Question No: 16 (Marks: 1) - Please choose one

Which method is used to obtain the relative frequencies?

- Dividing the mid points by the class frequency
- Dividing the mid points by the total frequency
- Dividing the total frequency by the class frequency

- Dividing the frequency by the total frequency

- Negatively skewed
- U shape figure

Question No: 17 (Marks: 1) - Please choose one

The following data shows the number of hours worked by 200 statistics students.

Number of Hours

Frequency

0 - 9

10 - 19

20 - 29

30 - 39

What is its class interval?

- 9
- 10
- 11
- Varies from class to class

Question No: 18 (Marks: 1) - Please choose one

When the values are equally distributed at the both sides from the center then it is a

- Symmetrical distribution
- Positively skewed

Question No: 19 (Marks: 1) - Please choose one

Which one of the following scale of measurement is required for the median?

- Nominal
- Ordinal
- Interval
- Ratio

Question No: 20 (Marks: 1) - Please choose one

In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- The regression line will pass through the origin
- The regression line will pass through the point (0,10).
- The regression line will pass through the point (0,-10).
- The slope of the line will also be equal to 0.

1. A quantity obtained by applying certain rule or formula is known as
Select correct option:

Estimate

Estimator

2. Criteria to check a point estimator to be good involves
Select correct option:

Consistency

Unbiasedness

Efficiency

Above all

3. The F-distribution always ranges from:

Select correct option:

0 to 1

0 to -8

-8 to +8

0 to +8

4. $1-\alpha$ is the probability of

Select correct option:

Type 1 error

Rejection region

Acceptance

regionType 2

error

5. Parameter is aquantity.

Select correct option:

Constant

Variable

6. To find the estimate of a parameter.....methods are used.

Select correct option:

<http://en.wikipedia.org/wiki/F->

[page](#)

Two

Three

e

Four

Many

y

<http://orfe.princeton.edu/~jqfan/fan/classes/524/notes2.pdf>

7. A failing student is passed by an examiner. It is an example of:

Select correct option:

Type I error

Type II error

Correct

decision

No information regarding student exams

8. For two mutually exclusive events A and B, $P(A) = 0.2$ and $P(B) = 0.4$, then $P(A \cup B)$ is:

Select correct option:

0.8

0.2

0.6

0.5

$P(A \cup B) = P(A) + P(B) = 0.2 + 0.4 = 0.6$

9. An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

Select correct option:

Binomial distribution

Hypergeometric

distribution Poisson

distribution Exponential

distribution

10. A standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:

Select correct option:

1/5

5/52

1/52

11. If $P(A \cap B) = 0.12$ $P(A) = 0.3$, find $P(B)$ where 'A' and 'B' are independent:

Select correct option:

0.1

0.2

0.3

0.4

0.3X0.4=0.12

12. The mean deviation of the normal distribution is approximately:

Select correct option:

7/8 of the S.D

4/5 of the S.D

3/4 of the S.D

1/2 of the S.D

13. We use the Poisson approximation to the binomial when:

Select correct option:

p is 0.01 or less & n is 10 or more

p is 0.05 or less & n is 20 or more pg221

p is 0.04 or less & n is 15 or

more p is 0.02 or less & n is 10

or more

14. The conditional probability $P(A|B)$ is:

Select correct option:

$P(A \cap B)/P(B)$ pg157

$P(A \cap B)/P(A)$
 $P(A \cup B)/P(B)$
 $P(A \cup B)/P(A)$

15. We use the General Rule of Multiplication to combine:
Select correct option:

Events those are not
independent Mutually
exclusive events Events that
total more than 1.00
Events based on subjective probabilities

16. Which statement is NOT CORRECT?
Select correct option:

The sample standard deviation measures variability of our sample values
A larger sample will give answers that vary less from the true value than smaller samples
The sampling distribution describes how our estimate (answer) will vary if a new sample
is taken A large sample size always gives unbiased estimators regardless of how the
sample is chosen

17. Probability of an impossible event is always:

Select correct option: Less than one
Greater than one
Between one and zero
Zero

18. The number of parameters in uniform distribution is (are):
Select correct option:

1

2 pg 224

3
4

19. The probability can never be:
Select correct option:

1
1/2
1
-1/2

20. The conditional probability $P(A|B)$ is:
Select correct option:

$P(A \cap B)/P(B)$
 $P(A \cap B)/P(A)$
 $P(A \cup B)/P(B)$
 $P(A \cup B)/P(A)$

21. A random sample of $n=25$ values gives sample mean 83. Can this sample be regarded as drawn from a normal population with $\mu=80$ and $s=7$? In this question the alternative hypothesis will be:
Select correct option:

$H_1: \mu =$

$H_1: \mu ?$

$H_1: \mu >$

80

$H_1: \mu < 80$ pg 278

22. If $f(x)$ is a continuous probability function, then $P(X = 2)$ is:
Select correct option:
correct

1
0
1/2
2

23. The binomial distribution is negatively skewed when:

Select correct option:

$p > q$ pg 214

$p <$

q

$p =$

q

$p = q = 1/2$

24. If we roll three fair dices then the total number of outcomes is:

Select correct option:

6

36

216

1296

$6^3 = 216$

25. When we draw the sample with replacement, the probability distribution to be used is:

Select correct option:

Binomial

Hypergeometric

Binomial & hypergeometric pg 219

Poisson

26. The moment ratios of normal distribution come out to be:

Select correct option:

0 and 1

0 and 2

0 and 3 pg 226

0 and 4

27. The probability of an event is always:

Select correct option:

greater than 0
less than 1
between 0 and 1
greater than 1

28. Symbolically, a conditional probability is:
Select correct option:

$P(AB)$
 $P(A/B)$
 $P(A)$
 $P(A \cup B)$

29. Suppose the test scores of 600 students are normally distributed with a mean of 76 and standard deviation of 8. The number of students scoring between 70 and 82 is:
Select correct option:

272
164
260
328

30. If $P(A) = 0.3$ and $P(B) = 0.5$, find $P(A/B)$ where 'A' and 'B' are independent:
Select correct option:

$$P(A/B) = P(A \cap B) / P(B)$$

0.3
0.5
0.8
0.15

31. An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:
Select correct option:

Binomial distribution
Hypergeometric distribution
Poisson distribution
Exponential distribution

32. If the second moment ratio is less than 3 the distribution will be:

Select correct option:

- Mesokurtic
- Leptokurtic
- Platykurtic pg 226
- None of these

33. For the independent events A and B if $P(A) = 0.25$, $P(B) = 0.40$ then $P(A \text{ and } B) = \dots\dots$

Select correct option:

- 0.65
- 0.1
- 0.50
- 0.15
- $P(A) * P(B) 0.25 * 0.40 = 0.1$

34. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as_____.

Select correct option:

- Probability distribution
- The expected return
- The standard deviation
- Coefficient of variation

35. A random variable X has a probability distribution as follows: $X \mid 0 \ 1 \ 2 \ 3 \ P(X) \mid 2k \ 3k \ 13k \ 2k$
What is the possible value of k:

Select correct option:

- 0.01
- 0.03
- 0.05
- 0.07

36. The probability of drawing any one spade card is:

Select correct option:

1/52

4/52

13/52

52/52

37. The function abbreviated to d.f. is also called the.....

Select correct option:

18

Probability

Probability distribution function pg

172 Commutative distribution function

Discrete function

38. Binomial distribution is skewed to the right if:

Select correct option:

$p=q$

$p <$

q

$p >$

q

$p =$

n

39. A discrete probability function $f(x)$ is always:

Select correct option:

Zero

One pg 172

Negative

Non-

negative

40. For a binomial distribution, $n = 10$ & $q = 0.6$, the mean of the distribution is:

Select correct option:

0.6

6.0

10

4

$P = 1 - q$

$1 - 0.6 = 0.4,$

mean $= np = 10 * 0.4 = 4$

41. In the FA examination, 24 candidates offered Statistics. If the probability of passing the subject be $\frac{1}{3}$, what will be the mean of the distribution?

Select correct option:

- 7
- 8
- 6
- 5

42. The probability that a certain machine will produce a defective item is $\frac{1}{4}$. If a random sample of 6 items is taken from the output of this machine, what is the probability that there will be 5 or more defectives in the sample?

Select correct option:

- 3/4096
- 18/4096
- 19/4096
- 4/4096

43. Probability of type II error is

Select correct option:

a

B pg 276

1-a

1-B

44. If the values of variables are increasing or decreasing in the same direction then such kind of correlation is referred as

Select correct

option: Perfect

increasing or decreasing in the same direction,
http://en.wikipedia.org/wiki/Correlation_and_dependence

Correlation

Positive Correlation

Negative Correlation

45. The moving averages of the Prices 55,60,65,70 are

Select correct option:

70, 75

[STA301 has no](#)

60, 65

65,65

70,60

46. The best measure of variation is

Select correct option:

Range

Quartile deviation

Variance

Coefficient of variance

47. Ms. Christian calculated a correlation coefficient of .75. Which of the following reflects the best interpretation of this?

Select correct

option: Weak

negative.

Strong negative.

Weak positive.

Strong positive.

<http://www.nvcc.edu/home/elanthier/methods/correlation.htm>
Strong positive is at 1 always

48.....use the division of a circle into different sectors. **Select correct option:**

Line graph

Sector graphs

Frequency Polygon

Conversion

Graphs

49. The measurement of measure of degree of to which any two variables vary together is called

Select correct option:

Regression Coefficient

Correlation

Both (a) and (b)

None of these

50. Analysis of Variance (ANOVA) is a test for equality of:

Select correct option:

variances

means

proportio

ns

only two parameters

51. For some data you are given Maximum value = 96 , Minimum Value = 23 , Range = 73, number of classes selected between 5 and 15 Then class width will be Select correct option:

15

8

5

All options 1,2 ,3 are possible

52. If strength of association is weak, then $r = ?$ Select correct option:

$r = -1$

$r = 0$

$r = 1$

$r = 2$

53. The moving averages of the Prices 90,70,30,110 are Select correct option:

63.33, 70

73.33, 80

45.45, 68

65.50, 75

54. With increase in sample size, distribution tends to be a.....

Select correct option:

Meso

kurtic

Normal

Bell

shaped

Above all

55. In the central tendency Mean, Median and Mode

Select correct option:

Mean is better than Median

Median is better than Mode

Mean is better than Mode

All of these are true

56. The degree to which numerical data tend to spread about an average is called

Select correct

option: The

dispersion

Standard deviation

Correlation
None of these

57.graphs are similar to bar graphs.

Select correct option:

column
line
conversion
sector

58. A pattern of variation of a time series that repeats every year is called:

Select correct option:

Cyclical
Seasonal
Trend
Secular

59. In the central tendency Mean, Median and Mode

Select correct option:

Mean is better than Median
Median is better than Mode
Mean is better than Mode
All of these are true

60. The degree to which numerical data tend to spread about an average is called

Select correct option:

The dispersion
Standard
deviation
Correlation
None of these

61.graphs are similar to bar graphs.

Select correct option:

column
line

conversion sector

62. A pattern of variation of a time series that repeats every year is called:

Select correct option:

- Cyclical
- Seasonal
- Trend
- Secular

63. You have measured the systolic blood pressure of a random sample of 22 employees of a company. A 95% confidence interval for the mean systolic blood pressure for the employees is computed to be (120,138). Which of the following statements gives a valid interpretation of this interval?

Select correct option:

About 95% of the sample of employees has a systolic blood pressure between 120 and 138.

About 95% of the employees in the company

have a systolic blood pressure between 120 and 138.

If the sampling procedure were repeated many times, about 95% of the samples would have a mean systolic blood pressure between 120 and 138.

If the sampling procedure were repeated many times, about 95% of the samples would have a mean systolic blood pressure between 120 and 138.

64. Assume that a population consists of 7 similar containers having the following weights (kg): 9.8, 10.2, 10.4, 9.8, 10.0, 10.2, 9.6. What is the second moment about mean?

Select correct option:

0.262 kg

0.069kg

0.521 kg

0.313kg

65. How many numbers of parameter(s) are in t-distribution?

Select correct option:

This distribution has only one parameter v , which is known as the degrees of freedom of the t-distribution.

option:0

2

3

66. With increase in sample size, distribution tends to be a.....

Select correct option:

Meso kurtic

Normal

Bell

shaped

Above all

67. F-distribution is a..... distribution.

Select correct option:

Unimodal pg 312

Bimodal

l

Discrete

e

Negatively skewed

68. Which one of the following sampling methods would give unbiased results, if you need to find out the number of people in your town liking vanilla or chocolate ice creams?

Select correct option:

Ask my neighbors

Randomly select a few ice cream shops in town, and

Ask my friends

Ask my

classmates

69. A standard deviation obtained from sampling distribution of sample statistics is known as

Select correct

option: Sampling

Error

Standard error pg 240

70. If the regression line: $Y = 3 + 5X$ meets y-axis at '8' units distance from origin then the value of x-intercept is

Select correct option:

-(3/5)

3/5

11/5

1

71. Sum of three terms whose mean is equal to 90 is

Select correct option:

270

30

Also 90

None of these

according to this the Answer is 3

270/3=90

72. FREQUENCY Function calculates how often values occur within a range of values. Select correct option:

true

False

73. Which of the following correlation coefficients represents the weakest correlation between two variables?

Select correct option:

0.15

-0.15

0.02

-1.00

74. If the value of r is 0.8 ,then the coefficient of determination isSelect correct option:

67%

64%

80%

75%

75. If the dependent variable increases with the independent variable then the coefficient of correlation is

Select correct option:

0 to -1

0 to - 0.5

0 to -2

0 to 1

76. F- distribution tends to normality, if

Select correct option:

V1~8

V2~8

V1 and V2 ~8

Sample size is large

77. Consistency of an estimator can be checked by comparing

Select correct option:

Mea

n

MSE

Variance

Standard deviation

78. A standardized estimate has mean and variance

Select correct option:

(1, 0)

(0, 1)

(μ , s^2)

(μ , s)

79. A composite hypothesis comprises of

Select correct option:

Equality

Not equal

to

Less than/greater

(b) and (c)

80. For a particular hypothesis test, $\alpha=0.05$ and $\beta=0.05$. The power of test is equal to:

Select correct option:

0.14

0.90

0.95

0.25

81. In a t-distribution

Select correct option:

Mean=median=mode

Mean>Median<Mode

Mean<Median

Mean>Mode

Median<Mode<Mean

82. If we reject the null hypothesis, we might be making

Select correct

option: Type I error

Type II error

A correct decision

Unpredictable

83. The Central Limit Theorem is important in Statistics because it allows us to use the normal distribution to make inferences concerning the population mean:

Select correct option:

Provided that the population is normally distributed a Provided that the population is normally distributed (Provided that the sample size is reasonably large (fo

Provided that the population is nor mally distributed and the population variance is known (for any sample size)

84. Herbicide A has been used for years in order to kill a particular type of weed, but an experiment is to

be conducted in order to see whether a new herbicide, Herbicide B, is more effective than Herbicide A.

Herbicide A will continue to be used unless there is sufficient evidence that Herbicide B is more effective.

The alternative hypothesis in this problem is that

Select correct option:

Herbicide A is more effective than Herbicide B

Herbicide B is more effective than Herbicide A

Herbicide A is not more effective than

Herbicide B Herbicide B is not more effective than Herbicide A

85. A data in which we study about Regions is called

Select correct option:

Qualitative

Quantitative

Geographical

Chronological

86. If the median of an arrangement of numbers is equal to the mean of its middle terms then the arrangement contains

Select correct option:

Odd number of terms

Even number of terms

Unlimited number of

terms Prime number

87. If the graph is very much scattered, then what can be the suitable value of r ?

Select correct option:

$r = - 0.9$

$r = -0.5$
 $r =$ page
 $r = 0.8$

88. In scatter
diagram, clustering
of points around a
straight line
indicates Select
correct
option: Linear
regression Non-
linear regression
Curvilinear linear regression
Both a and b

if there is straight line then it is linear regression,
but if points around it then this is Non-linear

89. If the standard deviation of a population is 9, the population variance
is Select correct option:

3
9
21.35
81

90. How many steps are involved in general procedure for testing hypothesis:

4
5
6
7

91. When testing for independence in a contingency table with 2 rows and 5 columns, there are
_____ degrees of freedom.

4
10
7
5
 $r-1 \times c-1$
 $2-1 \times 5-1=4$

92. The critical region is in:

The middle of a distribution
The tails of a distribution
Either the middle or the tails of a distribution
Neither the middle nor the tails of a distribution

93. t-distribution is used to test the hypothesis about.....

Mean
proportion

The term 1-B is called
Level of the test

power of the test
Size of the test
Critical region

94. The asymptotic distribution of t-statistic with n-degree of freedom is
F

Normal

Z **T**

95. The Gallup Poll has decided to increase the size of its random sample of Canadian voters from about 1200 people to about 4000 people. The effect of this increase is to:

Reduce the bias of the estimate

Increase the standard error of the

estimate
Reduce the variability of the
estimate

Increase the confidence interval width for the parameter

96. The value of chi square can never be :

Zero

Less than 1

Greater than 1

Negative

97. The curve of the F- distribution depends upon:

Mean

Variance

Standard

Deviation
Sample

Size

98. We want to test $H_0 : \mu = 1.5$ vs. $H_1 : \mu \neq 1.5$ at $\alpha = .05$. A 95% confidence interval for μ calculated from a given random sample is (1.4, 3.6) Based on this finding we:

Fail to reject H_0

Reject H_0

Cannot make any decision at all because the value of the test statistic is not available

Cannot make any decision at all because (1.4, 3.6) is only a 95%

99. When we want to test the equality of two variances we usually use

F-test

Chi-square test

ANOVA

Z-test

100. To find the estimate of a parametermethods are used.

Two

Three

FO already
Many

101. In testing hypothesis, we always begin it with assuming

Null hypothesis is true that: Alternative hypothesis is true

Sample size is
large Population is
normal

102. t-distribution is applicable in case of

Independent samples

Dependent samples

Both (a) and (b)

Normal populations

103. When testing for independence in a contingency table with 3 rows and 4 columns, there are

_____ **degrees of freedom.**

5

6

7

12

it means 0 to infinity

104. The Chi-Square distribution is continuous distribution ranging from: $0 = \infty = 8$

$-8 = \infty = 0$

$-8 = \infty = 1$

$-8 = \infty = 8$

105. The location of the critical region depends upon:

Null hypothesis

Alternative

hypothesis Value of

alpha

Value of test-statistic

106. A random sample of $n = 6$ has the elements 6, 10, 13, 14, 18 and 20. What is the point estimate of the population mean?

12

13.5

11

11.5

107. ML estimators may not

..... **Consistent**

Efficient

Unbiased

Biased

108. Which of the following reveals the weakest fact.

Select correct option:

The measure of central tendency measures that value which depends only on the extreme values
The measure of central tendency measures that value in the data which occurs in the data most frequent times.

The measure of central tendency measures the value which has tendency to lie in the central part of the data.

109. The measure of central tendency measures the distance of values from means

Frequency polygon is

Select correct option:

Bar Charts

A line graph

Pareto

Charts None

of these

110. A bar graph usesto show data.

Select correct option:

Points

Bars

Lines

Picture

s

111. Geographical data deals with...

Select correct option:

Religio

n

Height

Income

Region

s

112. Which one provides the basis for hypothesis

testing?**Null hypothesis**

Alternative hypothesis

Critical value

Test-statistics

113. The test statistic to test the $U_1 = U_2$ (U represent the mean of population) for normal population for $n > 30$.

F-

test

Z-

test

T-

test

Chi-Square test

114. In a t-distribution

Mean=median=mode

Mean>Median<Mode
de Median
>Mean>Mode
Media<Mode<Mean

115. 1-a is the probability of

Type 1 error

Rejection region

Acceptance region

Type 2 error

116. Inferential statistics involves

Testing

Confidence interval

Estimation

Above all

117. Probability of type II error

is a

B

1-a

1-B

118. if the equation of regression line is $y = 5$, then what result will you take out from it?

Select correct option:

The line passes through

origin. The line passes

through (5, 0) The line is

parallel to y-axis.

The line is parallel to x-axis.

119. If the estimating equation is $Y = a - bX$, Which of the following is true

Select correct option:

a) The y intercept is 'b'

b) Slope of line is negative

c) There is inverse relationship

d) b & c

120. The variance of t-distribution, for $v > 2$, is always:

Select correct option:

Greater than

zero Less than

one Equal to one

Greater than

one

121. Alpha is the probability of

Select correct option:

http://en.wikipedia.org/wiki/Probability_of_error

Reject H_0

Accepting

H_0 Rejecting

H_1

Accepting

H_1

122. What type of data is collected in population census?

Select correct option:

Two Types

123. The collection of all outcomes for an experiment is called

Select correct option:

a sample space

the intersection of

events joint probability

population

124. Which of the graph is used for a time series data:

Select correct option:

but Time series isn't in

Frequency curve

Frequency polygon

HISTOGRAM

Histogram

125. A histogram consists of a set of adjacent rectangles whose bases are marked off by:

Select correct option:

Class boundaries

Class limits

Class

frequency

Class marks

126. The value that has half of the observations above it and half the observations below it is known as:

Select correct option:

Mean
Media
nMode
Standard deviation

127. The height of a student is 60 inches. This is an example of..... ?
Select correct option:

Continuous data
Qualitative data
Categorical data
Discrete data

128. Range of the values -2,-3,-4,-3,-9,-2,-8,-1,0 is
Select correct option:

0
-9
8
9
0-9=-9

129. If the both tails of the distribution are equal, then distribution is called:
Select correct option:

J-shaped
Symmetrical
Positively Skewed
Negatively Skewed

130. Ranking scale also include the properties of which scale?
Select correct option:

Nominal
scale
Interval
scale Ratio
scale All of
these

131. Range of the values -2.50,-3.70,-4.80,-3.10,-9.70,-2.20,-8.90,-1.60, 0.60 is
Select correct option:

10.03
10.30

9.10

9.00

0.60+9.70=10.30

132. What is/are the mode for the following data: 1,m,d,n,,2,d,2,d,s,5,5,7

Select correct option:

2 d is max so it is d

d

5

2,d,5

133. If the standard deviation of a population is 5.5, the population variance is:

Select correct option:

5.5

31

25

30.25

134. What we commonly called a bell shaped distribution:

Select correct option:

syme

bi

moderu

shap

skewed

135. The beginnings of a cumulative frequency distribution are presented below. What is the next number in the Cumulative Frequency column? Classes Frequency Cumulative Frequency
6.1 to 8 1 1 8.1 to 10 2 10.1 to 12 3

Select correct option:

0

1

2

3

136. Range of the values -10,- 19, -9, -15, -28, -26, -25 is:

Select correct option:

+18

-18

-19

+19

137. Which one of the following is less than median for a symmetrical distribution:

Select correct option:

- 50percentile
- 51 percentile
- 2quartile
- 4decile

138. The value of the middle term in a ranked (ordered) data set is called the
Select correct option:

- mode
- mean
- median
- harmonic mean

139. Sum of absolute deviations of the values is least when deviations are taken from
Select correct option:

- mean
- media
- n
- mode
- g.m

140. Statistic is a numerical quantity, which is calculated from
Select correct option:

- data
- observation
- sample
- population

141. The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:
Select correct option:

- descriptive
- advance
- infernetial
- sample

142. How to find the class midpoint?

Select correct option:

Half the sum of upper class limit and lower class

limit Find the difference between consecutive lower

limits Count the number of observations in the class

Divide the class frequency by the number of observ

143. For given data, discuss the shape of the distribution: X f 0.2 8 1.2 15 2.2 23 3.2 40

Select correct option:

Positively skewed

Negatively
skewed

Symmetric curve

U- Shaped curve

144. Data classified by attributes are called:

Select correct option:

group

qualitative

quantitative

array

if '2' is a leading digit in 24335, than what are the trailing digits in the observation
todisplay a 'Stem-and –Leaf display'.

Select correct option:

4

335

4335

43

145. A frequency polygon is obtained by plotting the class frequencies against what?

Select correct option:

classboundary

cumulative

frequencyrelative

frequency

mid point

146. When more values are lying at the start of the distribution, it is:

Select correct option:

u shape

positive

negative

symmetrica

147. The data for an ogive is found in which distribution:

Select correct option:

A cumulative frequency distribution

**A joint frequency
distribution**

**A frequency
distribution**

A relative frequency distribution

148. Which one of the following is greater than median for a symmetrical distribution:

Select correct option:

1st Decile

7th Decile

44th

Percentile

14th

Percentile

149. Statistics deals with

Select correct option:

Individuals

Isolated items

Isolated items

Aggregates of facts

150. Data classified by attributes are

called:**Select correct option:**

Grouped data

Qualitative

data

Quantitative

data

**Arrayed
data**

151. As a general rule, statisticians tends to use which of the following number of classes when arranging the data

Select correct option:

Fewer than 5

Between 5 & 20

Between 8 & 15

More than 20

152. The collection of all outcomes for an experiment is called
Select correct option:

a sample space

the intersection of

events joint probability

population

153. If $P(E)$ is the probability that an event will occur, which of the following must be false:

Select correct option:

$P(E) = -1$

$P(E) = 1$

$P(E) = 1/2$

$P(E) = 1/3$

154. If we roll a die then probability of getting a '2' will be

Select correct option:

$2/6$

$1/6$

$4/6$

1

155. In a multiplication theorem $P(A \cap B)$ equals:

Select correct option:

$P(A) P(B)$

$P(A) + P(B)$

$P(A) * P(B|A)$ pg 158

$P(B|A) * P(B)$

156. If $Y = 3X + 5$, then S.D of Y is equal to

Select correct option:

9 s.d(x)

3 s.d(x)

s.d(x) + 5

$3s.d(x) + 5$

157. In regression line $Y = a + bX$, X is called:

Select correct option:

Dependent variable

Independent variable

**Explained
variable
Regressand**

158. Symbolically, a marginal probability is:

Select correct option:

P(AB)

P(AUB)

P(A/B)

P(A)

159. Which formula represents the probability of the complement of event A:

Select correct option:

1 + P (A)

1 - P (A)

P (A)

P (A) -1

160. If A and B are independent events with $P(A) = 0.05$ and $P(B) = 0.65$, then $P(A|B) =$:

Select correct option:

0.65

0.05

0.03

0.07 not sure

161. The probability of drawing a 'white' ball from a bag containing 4 red, 8 black and 3 white balls is:

Select correct option:

0

3/15

1/12

1/2

Total ball= 4+8+3=15

P=3/15

162. An expected value of a random variable is equal to:

Select correct

option: Variance

Mean

Standard

deviation

Quartile

163. When we toss a fair coin 4 times, the sample space consists of....points.

Select correct option:

4

8

12

16

164. ${}^5C_5 =$

Select correct option:

5

1

10

25

165. In a probability distribution, the sum of the probabilities is equal to:

Select correct option:

0

0.1

0.5

1

166. The simultaneous occurrence of two events is called:

Select correct option:

Joint probability

Subjective probability

Prior probability

Conditional

probability

167. Let E and F be events associated with the same experiment. Suppose the E and F are independent and that $P(E) = 1/4$ and $P(F) = 1/2$ Then $P(E \cup F)$ is:

Select correct

option: $1/8$

$3/4$

$7/8$

$5/8$

$$\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

MidTerm Papers Solved MCQS with Reference (1 to 22 lectures)

1. The probability of drawing a "white" ball from a bag containing 4 red, 8 black and 3 white balls is:

- 0
- $\frac{3}{15}$
- $\frac{1}{12}$
- $\frac{1}{2}$

The total probability of choosing white ball is $4 + 8 + 3 = 15$, $3/15$

2. If the probability of one event does not affected by the occurrence of another event then both events are:

- Dependent
- Not Mutually Exclusive
- Mutually Exclusive

- **Independent**

PG # 162 (Lec # 22)

3. A die is rolled. What is the probability of getting a number that is an even number and greater than 2.

- ☐ 1/2
- ☐ 1/3
- ☒ 2/3
- ☐ 5/6

2, 4, 5, 6: even or greater than 2. That means 2 out of 6 is 2/3

4. The collection of all the elements under study is called:

- ☐ Population PG # 12 (Lec # 2)
- ☐ Sample
- ☐ Data
- ☐ Registration

5. The mode value from raw data can be obtained by the help of

- ☐ Dot plot PG # 53 (Lec # 6)
- ☐ Stem and leaf plot
- ☐ Bar chart
- ☐ None of these

6. A student solved 25 questions from first 50 questions of a book. The probability that he will solve the remaining all questions is:

- ☐ 0.25
- ☒ 0.5
- ☐ 1
- ☐ 0

7. The difference between the largest and the smallest data values is called the

- Variance
- Interquartile range
- **Range**
- Coefficient of variation

PG # 83 (Lec # 10)

8. Which one is the formula of mid-quartile range:

- $Q_3 - Q_1$
- $\frac{Q_3 - Q_1}{2}$
- $\frac{Q_1 - Q_3}{2}$
- $Q_1 - Q_3$

2

PG # 80 (Lec # 9)

9. For the given data 2, 3, 7, 0, -8; G. M will be:

- Negative
- Positive
- Zero

○ **Undefined**

Geometric mean cannot be calculated for negative values

10. If X and Y are independent variables then $\text{Var}(X-Y) =$ ____

- **$\text{Var}(X) - \text{Var}(Y)$**
- $\text{Var}(X) + \text{Var}(Y)$
- $\text{Var}(X+Y)$
- $\text{Var}(X) \times \text{Var}(Y)$

11. Rang measures the dispersion around the ____

- Extreme values
- Minimum value
- Maximum value

○ **Mid-range value** PG # 86 (Lec # 10)

12. The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

- Scatter variable

○ **Independent variable** PG # 121

- Dependent variable
- Correlation variable

13. In a scatter diagram we plot:

○ **X and Y** PG # 121 (Lec #15)

- X and $\hat{Y}$

- $\hat{Y}$ and $\hat{Y}$

- $\hat{Y}$ and X

14. If $P(A \cap B) = P(A|B) \times P(B)$ then A and B will be:

○ **Independent**

- Mutually Exclusive
- Dependent
- Equally Likely

15. Which one of the following is the class frequency?

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

16. How to convert a frequency distribution to a relative frequency distribution?

- Find the difference between consecutive lower class limits
- Find the difference between consecutive lower limits
- Count the number of observations in the class
- **Divide the class frequency by the total number of observations** PG # 31 (Lec # 4)

17. In statistics, we deal with:

- Individuals
- Isolated items
- **Aggregates of facts** PG # 7
- Qualitative data

18. Pie chart consists of

- A rectangular
- A triangle
- A square
- **A circle** PG # 23 (Lec # 3)

19. When $Q_1 = 2$ and $Q_3 = 4$, what is the value of Median, if the distribution is symmetrical:

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4

PG # 112 (Lec # 14)

20. A letter is chosen at random from the word STATISTICS. The probability of getting a vowel is:

- ☐ **3/10**
- ☐ 4/10
- ☐ 5/10
- ☐ 6/10

21. When two coins are tossed simultaneously, P (one head) is:

- ☐ $\frac{1}{4}$
- ☐ $\frac{1}{2}$
- ☐ $\frac{3}{4}$
- ☐ 1

Total Sample HH,TH,HT,TT Probability One Head TH,HT So $\frac{2}{4}=\frac{1}{2}$

22. A value which has half of the observations above it and half of the observations below it; is known as.....

- ☐ Rang
- ☐ Mean
- ☐ Mode

☐ **Median**

PG # 64 (Lec # 7)

23. In a set of 20 values all the values are 2, what will be the Geometric Mean?

- ☐ 2
- ☐ 5
- ☐ 10
- ☐ 20

24. Which of the following measure of dispersion is least affected by the extreme values in a data?

- ☐ **Range** **PG # 83 (Lec # 10)**
- ☐ Quartile deviation
- ☐ Mean absolute deviation
- ☐ Standard deviation

25. Which is appropriate average for finding the average speed of a car:

- ☐ Mean
- ☐ Geometric mean
- ☐ **Harmonic mean** **PG # 78 and 79 (Lec # 9)**
- ☐ Weighted mean

26. What will be the value of the total amount of dispersion obtained by summing the $(X - \bar{X})$?

- ☐ **0** **PG # 88 (Lec # 11)**
- ☐ -1
- ☐ 1
- ☐ **100**

27. The deviation of a distribution from symmetry is called:

○ Kurtosis

○ **Skewness**

PG # 86 (Lec # 10)

○ Dispersion

○ Flatness

28. For the independent events A and B if $P(A) = 0.25$, $P(B) = 0.40$ then $P(A \cap B) = \dots\dots\dots$

○ 0.65

○ **0.1**

○ 0.50

○ 0.15

$$P(A \cap B) = P(A) \times P(B) = 0.25 \times 0.40 = 0.1$$

29. The variable plotted on the vertical or Y-axis in a scatter diagram is called the:

○ Scatter variable

○ Independent variable

○ **Dependent variable**

PG # 121

○ Correlation variable

30. In a multiplication theorem $P(A \cap B)$ equals:

○ $P(A) P(B)$

○ $P(A) + P(B)$

○ **$P(A) \cap P(B|A)$**

PG # 162 (Lec # 21)

○ $P(B|A) \cap P(B)$

31. For two independent events A and B, if $P(A) = 0.2$ and $P(B) = 0.4$, then what is $P(A \cap B)$?

- ☐ 0.06
- ☒ 0.08
- ☐ 0.02
- ☐ 0.04

$$P(A \cap B) = P(A) \times P(B) = 0.2 \times 0.4 = 0.08$$

32. The level of satisfaction for a certain facility which Govt. is providing is the example of:

- ☐ Constant
- ☐ Qualitative variable
- ☐ Variable
- ☐ Quantitative variable

Quantitative variable when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye color, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a **qualitative variable**.

33. Quantitative variable is further divided into:

- ☐ Continuous variable
- ☐ Discrete variable
- ☐ Continuous & Discrete variable
- ☐ None of the above

PG # 9 (Lec # 1)

34. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ☐ Ogive
- ☐ Frequency polygon
- ☐ Frequency curve
- ☐ Histogram

PG # 32 (Lec # 4)

35. Which type of data is collected in population census?

- Grouped data
- Secondary data
- **Primary data**
- Array data

36. Data obtained from the Bureau of Statistics, is an example of:

- None of these
- Primary data
- **Secondary data** **PG # 12 (Lec # 2)**
- Qualitative data

37. A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set** **PG # 134**
- None of these

38. In positively skewed distribution, which of the following relationship exists:

- The mean, median, and mode are all equal
- **The mean is larger than the median** [Click Here For More Detail](#)
- The median is larger than the mean
- The standard deviation must be larger than the mean or the median

39. For two mutually exclusive events A and B, $P(A) = 0.2$ and $P(R) = 0.4$, then $P(A \cup B)$ is:

☐ 0.8

☐ 0.2

☐ 0.6

PG # 154

☐ 0.5

$$P(A \cup B) = P(A) + P(B) = 0.2 + 0.4 = 0.6$$

40. When two coins are tossed the probability of at most one head is:

☐ 1/4

☐ 2/4

☒ 3/4

☐ 4/4

41. In a set of 20 values all the values are 10, what will be median?

☐ 2

☐ 5

☒ 10

☐ 20

42. Which one of the following measure is not based on all the observations?

☐ Arithmetic Mean

☐ Geometric Mean

☐ Harmonic Mean

☒ Mode

43. A five-number summary consists of :

- X_0 , Q_1 Median, Q_3 and X_m
- X_m , Q_1 , Mean, O_3 , and X_0
- X_m , Q_1 Mode, Q_3 , and X_0
- X_0 , Q_1 , Median, Q_2 , and X_m

PG # 97 (Lec # 12)

44. For what condition of k , at least $1 - 1/k^2$ of the data-values fall within k standard deviations of the mean:

- **Greater than 1**
- Less than 1
- Greater or equal to 1
- Less or equal to 1

PG # 94 (Lec # 12)

45. Which of the following can never be taken as the probability of an event?

- 1
- 0
- **0.5**
- **-0.5**

Probability always lie between 0 and 1

46. If E is an impossible event, then $P(E)$ is:

- 1
- **0.15**
- **0**
- **0.5**

47. The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

- Geometric Mean
- **Harmonic Mean** **PG # 77 (Lec # 9)**
- Mode
- Median

48. An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is more appropriate to calculate the average speed?

- Median
- Arithmetic mean
- **Harmonic mean** **PG # 78 (Lec # 9)**
- Geometric mean

49. What is the standard deviation of the following values? 7, 7, 7, 7, 7, 7, 7

- 7
- 1
- 49
- **0** [Click Here For Reference](#)

50. Which one of the following is a meso-kurtic curve?

- Negatively skewed
- Positively skewed
- J-shaped
- **Normal** **PG # 114 (Lec # 14)**

The **NORMAL curve** is a curve which is neither very peaked nor very flat, and hence it is taken as A BASIS FOR COMPARISON. The normal curve itself is called **MESOKURTIC**.

51. Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct?

- The correlation coefficient is -5
- The correlation coefficient is 5
- **The correlation coefficient is 1**
- The correlation coefficient is 0

The relationship is perfect, with no error

52. Which of the following comes first to make frequency distribution.

- Number of Groups
- Class interval
- **Range** **PG # 28**
- Tally marks

53. Data obtained from the Ministry of food, is an example of:

- Primary data
- **Secondary data** **PG # 12 (Lec # 2)**
- Qualitative data
- None of these

54. Total angle of a Pie Chart is:

- **90"**
- **180°**
- **270°**
- **360°** **PG # 23 (Lec # 3)**

55. Which one is the formula of range:

- ☐ $\frac{x_m - x_0}{x_0}$
- ☐ $\frac{x_0 - x_m}{x_m}$
- ☐ $\frac{x_0 - x_m}{2}$
- ☐ $\frac{x_0 - x_m}{x_m^2}$

PG # 28 (Lec # 4)

56. If a player well shuffles the pack of 52 playing card, then the probability of a black card from 52 playing cards is:

- ☐ 1/52
- ☐ 13/52
- ☐ 4/52
- ☒ 26/52

57. Pearsons coefficient of skewness is not possible if standard deviation is:

- ☐ 1
- ☒ 1.5

☐ 0

PG # 109 (Lec # 13)

- ☐ 3

58. What is the difference between permutation and combination?

- ☒ In permutation order is important and in combination order is not important

[Click Here For Reference](#)

- ☐ In permutation order is not important and in combination order is important
- ☐ Only combination is based on the classical definition of probability
- ☐ Only permutation is based on the classical definition of probability

59. If data is categorized about any characteristic, then which measurement scale should be used?

- Ordinal
- Interval
- **Nominal**
- Ratio

PG # 9 (Lec # 1)

60. Which of the following scale has true zero point?

- **Ratio Scale**
- Interval scale
- Nominal scale
- Ordinal scale

PG # 9 (Lec # 1)

61. In a simple bar chart:

- Width of bar is meaningful
- Height of bar is meaningful
- Height of bar is not meaningful
- **Height and width both are meaningful**

PG # 24

A simple bar chart consists of horizontal or vertical bars of equal width and lengths proportional to values they represent

62. What is the name of following curve?

- **Positively skewed curve**
- Symmetrical curve
- Normal curve
- Negatively skewed curve

PG # 39

63. If $A = \{1, 2, 3, 4, 5, 10\}$ and $B = \{1, 3, 5\}$ then $B \subseteq A$ means:

- ☐ A is less than B
- ☐ A is contained in B
- ☐ **B is contained in A.** PG # 134 (Lec # 16)
- ☐ B is less than A

64. The graph of cumulative frequency distribution is called:

- ☐ Frequency polygon
- ☐ **Ogive** PG # 43 (Lec # 5)
- ☐ Frequency curve
- ☐ Histogram

65. Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given series:

- ☐ 4
- ☐ 3
- ☐ **3**
- ☐ 2
- ☐ 3
- ☐ 2
- ☐ 1
- ☐ 3
- ☐ 2
- ☐ □

66. Which one of the following is equal to the 2nd quartile:

- ☐ P₃₃
- ☐ D₃
- ☐ **Median** PG # 69 (Lec # 8)
- ☐ Mode

67. The variance of a sample of 81 observations equal to 64. The standard deviation of the sample is equal to:

- ☒ 256
- ☒ 4096
- ☐ 8
- ☒ 6,561

68. Under which condition standard deviation could assume a negative value:

- ☐ When more than half of the data values are negative
- ☐ If all the data values are same
 - ☐ The standard deviation cannot be negative
- ☐ When all the data values are negative

Standard deviation cannot be negative because it is square rooted variance.

69. In a set of 10 values all the values are 5, what will be the D_5 ?

- ☐ 2
 - ☐ 5
 - ☐ 2
 - ☐ 5
 - ☐ 10
 - ☐ 20
- [Click Here For Reference](#)

70. When three, coins are tossed simultaneously, P (3 Heads) is:

- ☒ 3/8
- ☒ 1/8
- ☒ 4/8
- ☒ 2/8

71. The distribution is mesokurtic if the moment ratio i. e. b_2 is:

- Equal to 0
- **Equal to 3** PG # 119 (Lec # 14)
- Less than 3
- Greater than 3

72. A set is any well-defined collection of

- Positive Objects
- Negative Objects
- Same Objects
- **Distinct Objects** PG # 133 (Lec # 16)

73. If an observation in the data set is negative, then the geometric mean is:

- Positive
- negative
- Zero
- **undetermined**

74. Mean deviation is least when it is calculated from_____.

- Mean
- Mode
- **Median**
- Geometric mean

Mean deviation is minimum if the deviations are taken from the median.

75. For a positively skewed distribution m_3 will be:

- ☐ Positive PG # 119 (Lec # 14)
- ☐ Negative
- ☐ Zero
- ☐ 1

76. Which of the following value could not represent a coefficient of correlation?

- ☒ $r = 0.93$
- ☐ $r = 1.09$ PG # 128 (Lec # 15)
- ☒ $r = -0.73$
- ☐ $r = -1$

r is a pure number that lies between -1 and 1 i.e. $-1 < r < 1$

77. The conditional probability is $P(A|B)$ is given by formula:

- ☐ $\frac{P(A \cap B)}{P(A)}$
- ☐ $\frac{P(A \cap B)}{P(B)}$ PG # 159 (Lec # 20)
- ☐ $\frac{P(A \cup B)}{P(A)}$
- ☐ $\frac{P(A \cup B)}{P(B)}$

78. If a curve has a longer tail to the right, it is called :

- ☐ Positively skewed PG # 39 (Lec # 5)
- ☐ Negatively skewed
- ☐ J-shaped
- ☐ Symmetric

79. Which of the following scale has not true zero point?

- ☐ Nominal scale
- ☐ Ordinal scale

☐ Interval scale

PG # 9 (Lec # 1)

- ☐ Ratio scale

80. Mean is affected by the change of:

- ☐ Origin Only
- ☐ Scale Only

☐ Origin and Scale

- ☐ Not affected

Arithmetic Mean is affected due to the change of origin and scale.

81. What will be the mode for the data 2,3,4,5,4,6,4?

- ☐ 2
- ☐ 3

☐ 4

- ☐ 5

The mode is the value which appears the most often in the data. It is possible to have more than one mode if there is more than one value which appears the most.

82. In a set of 20 values all the values are 5, what will be the mean?

☐ 4

☐ 5

☐ 10

☐ 20

83. It is easy to find the median when data is.....

- ☒ Arranged
- ☐ Discrete
- ☐ Attributed
- ☐ Continuous

Question No: 1
Marks: 1)
Please choose one

The median of a set of data values is the middle value of the data set when it has been arranged in ascending order. That is, from the smallest value to the highest value.

For a

particular data the value of Pearson's coefficient of skewness is greater than zero. What will be the shape of distribution?

► Negatively skewed

- J-shaped
- Symmetrical
- Positively skewed

Question No: 2 (Marks: 1) - Please choose one

In measures

of relative dispersion unit of measurement is:

► Changed

- Vanish
- Does not change
- Dependent

The F-

Question No: 3 (Marks: 1) - Please choose one

distribution always ranges from:

In chi-square

- 0 to 1
- 0 to $-\infty$
- $-\infty$ to $+\infty$
- 0 to $+\infty$

Question No: 4 (Marks: 1) - Please choose one

test of independence the degrees of freedom are:

- ▶ $n - p$
- ▶ $n - p - 1$
- ▶ $n - p - 2$
- ▶ $n - 2$

The Chi-

Question No: 5 (Marks: 1) - Please choose one

Square distribution is continuous d istribution ranging from:

If X and Y are

- ▶ $-\infty \leq \chi^2 \leq \infty$
- ▶ $-\infty \leq \chi^2 \leq 1$
- ▶ $-\infty \leq \chi^2 \leq 0$
- ▶ $0 \leq \chi^2 \leq \infty$

Question No: 6 (Marks: 1) - Please choose one

$$E(X \pm Y)$$

random variables, then is equal to:

$$E(X) \pm E(Y)$$



$$E(X) - E(Y)$$

$$X \pm E(Y)$$



$$E(X) \pm Y$$



Question No: 7 (Marks: 1) - Please choose one

If $\hat{y}$ is the predicted value for a given x-value and b is the y-intercept then the equation of a regression line for an independent variable x and a dependent variable y is:

- ▶ $\hat{y} = mx + b$, where m = slope
- ▶ $x = \hat{y} + mb$, where m = slope
- ▶ $\hat{y} = x/m + b$, where m = slope
- ▶ $\hat{y} = x + mb$, where m = slope

Question No: 8 (Marks: 1) - Please choose one

▶ Value of test-statistic

the critical region depends upon:

- ▶ Null hypothesis
- ▶ Alternative hypothesis
- ▶ Value of alpha

Question No: 9 (Marks: 1) - Please choose one
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The location of

The variance of the t-distribution is give by the formula:

$$\sigma^2 = \sqrt{\frac{v}{v-2}}$$



$$\sigma^2 = v^2$$



$$v = 2$$

$$\sigma^2 = \frac{v}{v-1}$$



$$\sigma^2 = \frac{v}{v-2}$$



Question No: 10 (Marks: 1) - Please choose one

Which one is the correct formula for finding desired sample size?

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{e} \right)^2$$



$$n = \frac{Z_{\alpha/2} \sqrt{\sigma}}{e}$$



$$n = \frac{Z_{\alpha/2} \cdot \bar{X}}{e}$$



$$n = \frac{Z_{\alpha} \sigma}{e}$$



Question No: 11 (Marks: 1) - Please choose one

A discrete probability function f(x) is always:

▶ Non-negative

► Negative

► One

► Zero

Question No: 12 (Marks: 1) - Please choose one

_____ $E(4X + 5)$
= _____

► $12 E(X)$

► $4 E(X) + 5$

► $16 E(X) + 5$

► $16 E(X)$

Question No: 13 (Marks: 1) - Please choose one

_____ How $P(X + Y < 1)$ can be find:

► $f(0, 0) + f(0, 1) + f(1, 2)$

► $f(2, 0) + f(0, 1) + f(1, 0)$

► $f(0, 0) + f(1, 1) + f(1, 0)$

► $f(0, 0) + f(0, 1) + f(1, 0)$

Question No: 14 (Marks: 1) - Please choose one

_____ The
 $f(x|1)$ _____:

► $f(1,1)$

► $f(x,1)$

►

► $\frac{f(x,1)}{h(1)}$

► $\frac{f(x,1)h}{x}$

►

Question No: 15 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

► .0401

► .5500

► .4599

► .9599

Question No: 16 (Marks: 1) - Please choose one

In normal

distribution M.D. =

► 0.5σ

► 0.7979σ

► 0.6445σ

Question No: 17 (Marks: 1) - Please choose one

In an

ANOVA test there are 5 observations in each of three treatments. The degrees of freedom in the numerator and denominator respectively are.....

► 2, 4

► 3, 15

► 3, 12

► 2, 12

Question No: 18 (Marks: 1) - Please choose one

A set that contains all possible outcomes of a system is known as

► Finite Set

► Infinite Set

► Universal Set

► No of these

Question No: 19 (Marks: 1) - Please choose one

Stem and leaf

is more informative when data is :

► Equal to 100

► Greater Than 100

► Less than 100

► In all situations

Question No: 20

(Marks: 1)

- Please choose one

A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:

- ▶ Infinite population
- ▶ Finite population
- ▶ Concrete population
- ▶ **Hypothetical population**

In a multiplication theorem $P(A \text{ and } B)$ equals:

▶ **Select correct option:**

- ☐ $P(A) P(B)$
- ☒ $P(A) + P(B)$
- ☐ $P(A) * P(B|A)$
- ☐ $P(B|A) * P(B)$

The probability can never be:

▶ **Select correct option:**

- ☐ 1
- ☐ $1/2$
- ☐ 1
- ☒ $-1/2$

If two fair die are thrown, the probability of getting a double six is:

Select correct option:

- ☐ $1/6$
- ☒ $2/36$
- ☐ $1/36$
- ☐ $1/12$

If A and B are independent events with $P(A) = 0.05$ and $P(B) = 0.65$, then $P(A|B) =$:

Select correct option:

- ☐ 0.65
- ☐ 0.05
- ☒ 0.03
- ☐ 0.07

Twenty percent of the students in a class of 100 are planning to go to graduate school. The standard deviation of this binomial distribution is:

► **Select correct option:**

- ☒ 20
- ☐ 2
- ☐ 4
- ☐ 16

If $f(x)$ is a continuous probability function, then $P(X = 2)$ is:

Select correct option:

- ☐ 1
- ☒ 0
- ☐ 1/2
- ☐ 2

Probability of an impossible event is always:

Select correct option:

- ☐ Less than one
- ☐ Greater than one
- ☐ Between one and zero
- ☒ Zero

Question # 8 of 10 (**Start time: 01:38:25 PM**) **Total Marks: 1**

$E(4X + 5) =$ _____

▶ **Select correct option:**

12 E (X)

☐ $4 E (X) + 5$

☒ $16 E (X) + 5$

☐ $16 E (X)$

The location and shape of the normal curve is (are) determined by:

▶ **Select correct option:**

☐ Mean

☐ Variance

☐ Mean & variance

☒ **Mean & standard deviation**

The probability of success changes from trial to trial, is the property of:

▶ **Select correct option:**

☒ **Binomial experiment**

☐ Hypergeometric experiment

☐ Both binomial & hypergeometric experiment

☐ Poisson experiment

Question No: 1 (Marks: 1) - Please choose one

_____ **Mean deviation is always:**

- ▶ **Less than S.D**
- ▶ Greater than S.D
- ▶ Greater or equal to S.D
- ▶ Less or equal to S.D

Question No: 2 (Marks: 1) - Please choose one

_____ **The value of χ^2 can never be :**

- ▶ Zero
- ▶ Less than 1
- ▶ Greater than 1
- ▶ **Negative**

Question No: 3 (Marks: 1) - Please choose one

The mean of the F-distribution is:

▶ $\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$

▶ $\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$

▶ $\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$

▶ $\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$

▶

Question No: 4 (Marks: 1) - Please choose one

If X and Y are random variables, then $E(X + Y)$ is equal to:

$E(X) + E(Y)$

▶ $E(X) - E(Y)$

▶ $X + E(Y)$

▶ $E(X) + Y$

▶

Question No: 5 (Marks: 1) - Please choose one

Evaluate: $(9-4)!$

▶ 362880

▶ 120

▶ 24

▶ 6

Question No: 6 (Marks: 1) - Please choose one

Which formula represents the probability of the complement of event A:

▶ $1 + P(A)$

▶ $1 - P(A)$

▶ $P(A)$

▶ $P(A) - 1$

Question No: 7 (Marks: 1) - Please choose one

Ideally the width of confidence interval should be:

- ▶ 0
- ▶ 1
- ▶ 99
- ▶ 100

Question No: 8 (Marks: 1) - Please choose one

If the sampling distribution of $\bar{X}$ is normal, the interval $\mu_{\bar{X}} \pm 3\sigma_{\bar{X}}$ includes:

- ▶ 99% of the sample means
- ▶ 99.73% of the sample means
- ▶ 98% of the sample means
- ▶ 95% of the sample means

Question No: 9 (Marks: 1) - Please choose one

The probability distribution of a statistic is called the:

- ▶ Population distribution
- ▶ Frequency distribution
- ▶ Sampling distribution
- ▶ Sample distribution

Question No: 10 (Marks: 1) - Please choose one

An estimator T is said to be unbiased estimator of θ if

- ▶ $E(T) = \theta$
- ▶ $E(T) = T$
- ▶ $E(T) = 0$
- ▶ $E(T) = 1$

Question No: 11 (Marks: 1) - Please choose one

If the following is a probability distribution, then what is the value of 'a':

X	1	2	3
P(X)	0.1	a	0.1

- ▶ 0.6
- ▶ 0.8
- ▶ 0.2
- ▶ 0.4

Question No: 12 (Marks: 1) - Please choose one

A discrete probability function $f(x)$ is always:

- ▶ Non-negative
- ▶ Negative
- ▶ One
- ▶ Zero

Question No: 13 (Marks: 1) - Please choose one

An expected value of a random variable is equal to:

- ▶ Variance
- ▶ Mean
- ▶ Standard deviation
- ▶ Covariance

Question No: 14 (Marks: 1) - Please choose one

The $f(x|1)$ _____:

- ▶ $f(1,1)$
- ▶ $f(x,1)$
- ▶ $\frac{f(x,1)}{h(1)}$

$$\frac{f}{x, 1} h$$



Question No: 15 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

- ▶ .0401
- ▶ .5500
- ▶ .4599
- ▶ .9599

Question No: 16 (Marks: 1) - Please choose one

The continuity correction factor is used when:

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Question No: 17 (Marks: 1) - Please choose one

Which of the following is impossible in sampling:

- ▶ Destructive tests
- ▶ Heterogeneous
- ▶ To make voters list
- ▶ None of these

Question No: 18 (Marks: 1) - Please choose one

Which of the following is a systematic arrangement of data into rows and columns?

- ▶ Classification
- ▶ Tabulation
- ▶ Bar chart
- ▶ Component bar chart

Question No: 19 (Marks: 1) - Please choose one

Which one of the following statements is true regarding a sample?

- ▶ It is a part of population

- It must contain at least five observations
- It produces True value

Solved MCQs

Question No: 20 (Marks: 1) - Please choose one

an ogive is found in which distribution?

- A relative frequency distribution
- A frequency distribution
- A joint frequency distribution
- **A cumulative frequency distribution**

The data for

1- Hypothesis refers to

The outcome of an experiment

A conclusion drawn from an experiment

A form of bias in which the subject tries to outguess the experimenter

A tentative statement about the relationship

Statistics is used by researchers to

Analyze the empirical data collected in a study

Make their findings sound better

Operationally define their variables

Ensure the study comes out the way it was intended

3 - A literature review requires

Planning

Good & clear writing

Lot of rewriting

All of the above

4 - A literature review is based on the assumption that Copy from the work of others

Knowledge accumulates and learns from the work of others

Knowledge disaccumulates

None of the above option

5 - A theoretical framework

Elaborates the r/s among the variables

Explains the logic underlying these r/s

Describes the nature and direction of the r/s

All of the above

6 - Which of the following statement is not true?

A research proposal is a document that presents a plan for a project

A research proposal shows that the researcher is capable of successfully conducting the proposed research project

A research proposal is an unorganized and unplanned project

A research proposal is just like a research report and written before the research project

7 - Preliminary data collection is a part of the

Descriptive research

Exploratory research Applied

research Explanatory research

8 - Conducting surveys is the most common method of generating

Primary data

Secondary data

Qualitative data None of
the above

9 - After identifying the important variables and establishing the logical reasoning in theoretical framework, the next step in the research process is

To conduct surveys

To generate the hypothesis

To focus group discussions

To use experiments in an investigation

10 - The appropriate analytical technique is determined by The
research design

Nature of the data collected

Nature of the hypothesis Both

A & B

Personal interviews conducted in shopping malls are known as:

Mall interviews

Mall intercept interviews

Brief interviews

None of the given options

WATS lines provided by long distance telephone service at fixed rates. In this regard, WATS is the abbreviation of:

West Africa Theological Seminary Washtenaw

Area Transportation Study **Wide Area**

Telecommunications Service World Air

Transport Statistics

A list of questions which is handed over to the respondent, who reads the questions and records the answers himself is known as the:

Interview schedule

Questionnaire

Interview guide

All of the given options

One of the most critical stages in the survey research process is: Research

design

Questionnaire design

Interview design
Survey design

Question that consists of two or more questions joined together is called a:

Double barreled question

General question
Accurate question
Confusing question

The number of questionnaires returned or completed divided by the total number of eligible people who were contacted or asked to participate in the survey is called the:

Response rate

Participation rate
Inflation rate
None of the given options

To obtain the freest opinion of the respondent, when we ask general question before a specific question then this procedure is called as the:

Research technique
Qualitative technique
Funnel technique
Quantitative technique

A small scale trial run of a particular component is known as: Pilot testing

Pre-testing
Lab experiments
Both A & B

Field testing of the questionnaire shows that:

Respondents are willing to co-operate

Respondents are not willing to co-operate
Respondents do not like any participation
All of the given options

Service evaluation of hotels and restaurants can be done by the:

Self-administered questionnaires

Office assistant
Manager
None of the given options

The _____ analysis is a technique for gathering and analyzing the content of a text:

Content analysis

Graphical analysis
Field analysis
Mathematical analysis

_____ is the strength or power of a message in a direction _____ :

Intensity

Frequency
Direction
Space

_____ data refers to information _____ gathered by someone other than the researcher conducting the present study:

Primary data
Secondary data
Qualitative data
Quantitative data

_____ research produces soft data:

Qualitative research

Quantitative research
Descriptive research
Applied research

_____ research produces hard data:

Qualitative research
Quantitative research
Descriptive research
Applied research

_____ research is based on naturalism:

Field research
Descriptive research
Basic research
Applied research

_____ interview is always with one respondent alone:

Survey interview Field
interview Telephone
interview Electronic
interview

Historiography is the method of doing _____ research or of gathering & analyzing historical evidence:

Historical research
Basic research Applied
research Action research

Historical comparative researches often use _____ sources or different data types in combination:

Primary sources
Secondary sources
Internal sources
External sources

_____ is a comprehensive description and analysis of a single situation in a number of specific situations or cases:

Case study
Content analysis
Field research
None of the given options

Q1. All the persons involved in the collection of data and supervision of data collection process are called

- i. Fieldworkers
- ii. Researchers
- iii. Research assistants
- iv. None of the given options

Q2. While terminating the interview, the fieldworker should **not** do one of the following:

- i. He should record all the responses made by the interviewee before leaving.
- ii. He should thank the interviewee.
- iii. He should close the interview hastily.
- iv. He should answer all the questions the respondent asks concerning the nature and purpose of the study.

Q3. Which one of these is a type of Interviewee bias?

- i. The respondent does not tell his true income, age, or contact information.
- ii. The fieldworker fails to probe the interviewee properly.
- iii. The fieldworker contacted the wrong person for interview.
- iv. The fieldworker asks the questions in wrong order.

Q4. A magazine conducts a survey and asks its readers to cut the questionnaire from the magazine, fill it and send it via mail. It is a type of

- i. Purposive sampling
- ii. Snowball sampling
- iii. Sequential sampling
- iv. Convenience sampling

Q5. The height distribution of a few students in a school is an example of

- i. Statistic
- ii. Population
- iii. Parameter
- iv. Element

Q6. A researcher wants to conduct a survey of the drug users. Which type of sampling technique will be most appropriate here?

- i. Sequential sampling

- ii. Snowball sampling
- iii. Quota sampling
- iv. Convenience sampling

Q7. When there is a need to apply different data collection methods to different parts of the population, the best sampling method would be

- i. Double sampling
- ii. Cluster sampling
- iii. Stratified random sampling
- iv. Systematic random sampling

Q8. The sampling technique in which every element of the population has an equal, non-zero probability of being selected in a sample, is called

- i. Probability sampling
- ii. Convenience sampling
- iii. Purposive sampling
- iv. Quota sampling

Q9. Target population is also called

- i. Population
- ii. Survey population
- iii. Population element
- iv. Population frame

Q10. Which one of them is the method for probing the respondent?

- i. Repeat the question
- ii. Give an expectant pause
- iii. Repeat the respondent's reply
- iv. All of the given options

1. Which one of the following sets is the measure of central tendency?

Mean, standard deviation, mode
 Mean, median, standard deviation
Arithmetic mean, median, mode

Standard deviation, internal validity, mode

In lab experiment the effect of _____ Variables is controlled to evaluate the causal relationship.

Extraneous
 Moderate
 Intervening
All of the above

Internal validity refers to _____.

Researcher's degree of confidence.
 Generalizability
 Operationalization
 All of the above

Which of the following is the weakest experimental design?

One group pretest-posttest design

Quasi - experimental design Two group posttest only design Ex post facto design

How many times the students appear in the research class is the example of

Intensity
Space
Frequency
Direction

Disadvantage of content analysis is_____.

Researcher can increase the sample size

Provides access on the subjects to which researcher does have physical access. **Sometime documents provide incomplete account to the researcher** Spontaneous feelings can be recorded when they occurred

7. Which of the following statement is incorrect with respect to “An experimental design is a set of procedures specifying:”

How the test units (subjects) are to be divided into homogenous sub samples.

What independent variables or treatments are to be measured?

What dependent variables are to be measured? How the extraneous variables are to be controlled?

8. Time consumed in mall intercept interview is _____.

High
Moderate
Low
Nil

9. “Teacher should create a friendly environment in the classroom” this is the type of ____.

Leading question **Loaded question** Double Barreled
Burdenome question

10. Departmental stores selected to test a new merchandising display system is the example of

_____.
Quota sampling Convenience sampling **Judgmental sampling**
Purposive sampling

1. Which is not a source of existing statistics?
 - A. Government
 - B. International agencies
 - C. **Personal interviews**
 - D. Private sources
2. Followings are the advantages of secondary data except
 - A. Non-Reactivity
 - B. **Selective survival**
 - C. Low cost
 - D. Spontaneity
3. Following are the disadvantages of secondary data except
 - A. **Longitudinal analysis**
 - B. Sampling bias
 - C. Coding difficult
 - D. Incompleteness
4. The coding of the secondary data may be difficult because of
 - A. Differences in content or subject matter
 - B. Lack of standardization
 - C. Differences in length and format
 - D. **All of these**
5. The researcher must be concerned about the following problems while using secondary data in research.
 - A. Validity
 - B. Reliability
 - C. **Both of these**
 - D. None of these

Discrete variable is also called.....

Categorical variable
Discontinuous variable
Both A & B
None of the above

“Officers in my organization have higher than average level of commitment”Such a hypothesis is an example of.....

Descripti ve Hypothesis
 Directional Hypothesis
 Relational Hypothesis All
 of the above

‘Science’ refers to.....

A system for producing knowledge
The knowledge produced by a system
Both A & B

None of the above

Which one of the following is not a characteristic of scientific method?

Deterministic
Rationalism
Empirical
Abstraction

The theoretical framework discusses the interrelationships among the.....

Variables
Hypothesis
Concept Theory

.....research is based on naturalism.
Field research

Descriptive research
Basic research
Applied research

Personal interviews conducted in shopping malls are known as.....

Mall interviews
Mall intercept interviews
Brief interviews
None of the given options

..... is used to obtain the freest opinion of the respondent, by asking general question before a specific question.

Research technique
Qualitative technique
Funnel technique
Quantitative technique

In, the interviewer and members jointly control the pace and direction of the interview.

Field interview

Telephonic interview

Both A and B

None of the given options

Randomization of test units is a part of

Pretest Posttest

Matching

Experiment

The independent variable is;

☐ The variable manipulated in order to observe its effects

☐ The variable that is measured

☐ free spirited variable

☐ A confounding variable

If a researcher was studying the use of various instructional approaches to the "multiple intelligences" of his students, he is likely to be conducting which type of research?

☐ Basic

☐ Applied

☐ Evaluation

☐ Grounded theory

Research may differ along a series of dimensions. Which of the following may be applied to this statement?

☐ The type of data collected.

☐ The data-collection technique.

☐ The level of manipulation used to elicit data collection.

Which of the following is least likely to occur as an ethical problem with e -research?

People are not likely to be able to stop once they have begun participating.

☐ Informed consent cannot be completely monitored.

☐ Debriefing could be avoided.

☐ Privacy could be invaded.

☐ All of the given options

Which of the following is the least obtrusive and most accurate method for recording data in an interview?

☐ Note taking

☐ Videotaping

- ☐ Audio taping
- ☐ Writing notes after the interview

The most critical areas of an article to read is;

Results section

- ☐ Introduction
- ☐ Abstract
- ☐ Limitations

Quantitative social researchers rarely claim to have established causality because:

- ☐ They are more concerned with publishing the results of their reliability tests. ☐ They do not believe that this is an appropriate goal to be striving for.
- ☐ They keep forgetting which of the variables they have manipulated.
- ☐ They tend to use cross-sectional designs, which produce only correlations.

Which of the following is most beneficial to read in an article?

- ☐ Methods
- ☐ Introduction
- ☐ Figures
- ☐ References

If a nominal scale is used, it is permissible to calculate which of the following statistics?

- ☐ Mean
- ☐ Range
- ☐ Percentile
- ☐ Mode

One of the preoccupations of quantitative researchers is with generalization, which is a sign of:

- ☐ External validity
- ☐ Internal reliability
- ☐ External reliability
- ☐ Internal validity

Which of the following is the least appropriate research problem?

Does studying Latin improve the standardized vocabulary test scores of seventh grade students?

- ☐ Does a drilling fifth grade student with multiplication facts improve their standardized test scores?

What is the relationship between students' math attitudes and math

achievement? ☐ Should students have access to controversial novels in school? The following journal article would be an example of _____ research; "The benefits of florescent lighting on production in a factory setting"

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- ☐ Applied
- ☐ Interview
- ☐ Basic
- ☐ Stupid

The purpose of a literature review is to:

- ☐ Help you find out what is already known about this area.
- ☐ Identify any inconsistencies or gaps in the literature.
- ☐ Demonstrate an awareness of the theoretical context in which the current study can be located.
- ☐ Find what is already known, identify gaps demonstrate awareness.

Quantitative research has been criticized because:

- ☐ The measurement process suggests a spurious and artificial sense of accuracy.
- ☐ The reliance on instruments and procedures makes it high in ecological validity. ☐ It underestimates the similarities between objects in the natural and social worlds. ☐ I

has no validity.

The _____ is only useful if the concepts, ideas, questions, etc. to be investigated are both testable and falsifiable.

Independent Variable

☐ Dependent Variable Experimental

☐ Method

☐ Scientific Method

A _____ scale only assigns numbers to objects to classify the objects according to the characteristic of interest.

☐ Ratio

☐ Nominal

☐ Interval

☐ Dichotomous

12. Which of the following true about the relationship between theory building and data collection?

- ☐ When studies come out as expected, inductive support for the theory is gained.
- ☐ If an experiment fails, discarding the experiment is an example of affirming the consequent.
- ☐ When a hypothesis is not supported, virtually nothing has been learned about the theory.
- ☐ A good theory will be inclusive enough to explain every possible research outcome.
- ☐ Laiba has decided to use the test at the end of the textbook to measure the achievement levels of the students in her study. Which of the following BEST describes the chapter test?

Definition

Construct

☐ Variable

☐ Operationalized variable

☐

What is the reason for consulting handbooks, yearbooks, encyclopedias, or reviews in the initial stages of identifying a research topic?

- ☐ They are readily available.
- ☐ They provide an overview of the issues related to a topic.
- ☐ They are primary sources.
- ☐ They avoid reporting statistical data so one can interpret the results more easily.

What type of process research is? A process to discover the truth

- ☐ A process to invent the machines
- ☐ A process to make new medicines
- ☐ A process to create theories

Which one of the following is not an assumption of science?

- ☐ There are reoccurring patterns in the world. Events happen because of preceding causes.
- ☐ We can discover solutions to problems of interest.
- ☐ Theoretical explanations must agree with common sense.

A good qualitative problem statement:

- ☐ Defines the independent and dependent variables
- ☐ Conveys a sense of emerging design
- ☐ Specifies a research hypothesis to be tested
- ☐ Specifies the relationship between variables that the researcher expects to find.

Why do you need to review the existing literature?

- ☐ , To give your dissertation a proper academic appearance, with lots of references
- ☐ , Because without it, you could never reach the required word -count
- ☐ , To find out what is already known about your area of interest
- ☐ , To help in your general studying

If a researcher was studying the use of various instructional approaches to the "multiple intelligences" of his students, he is likely to be conducting which type of research?

- ☐ Basic
- ☐ Applied

☐ **Evaluation**

☐ **Grounded theory**

Ms. Roshi has been coordinating the Annual Festival at her school for the last several years. She wants to be sure the students and parents enjoy the festival again this year. On which source is she LEAST likely to rely when making decisions about what to do?

Select correct option:

☐ **Tradition**

☐ **Personal experience**

☐ **Research**

☐ **Expert opinion**

A review that only demonstrates familiarity with an area is rarely published but it often is part of an educational program is which type of review?

☐ Integrative reviews

☐ Theoretical reviews

☒ **Self-study reviews**

☐ Historical reviews

A researcher developed a measure of shyness and is now asking whether this measure does in fact measure a person's true state of shyness. This is a question of:

☐ **Reactivity**

☐ **Construct validity**

☐ **Reliability Content**

☐ **validity**

When you are confident that the experimental manipulation produced the changes you measured in the dependent variable, your study probably has good _____ validity.

☐ **Construct**

☐ **Internal**

☐ **External**

☐ **Causal**

☐

If a researcher is studying the effect of using laptops in his classroom to ascertain their

merit and worth, he is likely conducting which type of research?

Select correct option

- ☐ **Basic**
- ☐ **Applied**
- ☐ **Evaluation**
- ☐ **Experimental**

When doing research involving deception with human subjects, researchers have an obligation to do which of the following?

- ☐ Tell subjects the truth about the study's purpose and methods after the study is completed
- ☐ Prevent mental and physical harm to subjects
- ☐ Let subjects withdraw from the study at any time if they don't want to keep participating
- ☐ All of the given options

literature review requires;

- ☐
- ☐ planning clear
- ☐ writing
- ☐ good writing
- ☐ **All of the given option**

A measure has high internal consistency reliability when:

- ☐ **Multiple observers make the same ratings using the measure.**
- ☐ **Participants score at the high end of the scale every time they complete the measure.**
- ☐ **Multiple observers obtain the same score every time they use the measure. Each of the**
- ☐ **items correlates with other items on the measure.**

Which of the following is not a function of clearly identified research questions?

- ☐ **They keep you focused throughout the data collection period.**
- ☐ **They make the scope of your research as wide as possible.**

- ☐ They are linked together to help you construct a coherent argument.

Procedures determining what two issues are rarely used in quantitative research?

- ☐ Objectivity and subjectivity
- ☐ Reliability and validity
- ☐ Accessibility and replicability
- ☐ Quality and quantity
- ☐

- ☐ When studies come out as expected, inductive support for the theory is gained.
- ☐ If an experiment fails, discarding the experiment is an example of affirming the consequent.
- ☐ When a hypothesis is not supported, virtually nothing has been learned about the theory.
- ☐ A good theory will be inclusive enough to explain every possible research outcome.
- ☐

Temperature is measured from which scale?

Nominal

Ordinal Interval

All the given options.

Which of the following is a threat to internal validity of an experimental design? Maturation

Interaction of setting and treatment

Interaction effects of pre-testing reactive effects of experimental design

Which of the following statements is correct about validity and reliability? When internal validity is high, external validity is low

When internal validity is high, there is no change in external validity When internal validity is high, external validity is also high

All of the given option

Which of the following effect in internal validity occurs when test units with extreme scores move closer to the average score during the course of the experiment?

Statistical Regression

Selection bias

Maturation

Instrumentation

Interval scale measures which of the following?

The distance between each participant's individual score

An individual's score from zero upwards
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more or less of some underlying assumptions
Absolute Zero

Which of the following best describes an assessment that examines students ability to conduct a chemistry experiment in the lab?

Aptitude

interest inventory Performance

Assessment **Standardized test**

Which of the following are legitimate frameworks for setting out a literature review?

(1) Constructing intertextual coherence

(2) Deconstruction of textual coherence

(3) Problematizing the situation

(4) Resolving the discovered problems?

1 & 2

2 & 3

1 & 3

2 & 4

If researcher was studying the use of various instructional approaches to the “multiple intelligence” of his students, he is likely to be conducting which of the following type of research?

Basic

Evaluation

Applied Ground

theory

Which of the following is most beneficial to read in an article? Methods

Introduction

Figures

reference

One of the preoccupations of quantitative researchers is with generalization, which is a sign of

External validity

Internal Reliability

External Reliability

Internal validity

Which of the following is likely to reduce the validity of a test? Unclear test directions

Ambiguous test items

Unclear, ambiguous and untaught items

Untaught items

1. Question # 1 of 10 (Start time: 08:23:14 PM)

Total Marks: 1

If $Y=bX$, then variance of Y is

Select correct option:

$b^2 \text{ var}(x)$

$\text{var}(x)$

$b \text{ var}(x)$

$b \text{ square root var}(x)$

2. Question # 2 of 10 (Start time: 08:24:38 PM)

Total Marks: 1

If $f(x)$ is a continuous probability function, then $P(X = 2)$ is:

Select correct option:1

0

1/2

2

3. Question # 3 of 10 (Start time: 08:25:52 PM)

Total Marks: 1

In regression line $Y=a+bX$, Y is called:

Select correct option:

Dependent variable

Independent variable

Explanatory variable

Regressor

4. Question # 4 of 10 (Start time: 08:26:51 PM)

Total Marks: 1

If A and B are mutually exclusive events with $P(A)=0.25$ and $P(B)=0.50$, Then $P(A \text{ or } B)$ =.....

Select correct option:

0.25

0.75

0.50

1

5. Question # 5 of 10 (Start time: 08:28:06 PM)

Total Marks: 1

Symbolically, a conditional probability is:

Select correct option:

$P(AB)$

$P(A/B)$

$P(A)$

$P(A \cup B)$

6. Question # 6 of 10 (Start time: 08:28:42 PM)

Total Marks: 1

In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card isSelect correct option:

1/52

4/52

13/52

52/52

7. Question # 7 of 10 (Start time: 08:30:13 PM)

Total Marks: 1

Probability of a sure event is

Select correct option:8

1

0

0.5

8. Question # 8 of 10 (Start time: 08:31:42 PM)

Total Marks: 1

If $Y=3X+5$, then S.D of Y is equal to

Select correct option:9

s.d(x)

3 s.d(x)

s.d(x)+5

3s.d(x)+5

9. Question # 9 of 10 (Start time: 08:33:16 PM)

Total Marks: 1

The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is Select correct option:

4/52

2/52

13/52

26/52

10. Question # 10 of 10 (Start time: 08:34:40 PM)

Total Marks: 1

If $P(B|A) = 0.25$ and $P(A \text{ and } B) = 0.20$, then $P(A)$ is

Select correct option:

0.05

0.80

0.95

0.75

11. Question # 1 of 10 (Start time: 08:57:45 PM)

Total Marks: 1

When a coin is tossed 3 times, the probability of getting 3 tails is

Select correct option:

1/8

3/8

3/6

2/8

12. Question # 2 of 10 (Start time: 08:59:14 PM)

Total Marks: 1

In how many ways can a team of 11 players be chosen from a total of 16 players? Select correct option:

4368(not confirmed)

2426

5400

2680

13. Question # 3 of 10 (Start time: 09:00:38 PM)

Total Marks: 1

The standard deviation of c (constant) is

Select correct option:c

c square

0

does not exist

14. Question # 4 of 10 (Start time: 09:01:46 PM)

Total Marks: 1

If P (E) is the probability that an event will occur, which of the following must be false:

Select correct option:

$P(E) = -1$ $P(E) = 1$

$P(E) = 1/2$

$P(E) = 1/3$

15. Question # 5 of 10 (Start time: 09:02:48 PM)

Total Marks: 1

Let E and F be events associated with the same experiment. Suppose the E and F are independent and that $P(E) = 1/4$ and $P(F) = 1/2$ Then $P(E \cup F)$ is:

Select correct option:

$1/8$

$3/4$

$7/8$

$5/8$

16. Question # 6 of 10 (Start time: 09:04:09 PM)

Total Marks: 1

A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

Select correct option:

0.25

0.5

1

0

17. Question # 7 of 10 (Start time: 09:05:31 PM)

Total Marks: 1

If $Y = bX$, then variance of Y is

Select correct option:

$b^2 \text{var}(x)$

$b \text{var}(x)$

$b \text{ square root var}(x)$

18. Question # 9 of 10 (Start time: 09:07:48 PM)

Total Marks: 1

The classical definition of probability assumes:

Select correct option: Exhaustive

events Mutually exclusive

events **Equally likely events**

Independent events

19. Question # 10 of 10 (Start time: 09:08:50 PM)

Total Marks: 1

In scatter diagram, the variable plotted along Y-axis is:

Select correct option:

Independent variable

Dependent variable

Continuous variable

Discrete variable

20. Which of the following measures of dispersion are based on deviations from the mean?

Select correct option:

Variance

Standard deviation

Mean deviation

All of the these

21. What does it mean when a data set has a standard deviation equal to zero? Select correct option:

All values of the data appear with the same frequency.

The mean of the data is also zero. All of the data have the same value. There are no data to begin with.

22. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____. Select correct option:

Probability distribution

The expected return
The standard deviation
Coefficient of variation

23. Which of the following can never be probability of an event? Select correct option:

0
1
0.5
-0.5

24. The standard deviation of -1, -1, -1, -1 will be Select correct option:

1
-1
0
Does not exist

25. Which formula represents the probability of the complement of event A? Select correct option:

$1 + P(A)$
 $1 - P(A)$
 $P(A)$
 $P(A) - 1$

26. The Special Rule of Addition is used to combine: Select correct option:

Independent Events
Mutually Exclusive Events
Events that total more than 1.00
Events based on subjective probabilities

27. _____ set which is the sub-set of every set is
Select correct option:

Empty Set

Power Set

Universal Set

Super Set

28. $E(4X + 5) =$ _____

Select correct option:

12 E (X)

4 E (X) + 5

16 E (X) + 5

16 E (X)

29. When two dice are rolled the number of possible sample points is :Select correct option:

6

12

24

36

30. Question # 1 of 10 (Start time: 09:43:04 PM) Total Marks: 1If two events A and B are not mutually exclusive then
Select correct option:

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$P(A \text{ or } B) = P(A) + P(B)$

$P(A \text{ or } B) = P(A) \times P(B)$

$P(A \text{ or } B) = P(A) + P(B)$

31.31.

Question # 2 of 10 (Start time: 09:43:59 PM) Total Marks: 1

Evaluate (10-4)!

Select correct option:

1000

720

480

32

32. Question # 3 of 10 (Start time: 09:45:01 PM) Total Marks: 1When E is an impossible event, then P(E) is:
Select correct option:

0

1

2

0.5

33. Question # 4 of 10 (Start time: 09:46:20 PM) Total Marks: 1 When we
toss a coin , we get only:
One & Join

Select correct option:

1 outcome

2 outcome

3 outcome

4 outcome

34.34.

Question # 5 of 10 (Start time: 09:47:15 PM) Total Marks: 1For exhaustive events, the $P(A \cup B \cup C)$ is equal to:

Select correct option:

$P(A)$

$P(S)$

$P(A) * P(B) * P(C)$

$P(B)$

35. Question # 6 of 10 (Start time: 09:48:21 PM) Total Marks: 1

A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

Select correct option:

0.25

0.5

1

0

36. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Select correct option:

Probability distribution

The expected return The

standard deviation

Coefficient of variation

37.37.

Question # 9 of 10 (Start time: 09:50:35 PM) Total Marks: 1If we roll a die then probability of getting a '6' will be

Select correct option:

2/6

1/6

4/6

1

38. Question # 10 of 10 (Start time: 09:51:36 PM) Total Marks: 1If $P(A)$

$= 0.45$, $P(B) = 0.35$, and $P(A \text{ and } B) = 0.25$, then $P(A | B)$ is:
Select correct option:

- 1.4
- 1.8

0.714

0.556

39. Question # 8 of 10 (Start time: 09:49:53 PM) Total Marks: 1 Which of the following is not a measure of central tendency?
Select correct option:

Percentile

Quartile

Standard deviation

Mode

40. Question # 1 of 10 (Start time: 09:56:49 PM) Total Marks: 1 Random experiment can be repeated any no. of times under the..... conditions. Select correct option:

Different

Similar

41. Question # 2 of 10 (Start time: 09:58:09 PM) Total Marks: 1 What is the probability of sure event?
Select correct option:

0

1

0.5

2

42. Question # 3 of 10 (Start time: 09:58:41 PM) Total Marks: 1 The simultaneous occurrence of two events is called:
Select correct option:

Joint probability

Subjective probability

Prior probability

Conditional probability

43. Question # 4 of 10 (Start time: 09:59:47 PM) Total Marks: 1 In regression analysis, the variable that is being predicted is the
Select correct option:

Dependent variable

Independent variable

Intervening variable

None of these

1. Question # 1 of 10 (Start time: 08:23:14 PM)
Y=bX, then variance of Y is

Total Marks: 1

Select correct option:

- b*2 var(x)
- var(x)
- b var(x)
- b square root var(x)**

2. Question # 2 of 10 (Start time: 08:24:38 PM)
f(x) is a continuous probability function, then P(X = 2) is:

Total Marks: 1

Select correct option:1

- 0
- 1/2
- 2**

3. Question # 3 of 10 (Start time: 08:25:52 PM)
line Y=a+bX, Y is called:

Total Marks: 1

Select correct option:

- | | |
|------------------|-----------------|
| Dependent | variable |
| Independent | variable |
| Explanatory | variable |
| Regressor | |

4. Question # 4 of 10 (Start time: 08:26:51 PM)

Total Marks: 1

If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P(A or B) =.....

Select correct option:

- 0.25
- 0.75**
- 0.50
- 1

5. Question # 5 of 10 (Start time: 08:28:06 PM) Total Marks: 1
Symbolically, a conditional probability is:

Select correct option:

- P(AB)
- P(A/B)**
- P(A)
- P(AUB)

6. Question # 6 of 10 (Start time: 08:28:42 PM)

Total Marks: 1

In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is

Select correct option:

- 1/52
- 4/52**

13/ 52

52/52

7. Question # 7 of 10 (Start time: 08:30:13 PM)
Probability of a sure event is
Select correct option:8

Total Marks: 1

1

0

0.5

8. Question # 8 of 10 (Start time: 08:31:42 PM)
 $Y=3X+5$, then S.D of Y is equal to
Select correct option:9

Total Marks: 1If

s.d(x)

3 s.d(x)

s.d(x)+5

3s.d(x)+5

9. Question # 9 of 10 (Start time: 08:33:16 PM)

Total Marks: 1

The probability of drawing a red queen card from well -shuffled pack of 52 playingcards is
Select correct option:

4/52

2/ 52

13/52

26/52

10. Question # 10 of 10 (Start time: 08: 34:40 PM)
 $P(B|A) = 0.25$ and $P(A \text{ and } B) = 0.20$, then $P(A)$ is
Select correct option:

Total Marks: 1If

0.05

0.80

0.95

0.75

11. Question # 1 of 10 (Start time: 08:57:45 PM)
When a coin is tossed 3 times, the probability of getting 3 tails is
Select correct option:

Total Marks: 1

1/ 8

3/8

3/6

2/8

12. Question # 2 of 10 (Start time: 08:59:14 PM)
In how many ways can a team of 11 players be chosen from a total of 16 players?
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Total Marks: 1

Select correct option:

4368 (not confirmed)

2426

5400

2680

13. Question # 3 of 10 (Start time: 09:00:38 PM)

Total Marks: 1

The standard deviation of c (constant) is

Select correct option:

c square

0

does not exist

14. Question # 4 of 10 (Start time: 09:01:46 PM)

Total Marks: 1

If P (E) is the probability that an event will occur, which of the following must be false:

Select correct option:

$P(E) = -1$

$P(E) = 1$

$P(E) = 1/2$

$P(E) = 1/3$

15. Question # 5 of 10 (Start time: 09:02:48 PM)

Total Marks: 1

Let E and F be events associated with the same experiment. Suppose the E and F are independent and that $P(E) = 1/4$ and $P(F) = 1/2$ Then $P(E \cup F)$ is:

Select correct option:

$3/4$

$7/8$

$5/8$

16. Question # 6 of 10 (Start time: 09:04:09 PM)

Total Marks: 1

A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

Select correct option:

0.25

0.5

1

0

17. Question # 7 of 10 (Start time: 09:05:31 PM)

Total Marks: 1

If $Y = bX$, then variance of Y is

Select correct option:

$b^2 \text{var}(x)$

$\text{var}(x)$

b var(x)

b square root var(x)

18. Question # 9 of 10 (Start time: 09:07:48 PM)
The classical definition of probability assumes:

Total Marks: 1

Select correct option: Exhaustive
events Mutually exclusive
events **Equally likely events**

Independent events

19. Question # 10 of 10 (Start time: 09:08:50 PM)
scatter diagram, the variable plotted along Y -axis is:

Total Marks: 1

Select correct option:
Independent variable
Dependent variable
Continuous variable
Discrete variable

20. Which of the following measures of dispersion are based on deviations from the mean?

Select correct option:

Variance

Standard deviation
Mean deviation

All of the these

21. What does it mean when a data set has a standard deviation equal to zero? Select correct option:

All values of the data appear with the same frequency.

The mean of the data is also zero. All of
the data have the same value. There are no
data to begin with.

22. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Select correct option:

Probability distribution

The expected return
The standard deviation
Coefficient of variation

23. Which of the following can never be probability of an event? Select correct option:

- 0
- 1
- 0.5
- 0.5**

24. The standard deviation of -1, -1, -1, -1 will be Select correct option:

- 1
- 1
- 0**
- Does not exist

25. Which formula represents the probability of the complement of event A? Select correct option:

- $1 + P(A)$
- $1 - P(A)$**
- $P(A)$
- $P(A) - 1$



26. The Special Rule of Addition is used to combine: Select correct option:

- Independent Events
- Mutually Exclusive Events**
- Events that total more than 1.00
- Events based on subjective probabilities

27. set which is the sub-set of every set is Select correct option:

- Empty Set**
- Power Set
- Universal Set
- Super Set

28. $E(4X + 5) = \underline{\hspace{2cm}}$

Select correct option:

- 12 E (X)
- $4 E (X) + 5$**
- $16 E (X) + 5$
- $16 E (X)$

29. When two dice are rolled the number of possible sample points is :Select correct option:

- 6
- 12
- 24
- 36**

30. Question # 1 of 10 (Start time: 09:43:04 PM)
two events A and B are not mutually exclusive then
Select correct option:

Total Marks: 1

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$P(A \text{ or } B) = P(A) + P(B)$

$P(A \text{ or } B) = P(A) \times P(B)$

$P(A \text{ or } B) = P(A) + P(B)$

31. Question # 2 of 10 (Start time: 09:43:59 PM)
Evaluate $(10 - 4)!$ Select
correct option:

Total Marks: 1

- 1000
- 720**
- 480
- 32

32. Question # 3 of 10 (Start time: 09:45:01 PM) Total Marks: 1
When E is an impossible event, then $P(E)$ is:
Select correct option:

- 0**
- 1
- 2
- 0.5

33. Question # 4 of 10 (Start time: 09:46:20 PM)
When we toss a coin , we get only:
Select correct option:

Total Marks: 1

- 1 outcome**
- 2 outcome
- 3 outcome
- 4 outcome

34. Question # 5 of 10 (Start time: 09:47:15 PM) Total Marks: 1
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For exhaustive events, the $P(A \cup B \cup C)$ is equal to:

Select correct option:

$P(A)$

$P(S)$

$P(A) * P(B) *$

$P(C)P(B)$

35. Question # 6 of 10 (Start time: 09:48: 21 PM) Total Marks: 1

A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:

Select correct option:

0.25

0.5

1

0

36. A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.

Select correct option:

Probability distribution

The expected return The

standard deviation

Coefficient of variation

37.

Question # 9 of 10 (Start time: 09:50:35 PM) Total Marks: 1 If we roll a die then probability of getting a '6' will be

Select correct option:

2/6

1/6

4/6

1

38. Question # 10 of 10 (Start time: 09:51:36 PM) Total Marks: 1 If

$P(A) = 0.45$, $P(B) = 0.35$, and $P(A \text{ and } B) = 0.25$, then $P(A | B)$ is: Select correct option:

1.4

1.8

0.714

0.556

39. Question # 8 of 10 (Start time: 09:49:53 PM) Total Marks: 1
Which of the following is not a measure of central tendency?

Select correct option:

Percentile

Quartile

Standard deviation

Mode

40. Question # 1 of 10 (Start time: 09:56:49 PM) Total Marks: 1
Random experiment can be repeated any no. of times under the..... conditions. Select correct option:

Different

Similar

41. Question # 2 of 10 (Start time: 09:58:09 PM) Total Marks: 1
What is the probability of sure event?

Select correct option:

0

1

0.5

2

42. Question # 3 of 10 (Start time: 09:58:41 PM) Total Marks: 1
The simultaneous occurrence of two events is called:

Select correct option:

Joint probability Subjective probability
Prior probability
Conditional probability

43. Question # 4 of 10 (Start time: 09:59:47 PM) Total Marks: 1
In regression analysis, the variable that is being predicted is the Select correct option:

Dependent variable

Independent variable
Covariate

Intervening variable

None of these

Thanks to Armaan Makhani for His Sharing

1. A quantity obtained by applying certain rule or formula is known as

Select correct option:

Estimate

Estimator

2. Criteria to check a point estimator to be good involves

Select correct option:

Consistency

Unbiasedness

Efficiency

Above all pg 258

3. The F -distribution always ranges from:

Select correct option:

0 to 1

0 to -8

-8 to +8

0 to +8

4. $1 - \alpha$ is the probability of

Select correct option:

Type 1 error

Rejection region

Acceptance region

Type 2 error

5. Parameter is a quantity.

Select correct option:

Constant

Variable

6. To find the estimate of a parameter methods are used.

Select correct option:

Two

Three

Four

Many

7. A failing student is passed by an examiner. It is an example of:

Select correct option:

Type I error **Type**

II error Correct

decision

No information regarding student exams

8. For two mutually exclusive events A and B, $P(A) = 0.2$ and $P(B) = 0.4$, then

$P(A \cup B)$ is:

Select correct option:

0.8

0.2

0.6

0.5

9. An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn without replacement. It is the example of:

Select correct option:

Binomial distribution

Hypergeometric distribution

Poisson distribution

Exponential distribution

10. A standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:

Select correct option:

1/5

1/13

5/52

1/52

11. If $P(A \cap B) = 0.12$, $P(A) = 0.3$, find $P(B)$ where 'A' and 'B' are independent:

Select correct option:

0.1

0.2

0.3

0.4

12. The mean deviation of the normal distribution is approximately:

Select correct option:

7/8 of the S.D

4/5 of the S.D

3/4 of the S.D

1/2 of the S.D

13. We use the Poisson approximation to the binomial when:

Select correct option:

p is 0.01 or less & n is 10 or more

p is 0.05 or less & n is 20 or more

p is 0.04 or less & n is 15 or more

p is 0.02 or less & n is 10 or more

14. The conditional probability $P(A \setminus B)$ is:

Select correct option:

$P(A \cap B)/P(B)$

$P(A \cap B)/P(A)$

$P(A \cup B)/P(B)$

$P(A \cup B)/P(A)$

15. We use the General Rule of Multiplication to combine:

Select correct option:

Events those are not independent

Mutually exclusive events

Events that total more than 1.00

Events based on subjective probabilities

16. Which statement is NOT CORRECT?

Select correct option:

The sample standard deviation measures variability of our sample values

A larger sample will give answers that vary less from the true value than smaller samples
The sampling distribution describes how our estimate (answer) will vary if a new sample is taken

A large sample size always gives unbiased estimators regardless of how the sample is chosen

17. Probability of an impossible event is always:

Select correct option:

Less than one

Greater than one

Between one and zero

18. The number of parameters in uniform distribution is (are):

Select correct option:

1

2 pg 224

3

4

19. The probability can never be:

Select correct option:

1

1/2

1

-1/2

20. The conditional probability $P(A \setminus B)$ is:

Select correct option:

$P(A \cap B)/P(B)$

$P(A \cap B)/P(A)$

$P(A \cup B)/P(B)$

$P(A \cup B)/P(A)$

21. A random sample of $n=25$ values gives sample mean 83. Can this sample be regarded as drawn

from a normal population with $\mu = 80$ and $s = 7$? In this question the alternative hypothesis will be:

Select correct option:

$H_1: \mu = 80$

$H_1: \mu ?$

$H_1: \mu > 80$

80

$H_1: \mu < 80$ pg 278

22. If $f(x)$ is a continuous probability function, then $P(X = 2)$ is:

Select correct option:

1

0

1/2

2

23. The binomial distribution is negatively skewed when:

Select correct option:

$p > q$ pg 214

$p < q$
 $p = q$
 $p = q = 1/2$

24. If we roll three fair dices then the total number of outcomes is:

Select correct option:

6
36
216
1296

25.

we draw the sample with replacement, the probability distribution used is:

When
to be

Select correct option:

Binomial
Hypergeometric
Binomial & hypergeometric pg 219
Poisson

26. The moment ratios of normal distribution come out to be:

Select correct option:

0 and 1
0 and 2
0 and 3 pg 226
0 and 4

27. The probability of an even t is always:

Select correct option:

greater than 0
less than 1
between 0 and 1
greater than 1

28. Symbolically, a conditional probability is:

Select correct option:

$P(AB)$
 $P(A/B)$
 $P(A)$
 $P(A \cup B)$

29.

600 students are normally distributed

Suppose the test scores of
ed with a mean

of 76 and

standard deviation of 8. The number of students scoring between 70 and 82 is:

Select correct option:

272
164
260
328

30. If $P(A) = 0.3$ and $P(B) = 0.5$, find $P(A/B)$ where 'A' and 'B' are independent:

Select correct option:

0.3
0.5
0.8
0.15

31. An urn contains 4 red balls and 6 green balls. A sample of 4 balls is selected from the urn

without replacement. It is the example of:

Select correct option:

Binomial distribution

Hypergeometric distribution

Poisson distribution

Exponential distribution

32. If the second moment ratio is less than 3 the distribution will be:

Select correct option:

Mesokurtic

Leptokurtic

Platykurtic pg 226

None of these

33. For the independent events A and B if $P(A) = 0.25$, $P(B) = 0.40$ then $P(A \text{ and } B)$ =

Select correct option:

0.65

0.1

0.50

0.15

34. A set of possible values that a random variable can assume and their associated probabilities of

occurrence are referred to as _____.

Select correct option:

Probability distribution

The expected return The

standard deviation

Coefficient of variation

35. A random variable X has a probability distribution as follows: X | 0 1 2 3 P(X) |
2k 3k 13k 2k

What is the possible value of k:

Select correct option:

0.01

0.03

0.05

0.07

36. The probability of drawing any one spade card is:

Select correct option:

1/52

4/52

13/52

52/52

37. The function abbreviated to d.f. is also called the.....

Select correct option:

Probability density function **Probability**

distribution function pg 172

Commutative distribution function

Discrete function

38. Binomial distribution is skewed to the right if:

Select correct option:

$p=q$

$p < q$

$p > q$

$p=n$

39. A discrete probability function $f(x)$ is always:

Select correct option:

Zero

One pg 172

Negative Non-negative

40. For a binomial distribution, $n = 10$ & $q = 0.6$, the mean of the distribution is:

Select correct option:

0.6

6.0

10

4

41. In the FA examination, 24 candidates offered Statistics. If the probability of passing the subject

be $1/3$, what will be the mean of the distribution?

Select correct option:

7

8

6

5

42. The probability that a certain machine will produce a defective item is $1/4$. If a random sample

of 6 items is taken from the output of this machine, what is the probability that there will be 5 or more defectives in the sample?

Select correct option:

$3/4096$

$18/4096$

$19/4096$

$4/4096$

43. Probability of type II error is

Select correct option:

a

B pg 276

$1-a$

$1-B$

44. If the values of variables are increasing or decreasing in the same direction then such kind of

correlation is referred as

Select correct option:

Zero Correlation

Perfect Correlation

Positive Correlation

Negative Correlation

45. The moving averages of the Prices 55,60,65,70 are

Select correct option:

70, 75

60, 65

65,65

70,60

46. The best measure of variation is

Select correct option:

Range

Quartile deviation

Variance

Coefficient of variance

47. Ms. Christian calculated a correlation coefficient of .75. Which of the following reflects the

best interpretation of this?

Select correct option :

Weak negative.

Strong negative.

Weak positive.

Strong positive.

48.use the division of a circle into different sectors. Select correct option:

Line graph

Sector graphs

Frequency Polygon

Conversion Graphs

49. The measurement of measure of degree of to which any two variables vary together is called

Select correct option:

Regression Coefficient

Correlation

Both (a) and (b)

None of these

50. Analysis of Variance (ANOVA) is a test for equality of:

Select correct option:

variances means

proportions

only two parameters

51. For some data you are given Maximum value = 96 , Minimum Value = 23 ,

Range = 73,

number of classes selected between 5 and 15 Then class width will be Select correct option:

15

8

5

All options 1,2 ,3 are possible

52. If strength of the association between X and Y is very weak, then $r = ?$ Select correct option:

$r = -1$

$r = 0$

$r = 1$

$r = 2$

53. The moving averages of the Prices 90, 70, 30, 110 are Select correct option:

63.33, 70

73.33, 80

45.45, 68

65.50, 75

54. With increase in sample size, distribution tends to be a.....

Select correct option:

Meso kurtic

Normal

Bell shaped

Above all

55. In the central tendency Mean, Median and Mode

Select correct option:

Mean is better than Median

Median is better than Mode

Mean is better than Mode

All of these are true

56. The degree to which numerical data tend to spread about an average is called

Select correct option:

The dispersion

Standard deviation

Correlation

None of these

57.graphs are similar to bar graphs.

Select correct option:

column

line conversion

sector

58. A pattern of variation of a time series that repeats every year is called:

Select correct option:

Cyclical

Seasonal

Trend

Secular

59. In the central tendency Mean, Median and Mode

Select correct option:

Mean is better than Median

Median is better than Mode

Mean is better than Mode

All of these are true

60. The degree to which numerical data tend to spread about an average is called

Standard deviation
Correlation
None of these

61.graphs are similar to bar graphs.

Select correct option:

column

line conversion
sector

62. A pattern of variation of a time series that repeats every year is called:

Select correct option:

Cyclical

Seasonal

Trend
Secular

63. You have measured the systolic blood pressure of a random sample of 22 employees of a

company. A 95% confidence interval for the mean systolic blood pressure for the employees is computed to be (120,138). Which of the following statements gives a valid interpretation of this interval?

Select correct option:

About 95% of the sample of employees has a systolic blood pressure between 120 and 138.

95% of the employees in the company have a systolic blood pressure between 120 and 138.
If the sampling procedure were repeated many times, about 95% of the samples would have a confidence interval containing the true mean.

64. Assume that a population consists of 7 similar containers having the following weights (kg):

9.8, 10.2, 10.4, 9.8, 10.0, 10.2, 9.6 What is the second moment about mean?

Select correct option:

0.262 kg

0.069kg

0.521 kg 0.313kg

65. How many numbers of parameter(s) are in the normal distribution?

Select correct option:

0

1 pg 292

2

3

66. With increase in sample size, distribution tends to be a.....

Select correct option:

Meso kurtic

Normal Bell

shaped

Above all

67.F-distribution is a distribution.

Select correct
option:

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BimodalDiscrete
Negatively skewed

68. Which one of the following sampling methods would give unbiased results, if you need to find out the number of people in your town liking vanilla or chocolate ice creams?

Select correct option:

Ask my neighbors

Randomly select a few ice cream shops in town, and

Ask my friends Ask my classmates

69. A standard deviation obtained from sampling distribution of sample statistics is known as

Select correct option:

Sampling Error

Standard error pg 240

70. If the regression line: $Y = 3 + 5X$ meets y-axis at '8' units distance from origin then the value of

x-intercept is

Select correct option:

$-(3/5)$

$3/5$

$11/5$

1

71. Sum of three terms whose mean is equal to 90 is

Select correct option:

270

30

Also 90

None of these

72. FREQUENCY Function calculates how often values occur within a range of values.

Select correct option:

true

False

73. Which of the following correlation coefficients represents the weakest correlation between two variables?

Select correct option:

0.15

-0.15

0.02

-1.00

74. If the value of r is 0.8, then the coefficient of determination is

Select correct option:

67%

64%

80%

75%

75. If the dependent variable increases with the independent variable then the coefficient of correlation is

Select correct option:

0 to -1

0 to -0.5

0 to -2

0 to 1

76. F - distribution tends to normality, if

Select correct option:

$V_1 \sim 8$

$V_2 \sim 8$

V_1 and $V_2 \sim 8$

Sample size is large

77. Consistency of an estimator can be checked by comparing

Select correct option:

Mean

MSE

Variance

Standard deviation

78. A standardized estimate has mean and variance

Select correct option:

(1, 0)

(0, 1)

(μ , s^2)

(μ , s)

79. A composite hypothesis comprises of

Select correct option:

Equality Not

equal to

Less than/greater

(b) and (c)

80. For a particular hypothesis test, $\alpha=0.05$ and $\beta=0.05$. The power of test is equal to:

Select correct option:

0.14

0.90

0.95

0.25

81. In a t -distribution

Select correct option:

Mean=median=mode

Mean>Median<Mode

Median >Mean>Mode

Median<Mode<Mean

82. If we reject the null hypothesis, we might be making

Select correct option:

Type I error

Type II error

A correct decision

Unpredictable

83. The Central Limit Theorem is important in Statistics because it allows us to use the normal

distribution to make inferences concerning the population mean:

Select correct option:

Provided that the population is normally distributed and

Provided that the population is normally distributed (

Provided that the sample size is reasonably large (for

Provided that the population is normally distributed and the population variance is known (for any sample size)

84. Herbicide A has been used for years in order to kill a particular type of weed, but an experiment is to

be conducted in order to see whether a new herbicide, Herbicide B, is more effective than Herbicide A. Herbicide A will continue to be used unless there is sufficient evidence that Herbicide B is more effective.

The alternative hypothesis in this problem is that Select correct option:

Herbicide A is more effective than Herbicide B

Herbicide B is more effective than Herbicide A

Herbicide A is not more effective than Herbicide B

Herbicide B is not more effective than Herbicide A

85. A data in which we study about Regions is called

Select correct option:

Qualitative

Quantitative

Geographical

Chronological

86. If the median of an arrangement of numbers is equal to the mean of its middle terms then the

arrangement contains

Select correct option:

Odd number of terms

Even number of terms

Unlimited number of terms

Prime number

87.

then what can be

Select correct option:

$r = -0.9$

$r = -0.5$

$r = 0.1$

$r = 0.8$

88. In scatter diagram, clustering of points around a straight line indicates Select correct option:

Linear regression Non-linear regression

Curvilinear linear regression

Both a and b

If the graph is very much scattered, what is the suitable value of r ?

89. If the standard deviation of a population is 9, the population variance is
Select correct option:

- 3
- 9
- 21.35

81

90. How many steps are involved in general procedure for testing hypothesis:

- 4
- 5
- 6
- 7

91. When testing for independence in a contingency table with 2 rows and 5 columns, there are _____ degrees of freedom.

- 10
- 7
- 5

92. The critical region is in:

The middle of a distribution

The tails of a distribution

Either the middle or the tails of a distribution Neither the middle nor the tails of a distribution

93. t-distribution is used to test the hypothesis about....

Mean

proportion

The term 1-B is called

Level of the test power
of the test Size of the test
Critical region

94. The asymptotic distribution of t-statistic with n-degree of freedom is

F

Normal

Z T

95. The Gallup Poll has decided to increase the size of its random sample of

Canadian voters from

about 1200 people to about 4000 people. The effect of this increase is to:

Reduce the bias of the estimate

Increase the standard error of the estimate

Reduce the variability of the estimate

Increase the confidence interval width for the parameter

96. The value of chi square can never be :

Zero

Less than 1

Greater than 1

Negative

97. The curve of the F - distribution depends up on:

Mean

Variance

Standard Deviation

Sample Size

98. We want to test $H_0 : \mu = 1.5$ vs. $H_1 : \mu \neq 1.5$ at $\alpha = .05$. A 95% confidence interval for μ

calculated from a given random sample is (1.4, 3.6) Based on this finding we:

Fail to reject H_0

Reject H_0

Cannot make any decision at all because the value of the test statistic is not available

Cannot make any decision at all because (1.4, 3.6) is only a 95%

99. When we want to test the equality of two variances we usually use

F-test

Chi-square test

ANOVA

Z_test

100. To find the estimate of a parametermethods are used.

Two

Three

Four

Many

101. In testing hypothesis, we always begin it with assuming that:

Null hypothesis is true

Alternative hypothesis is true

Sample size is large Population
is normal

102. t -distribution is applicable in case of

Independent samples

Dependent samples Both

(a) and (b) Normal

populations

103. When testing for independence in a contingency table with 3 rows and 4 columns, there are

_____degrees of freedom.

5

6

7

12

104. The Chi - Square distribution is continuous distribution ranging from:

0 = $\chi^2 = 8$

-8 = $\chi^2 = 0$

-8 = $\chi^2 = 1$

-8 = $\chi^2 = 8$

105. The location of the critical region depends upon:

Null hypothesis

Alternative hypothesis

Value of α

Value of test-statistic

106. A random sample of $n = 6$ has the elements 6, 10, 13, 14, 18 and 20. What is the point

estimate of the population mean?

12

13.5

11

11.5

107. ML estimators may not

Consistent

Efficient

Unbiased

Bised

108. Which of the following reveals the weakest fact.

Select correct option:

The measure of central tendency measures that value which depends only on the extreme values

The measure of central tendency measures that value in the data which occurs in the data most frequent times.

The measure of central tendency measures the value which has tendency to lie in the central part of the data.

109. The measure of central tendency measures the distance of values from means

Frequency polygon is

Select correct option:

Bar Charts **A**

line graph

Pareto Chats

None of these

110. A bar graph uses to show data.

Select correct option:

Points

Bars

Lines

Pictures

111. Geographical data deals with...

Select correct option:

Religion Height

Income **Regions**

112. Which one provides the basis for hypothesis testing?

Null hypothesis

Alternative hypothesis

Critical value

Test-statistics

113. The test statistic to test the $U_1 = U_2$ (U represent the mean of population) for normal

population for $n > 30$.

F-test

Z-test

T-test

Chi-Square test

114. In a t -distribution

Mean=median=mode

Mean>Median<Mode

Median >Mean>Mode

Media<Mode<Mean

115. $1 - \alpha$ is the probability of

Type 1 error

Rejection region

Acceptance region

Type 2 error

116. Inferential statistics involves

Testing

Confidence interval

Estimation

Above all

117. Probability of type II error is

a

B

1 - a

1 - B

118. if the equation of regression line is $y = 5$, then what result will you take out from it?

Select correct option:

The line passes through origin. The line passes through (5, 0) The line is parallel to y-axis.

The line is parallel to x-axis.

119. If the estimating equation is $Y = a - bX$, Which of the following is true

Select correct option:

a) The y intercept is 'b'

b) Slope of line is negative

c) There is inverse relationship

d) b & c

120. The variance of t -distribution, for $v > 2$, is always:

Select correct option:

Greater than zero

Less than one

Equal to one

Greater than one

121. Alpha is the probability of

Select correct option:

Rejecting H_0

Accepting H_0

Rejecting H_1

Accepting H_1

122. What type of data is collected in population census?

Select correct option:

Two Types

123. The collection of all outcomes for an experiment is called

Select correct option: Come &

a sample space

the intersection of events

joint probability population

124. Which of the graph is used for a time series data:

Select correct option:

Frequency curve

Frequency polygon

Historigram

Histogram

125. A histogram is consists of a set of adjacent rectangles whose bases are marked off by:

Select correct option:

Class boundaries

Class limits Class

frequencyClass

marks

126. The value that has half of the observations above it and half the observations below it is

known as:

Select correct option:

Mean

Median

Mode

Standard deviation



127. The height of a student is 60 inches. This is an example of ?

Select correct option:

Continuous data

Qualitative data

Categorical dataDiscrete data

128. Range of the values -2,-3,-4,-3,-9,-2,-8,-1,0 is

Select correct option:

0

-9

8

9

129. If the both tails of the distribution are equal, then distribution is called:

Select correct option:

J-shaped

Symmetrical

Positively Skewed

Negatively Skewed

130. Ranking scale also include the properties of which scale?

Select correct option:

Nominal scale

Interval scale

Ratio scale

All of these

Come &

131. Range of the values -2.50, -3.70, -4.80, -3.10, -9.70, -2.20, -8.90, -1.60, 0.60 is

Select correct option:

10.03

10.30

9.10

9.00

132. What is/are the mode for the following data: 1,m,d,n,,2,d,2,d,s,5,5,7

Select correct option:

2

d

5

2,d,5

133. If the standard deviation of a population is 5.5, the population variance is:

Select correct option:

5.5

31

25

30.25

134. What we commonly called a bell shaped distribution:

Select correct option:

syme

bi moderu

shap

skewed

135. The beginnings of a cumulative frequency distribution are presented below.

What is the

next number in the Cumulative Frequency column? Classes Frequency CumulativeFrequency

6.1 to 8 1 1 8.1 to 10 2 10.1 to 12 3

Select correct option:

0

1

2

3

136. Range of the values -10, -19, -9, -15, -28, -26, -25 is:

Select correct option:

+18

-18

-19

+19

137. Which one of the following is less than median for a symmetrical distribution:

Select correct option:

50percentile

51percentile

2quartile 4decile

138. The value of the middle term in a ranked (ordered) data set is called the

Select correct option:

mode

Come &

mean

median

harmonic mean

139. Sum of absolute deviations of the values is least when deviations are taken from

Select correct option:

mean

median

mode

g.m

140. Statistic is a numerical quantity, which is calculated from

Select correct option:

data observation

sample

population

141. The branch of Statistics that is concerned with the procedures and

methodology for

obtaining valid conclusions is called:

Select correct option:

descriptive

advance

inferential

sample

142. How to find the class midpoint?

Select correct option:

Half the sum of upper class limit and lower class limit

Find the difference between consecutive lower limits

Count the number of observations in the class Divide the class frequency by the number of observ

143. For given data, discuss the shape of the distribution: X f 0.2 8 1.2 15 2.2 23

3.2 40

Select correct option:

Positively skewed

Negatively skewed

Symmetric curve U-

Shaped curve

144. Data classified by attributes are called:

Select correct option:

group

qualitative

quantitative array

if '2' is a leading digit in 24335, then what are the trailing digits in the observation to display a 'Stem-and-Leaf display'.

Select correct option:

4

335

4335

43

Come &



145. A frequency polygon is obtained by plotting the class frequencies against what?

Select correct option:

class boundary

cumulative frequency

relative frequency mid

point

146. When more values are lying at the start of the distribution, it is:

Select correct option:

u shape positive

negative

symmetrical

147. The data for an ogive is found in which distribution:

Select correct option:

A cumulative frequency distribution

A joint frequency distribution

A frequency distribution

A relative frequency distribution

148. Which one of the following is greater than median for a symmetrical distribution:

Select correct option:

1st Decile

7th Decile

44th Percentile

14th Percentile

149. Statistics deals with

Select correct option:

Individuals

Isolated items

Aggregates of facts

150. Data classified by attributes are called:

Select correct option:

Grouped data

Qualitative data

Quantitative data

Arrayed data

151. As a general rule, statisticians tend to use which of the following number of

classes when arranging

the data Select correct

option:

Fewer than 5

Between 5 & 20

Between 8 & 15

More than 20

152. The collection of all outcomes for an experiment is called

Select correct option:

a sample space

the intersection of events

the intersection of events



**joint probability
population**

153. If $P(E)$ is the probability that an event will occur, which of the following must be false:

Select correct option:

$P(E) = -1$

$P(E) = 1$ $P(E) = 1/2$

$P(E) = 1/3$

154. If we roll a die then probability of getting a '2' will be

Select correct option:

$2/6$

$1/6$

$4/6$

1

155. In a multiplication theorem $P(A \cap B)$ equals:

Select correct option:

$P(A) P(B)$

$P(A) + P(B)$

$P(A) * P(B|A)$ pg 158

$P(B \setminus A) * P(B)$

156. If $Y = 3X + 5$, then S.D of Y is equal to

Select correct option:

9 s.d(x)

3 s.d(x)

s.d(x) + 5

$3s.d(x) + 5$

157. In regression line $Y = a + bX$, X is called:

Select correct option:

Dependent variable

Independent variable

Explained variable

Regressand

158. Symbolically, a marginal probability is:

Select correct option:

$P(AB)$

$P(A \cup B)$

$P(A/B)$

$P(A)$

159. Which formula represents the probability of the complement of event A:

Select correct option:

$1 + P(A)$

$1 - P(A)$

$P(A)$

$P(A) - 1$

160. If A and B are independent events with $P(A) = 0.05$ and $P(B) = 0.65$, then $P(A|B)$

= :

Select correct option:

0.65

0.05

Come &



0.03

0.07 not sure

161. The probability of drawing a 'white' ball from a bag containing 4 red, 8 black and 3 white

balls is:

Select correct option:

0

3/15

1/12

1/2

<http://www.vustudents.net>

162. An expected value of a random variable is equal to:

Select correct option:

Variance

Mean

Standard deviation

Quartile

163. When we toss a fair coin 4 times, the sample space consists of....points.

Select correct option:

4

8

12

16

164. $5C5 =$

Select correct option:

5

1

10

25

165. In a probability distribution, the sum of the probabilities is equal to:

Select correct option:

0

0.1

0.5

1

166. The simultaneous occurrence of two events is called:

Select correct option:

Joint probability

Subjective probability

Prior probability

Conditional probability

167. Let E and F be events associated with the same experiment. Suppose the E and F are independent and that $P(E) = 1/4$ and $P(F) = 1/2$ Then $P(E \cup F)$ is:

Select correct option:

1/8

3/4

7/8

5/8

Come &

Question No: 1 (Marks: 1) - Please choose one

_____ When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

- ▶ Mutually exclusive
- ▶ Equally likely
- ▶ Not mutually exclusive
- ▶ Exhaustive

Question No: 2 (Marks: 1) - Please choose one

_____ The mean of the F-distribution is:



▶ $\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$

▶ $\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$

▶ $\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$

▶ $\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$

Question No: 3 (Marks: 1) - Please choose one

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_____ The LSD test is applied only if the null hypothesis is:

- ▶ Rejected
- ▶ Accepted
- ▶ No conclusion
- ▶ Acknowledged

Question No: 4 (Marks: 1) - Please choose one

_____ Analysis of variance is a procedure that enables us to test the equality of several:

<http://.ning.com>

- ▶ Variances
- ▶ Means
- ▶ Proportions
- ▶ Groups



Question No: 5 (Marks: 1) - Please choose one

_____ ANOVA was introduced by :

- ▶ Helmert
- ▶ Pearson
- ▶ R.A Fisher
- ▶ Francis

Question No: 6 (Marks: 1) - Please choose one

_____ For testing of hypothesis about population proportion , we use:

- ▶ Z-test
- ▶ t-Test
- ▶ Both Z & T-test

- ▶ F test

Question No: 7 (Marks: 1) - Please choose one

_____ If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ▶ 0,1,2,3
- ▶ 1,3,3,1
- ▶ 1, 2, 3
- ▶ 3, 2

Question No: 8 (Marks: 1) - Please choose one

_____ If X and Y are independent variables, then $E(XY)$ is: <http://.ning.com>

- ▶ $E(XX)$
- ▶ $E(X).E(Y)$
- ▶ $X.E(Y)$
- ▶ $Y.E(X)$

Question No: 9 (Marks: 1) - Please choose one

_____ The parameters of the binomial distribution $b(x; n, p)$ are:

- ▶ x & n
- ▶ x & p
- ▶ n & p

► x, n & p

Question No: 10 (Marks: 1) - Please choose one

_____ If $P(E)$ is the probability that an event will occur, which of the following must be false:

- ▶ $P(E) = -1$
- ▶ $P(E) = 1$
- ▶ $P(E) = 1/2$
- ▶ $P(E) = 1/3$

Question No: 11 (Marks: 1) - Please choose one

_____ An estimator T is said to be unbiased estimator of θ if $E(T) = \theta$ if <http://.ning.com>

- ▶ $E(T) = \theta$
- ▶ $E(T) = T$
- ▶ $E(T) = 0$
- ▶ $E(T) = 1$



Question No: 12 (Marks: 1) - Please choose one

_____ The best unbiased estimator for population variance σ^2 is:

- ▶ Sample mean
- ▶ Sample median

► Sample proportion

► Sample variance

Question No: 13 (Marks: 1) - Please choose one

_____ The sample

variance $\frac{\sum (x_i - \bar{x})^2}{n}$ is:

► Unbiased estimator of σ^2

► Biased estimator of σ^2

► Unbiased estimator of μ

► None of these



Question No: 14 (Marks: 1) <http://.ning.com> - Please choose one

_____ When c is a constant, then E(c) is:

0

c

Come &

-C

STA301 Solved MCQz

by Muhammad Ishfaq

- ▶ 0
- ▶ 1
- ▶ c
- ▶ -c

Question No: 15 (Marks: 1) - Please choose one

_____ If $f(x, y)$ is bivariate probability density function of continuous r.v.'s X and Y then

$g(x)$ is:



☐

☐ $\int_{-\infty}^{\infty} f(x, y) dx$

▶

☐

☐ $\int_{-\infty}^{\infty} f(x, y) dy$

▶

☐

☐

☐

☐

☐ ☐

$$\int f(x,y) \, dx \, dy$$

$$\int_a^b \int_c^d f(x,y) \, dy \, dx$$



Come &

Question No: 16 (Marks: 1) - Please choose one

_____ The analysis of variance technique is a method for : <http://.ning.com>

- ▶ Comparing F distributions
- ▶ Comparing three or more means
- ▶ Measuring sampling error
- ▶ Comparing variances

Question No: 17 (Marks: 1) - Please choose one

_____ The continuity correction factor is used when:

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one

_____ Stem and leaf is more informative when data is :

- ▶ Equal to 100
- ▶ Greater Than 100
- ▶ Less than 100
- ▶ In all situations

Question No: 19 (Mark:1) - Please choose

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_____ The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:
<http://.ning.com>

- ▶ Descriptive Statistics
- ▶ Advance Statistics
- ▶ Inferential Statistics

- ▶ Sampled Statistics

Question No: 20 (Marks: 1) - Please choose one

_____ Which of the following is a systematic arrangement of data into rows and columns?

- ▶ Classification
- ▶ Tabulation
- ▶ Bar chart
- ▶ Component bar chart

Question No: 21 (Marks: 1) - Please choose one

_____ In normal distribution Q.D =

- ▶ 0.5σ
- ▶ 0.75σ

► 0.7979σ

► 0.6745σ

Question No: 22 (Marks: 1) - Please choose one

_____ In normal
distribution β_2 ☐

► 1

► 2

► 3

► 0

Question No: 23 (Marks: 1) - Please choose one

_____ If you connect
the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends,
what will you get?

► Ogive

► Frequency polygon

► Frequency curve

► Histogram

Question No: 24 (Marks: 1) <http://.ning.com> - Please choose one

_____ Which one
of the following statements is true regarding a population?

► It must be a large number of values

► It must refer to people

► It is a collection of individuals, objects, or measurements

► It is small part of whole

Question No: 25 (Marks: 1) - Please choose one

$Q_1 = 2$ and $Q_3 = 4$

,what is the value of Median, if the distribution is symmetrical:

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ 4

Question No: 26 (Marks: 1) - Please choose one

_____ In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- ▶ The regression line will pass through the origin
 - ▶ The regression line will pass through the point (0,10).
 - ▶ The regression line will pass through the point (0,-10).
 - ▶ The slope of the line will also be equal to 0.

Question No: 27 (Marks: 1) - Please choose one

_____ The degrees of freedom for a t-test with sample size 10 is:

- ▶ 5
- ▶ 8
- ▶ 9
- ▶ 10

Question No: 28 (Marks: 1) - Please choose one

_____ In testing

of hypothesis, we always begin it with assuming that:

- ▶ Null hypothesis is true

- ▶ Alternative hypothesis is true
- ▶ Sample size is large
- ▶ Population is normal

Question No: 29 (Marks: 1) - Please choose one

_____ A failing student is passed by an examiner is an example of:

- ▶ Type I error
- ▶ Type II error
- ▶ Correct decision
- ▶ No information regarding student exams

Question No: 30 (Marks: 1) - Please choose one

_____ How to find $P(X \leq Y \leq 1)$?

- ▶ $f(0, 0) + f(0, 1) + f(1, 2)$
- ▶ $f(2, 0) + f(0, 1) + f(1, 0)$
- ▶ $f(0, 0) + f(1, 1) + f(1, 0)$
- ▶ $f(0, 0) + f(0, 1) + f(1, 0)$

Question No: 1 (Marks: 1) - Please choose one

_____ The value of χ^2 can never be :

► Zero

- ▶ Less than 1
- ▶ Greater than 1
- ▶ Negative

Question No: 2 (Marks: 1) - Please choose one

_____ The mean of the F-distribution is:

$$\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$$



$$\frac{v_2}{v_2 - 2} \text{ for } v_2 > 2$$



$$\frac{v_1}{v_1 - 2} \text{ for } v_1 > 2$$





Question No: 3 (Marks: 1) - Please choose one

_____ The F-distribution always ranges from:

► 0 to 1

- ▶ 0 to $-\infty$
- ▶ $-\infty$ to $+\infty$
- ▶ 0 to $+\infty$

Question No: 4 (Marks: 1) - Please choose one

_____ The total number of samples when sampling is done with replacement :

- ▶ N^n
- ▶ C_n^N
- ▶ $\frac{N}{n}$
- ▶ 1

Question No: 5 (Marks: 1) - Please choose one

_____ ANOVA was introduced by :

- ▶ Helmert
- ▶ Pearson
- ▶ R.A Fisher
- ▶ Francis

Question No: 6 (Marks: 1) - Please choose one

_____ The test statistic

used in analysis of variance procedure follow the..... distribution.:

► χ^2

► T

► Z

► F

Question No: 7 (Marks: 1) - Please choose one

_____ For testing
of hypothesis about population proportion , we use:

► Z-test

► t-Test

► Both Z & T-test

► F test

Question No: 8 (Marks: 1) - Please choose one

_____ If X and Y are
random variables, then $E(X + Y)$ is equal to:

► $E(X) + E(Y)$

► $E(X) - E(Y)$

► $X + E(Y)$

► $E(X) + Y$

Question No: 9 (Marks: 1) - Please choose one

_____ A die is

rolled. What is the probability that the number rolled is greater than 2 and even:

► $\frac{1}{2}$

Come &

► $1/3$

► $2/3$

► $5/6$

Question No: 10 (Marks: 1) - Please choose one

_____ The probability of drawing a king of spade from a pack of 52 cards is:

► $1/4$

► $1/13$

► $1/26$

► $1/52$

Question No: 11 (Marks: 1) - Please choose one

_____ An estimator T is said to be unbiased estimator of θ if

► $E(T) = \theta$

► $E(T) = T$

► $E(T) = 0$

► $E(T) = 1$

Question No: 12 (Marks: 1) - Please choose one

_____ From point estimation, we always get:
 Come &

► Single value

► Two values

► Range of values

► Zero

Question No: 13 (Marks: 1) - Please choose one

_____ The best
unbiased estimator for population variance σ^2 is:

► Sample mean

► Sample median

► Sample proportion

► Sample variance

Question No: 14 (Marks: 1) - Please choose one

_____ When c is a
constant, then $E(c)$ is:

0

c

-c

▶ 0

▶ 1

▶ c

▶ -c

Question No: 15 (Marks: 1) - Please choose one

_____ $\text{Var}(4X + 5)$
= _____

▶ $16 \text{ Var}(X)$

▶ $16 \text{ Var}(X) + 5$

▶ $4 \text{ Var}(X) + 5$

▶ $12 \text{ Var}(X)$

Question No: 16 (Marks: 1) - Please choose one

_____ When $f(x)$

iscontinuous probability function, then $P(X = 1)$ is:

- ▶ 1
- ▶ ☐
- ▶ ☐ ☐
- ▶ 0

Question No: 17 (Marks: 1) - Please choose one

The hypergeometric random variable is a(an):

- ▶ Continuous variable
- ▶ Discrete variable
- ▶ Undefined
- ▶ Independent variable

Question No: 18 (Marks: 1) - Please choose one

From a sample of 200 people were asked whether they like a particular product. Fifty said 'yes' and remain said 'no', assuming 'yes' means a success, which of the following is correct?

- ▶ Sample proportion $p=0.33$
- ▶ Sample proportion $p=0.25$
- ▶ Population proportion $p= 0.33$

Question No: 19 (Marks: 1) - Please choose one

_____ In any data set,
what percent of values fall in the interval $Median \pm Q.D$?

- ▶ 50 per cent
- ▶ 68.5 per cent
- ▶ 95.4 per cent
- ▶ 99 per cent

Question No: 20 (Marks: 1) - Please choose one

_____ If
5

$\sum_{i=1}^n X_i = 20$ and $\sum_{i=1}^n X_i^2 = 0$, then $\bar{X} = \dots\dots$

- ▶ 0
- ▶ 20
- ▶ 5
- ▶ 25

Question No: 21 (Marks: 1) - Please choose one

_____ The height of a

student is 60 inches. This is an example of ?

► Continuous data Come &

- ▶ Qualitative data
- ▶ Categorical data
- ▶ Discrete data

Question No: 22 (Marks: 1) - Please choose one

_____ In Statistics, we have MSE which is abbreviation of.....

- ▶ Mean square error
- ▶ Measured square error
- ▶ Medical screening exam
- ▶ Major sampling error

Question No: 23 (Marks: 1) - Please choose one

_____ Which one is the formula of mid range:

- ▶ $x_m \square x_0$
- ▶ $x_0 \square x_m$
- ▶ $\frac{x_0 \square x_m}{2}$
- ▶ $\underline{x_0 \square x_m}$



Question No: 24 (Marks: 1) - Please choose one

_____ The deviation of a distribution from symmetry is called:

- ▶ Kurtosis
- ▶ Skewness
- ▶ Dispersion
- ▶ Flatness

Question No: 25 (Marks: 1) - Please choose one

_____ If E is an impossible event, then $P(E)$ is:

- ▶ 1
- ▶ 2
- ▶ 0
- ▶ 0.5

Question No: 26 (Marks: 1) - Please choose one

_____ If a data set has the even number of observations, the median :

- ▶ Is the average value of the two middle items
- ▶ Can not be determined

► must be equal to the mean

- Is the average value of the two middle items when all items are arranged in ascending order

Question No: 27 (Marks: 1) - Please choose one

_____ For the Poisson distribution $P(X = 1) = \frac{e^{-0.135} 0.135^1}{1!}$ the mean value is :

- 2
- 5
- 10
- 0.135

Question No: 28 (Marks: 1) - Please choose one

_____ In testing of hypothesis, we always begin it with assuming that:

- Null hypothesis is true
- Alternative hypothesis is true
- Sample size is large
- Population is normal

Question No: 29 (Marks: 1) - Please choose one

_____ Variance of the t-distribution is given by the formula:

$$\sigma^2 = \sqrt{\frac{V}{V-2}}$$



Come &

► $\sigma^2 = \frac{v^2}{v+2}$

► $\sigma^2 = \frac{v}{v+1}$

► $\sigma^2 = \frac{v}{v+2}$

Question No: 30 (Marks: 1) - Please choose one

_____ If a continuous probability distribution has $\beta_2 = 2.14$ then what will be peakedness of the distribution?

- Platykurtic
- Mesokurtic
- Leptokurtic
- Moderately skewed

Question # 1 of 10 (Start time: 07:10:27 PM) Total Marks: 1 When two dice are rolled the number of possible sample points is :Select correct option:

- 6
- 12
- 24
- 36

Question # 2 of 10 (Start time: 07:11:18 PM) Total Marks: 1

A fair coin is tossed three times, the probability that at least one head appear is:Select correct option:

- 1/8
- 7/8
- 3/8
- 5/8

Question # 3 of 10 (Starttime: 7:12:14PM)
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Total Marks: 1
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In scatter diagram, the variable plotted along Y -axis is:Select correct option:

Independent variable

Dependent variable

Continuous variable Discrete variable

of the following is not a measure of central tendency?Select correct option:

Percentile

Quartile

Standard deviation

Mode

Which of the following is NOT a possible probability?Select correct option:

25/100

1.25

0

1

Question # 6 of 10 (Start time: 07:16:06 PM) Total Marks: 1Evaluate (10-4)!

Select correct option:

1000

720

480

32

Question # 7 of 10 (Start time: 07:17:34 PM) Total Marks: 1When E is an impossible event, then $P(E)$ is:

Select correct option:

0

1

2

0.5

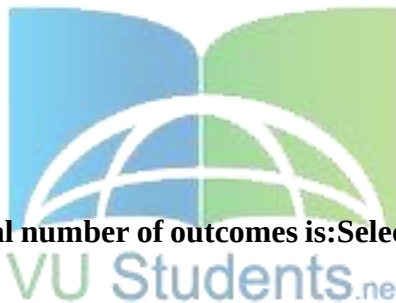
Question # 8 of 10 (Start time: 07:19:00 PM) Total Marks: 1

**The probability of drawing a 'white' ball from a bag containing 4 red, 8 black and 3 whiteballs is:
Select correct option:**

- 0
- 3/15
- 1/12
- 1/2

**standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:
Select correct option:**

- 1/5
- 1/13
- 5/52
- 1/52



If we roll three fair dices then the total number of outcomes is:Select correct option:

- 6
- 36
- 216
- 1296

Question No: 1 (Marks: 1) - Please choose one
Mean deviation is always:

- Less than S.D
- Greater than S.D
- Greater or equal to S.D
- Less or equal to S.D

Question No: 2 (Marks: 1) - Please choose one
The value of χ^2 can never be :

- Zero
- Less than 1
- Greater than 1
- Negative

Question No: 3 (Marks: 1) - Please choose one
The mean of the F -distribution is:

- ▶
- ▶
- ▶
- ▶

Question No: 4 (Marks: 1) - Please choose one
If X and Y are random variables, then is equal to:

- ▶
- ▶
- ▶
- ▶

Question No: 5 (Marks: 1) - Please choose one
Evaluate: $(9-4)!$

- ▶ 362880
- ▶ 120
- ▶ 24
- ▶ 6

Question No: 6 (Marks: 1) - Please choose one

Which formula represents the probability of the complement of event A:

- ▶ $1 + P(A)$
- ▶ $1 - P(A)$
- ▶ $P(A)$
- ▶ $P(A) - 1$

Question No: 7 (Marks: 1) - Please choose one
Ideally the width of confidence interval should be:

- ▶ 0
- ▶ 1
- ▶ 99
- ▶ 100

Question No: 8 (Marks: 1) - Please choose one

If the sampling distribution of is normal, the interval includes:

- ▶ 99% of the sample means
- ▶ 99.73% of the sample means
- ▶ 98% of the sample means
- ▶ 95% of the sample means

Question No: 9 (Marks: 1) - Please choose one

The probability distribution of a statistic is called the:

- ▶ Population distribution
- ▶ Frequency distribution
- ▶ Sampling distribution
- ▶ Sample distribution

Question No: 10 (Marks: 1) - Please choose one An estimator T is said to be unbiased estimator of θ if

- ▶ $E(T) = \theta$
- ▶ $E(T) = T$
- ▶ $E(T) = 0$
- ▶ $E(T) = 1$

Question No: 11 (Marks: 1) - Please choose one

If the following is a probability distribution, then what is the value of 'a':

X

1 2 3

$P(X)$

0.1 a 0.1

- ▶ 0.6
- ▶ 0.8
- ▶ 0.2
- ▶ 0.4

Question No: 12 (Marks: 1) - Please choose one A discrete probability function $f(x)$ is always:

- ▶ Non-negative
- ▶ Negative
- ▶ One
- ▶ Zero

Question No: 13 (Marks: 1) - Please choose one An expected value of a random variable is equal to:

- ▶ Variance
- ▶ Mean
- ▶ Standard deviation
- ▶ Covariance

Question No: 14 (Marks: 1) - Please choose one The:

- ▶
- ▶
- ▶
- ▶

Question No: 15 (Marks: 1) - Please choose one The area under a normal curve between 0 and -1.75 is

- ▶ .0401
- ▶ .5500
- ▶ .4599
- ▶ .9599

Question No: 16 (Marks: 1) - Please choose one The continuity correction factor is used when:

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Question No: 17 (Marks: 1) - Please choose one Which of the following is impossible in sampling:

- ▶ Destructive tests
- ▶ Heterogeneous
- ▶ To make voters list
- ▶ None of these

Question No: 18 (Marks: 1) - Please choose one

Which of the following is a systematic arrangement of data in rows and columns?
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- Classification
- Tabulation
- Bar chart
- Component bar chart

Question No: 19 (Marks: 1) - Please choose one

Which one of the following statements is true regarding a sample?

- It is a part of population
- It must contain at least five observations
- It refers to descriptive statistics
- It produces True value

Question No: 20 (Marks: 1) - Please choose one
The data for an ogive is found in which distribution?

- A relative frequency distribution
- A frequency distribution
- A joint frequency distribution
- A cumulative frequency distribution

Question No: 1 (Marks: 1) - Please choose one

When each outcome of a sample space has equal chance to occur as any other, the outcomes are called:

- Mutually exclusive
- Equally likely
- Not mutually exclusive
- Exhaustive

Question No: 2 (Marks: 1) - Please choose one
The mean of the F-distribution is:

-
-
-
-

Question No: 3 (Marks: 1) - Please choose one

The LSD test is applied only if the null hypothesis is:

- Rejected
- Accepted
- No conclusion
- Acknowledged

Question No: 4 (Marks: 1) - Please choose one

Analysis of variance is a procedure that enables us to test the equality of several:

- Variances
- Means
- Proportions
- Groups

Question No: 5 (Marks: 1) - Please choose one
ANOVA was introduced by :

- ▶ Helmert
- ▶ Pearson
- ▶ R.A Fisher
- ▶ Francis

Question No: 6 (Marks: 1) - Please choose one
For testing of hypothesis about population proportion , we use:

- ▶ Z-test
- ▶ t-Test
- ▶ Both Z & T-test
- ▶ F test

Question No: 7 (Marks: 1) - Please choose one
If a random variable X denotes the number of heads when three distinct coins are tossed, the X assumed the values:

- ▶ 0,1,2,3
- ▶ 1,3,3,1
- ▶ 1, 2, 3
- ▶ 3, 2

Question No: 8 (Marks: 1) - Please choose one
If X and Y are independent variables, then E (XY) is:

- ▶ $E(XY)$
- ▶ $E(X).E(Y)$
- ▶ $X.E(Y)$
- ▶ $Y.E(X)$

Question No: 9 (Marks: 1) - Please choose one
The parameters of the binomial distribution $b(x; n, p)$ are:

- ▶ x & n
- ▶ x & p
- ▶ n & p
- ▶ x, n & p

Question No: 10 (Marks: 1) - Please choose one
If P (E) is the probability that an event will occur, which of the following must be false:

- ▶ $P(E) = -1$
- ▶ $P(E) = 1$
- ▶ $P(E) = 1/2$
- ▶ $P(E) = 1/3$

Question No: 11 (Marks: 1) - Please choose one
Come & Join Us at

An estimator T is said to be unbiased estimator of if

- ▶ $E(T) =$
- ▶ $E(T) = T$
- ▶ $E(T) = 0$
- ▶ $E(T) = 1$

Question No: 12 (Marks: 1) - Please choose one
The best unbiased estimator for population variance is:

- ▶ Sample mean
- ▶ Sample median
- ▶ Sample proportion
- ▶ Sample variance

Question No: 13 (Marks: 1) - Please choose one
The sample variance is:

- ▶ Unbiased estimator of
- ▶ Biased estimator of
- ▶ Unbiased estimator of
- ▶ None of these

Question No: 14 (Marks: 1) - Please choose one
When c is a constant, then $E(c)$ is:

- 1
 - 0
 - c
 - $-c$
-
- ▶ 0
 - ▶ 1
 - ▶ c
 - ▶ $-c$

Question No: 15 (Marks: 1) - Please choose one

If $f(x, y)$ is bivariate probability density function of continuous r.v.'s X and Y then
is:

Come &



Question No: 16 (Marks: 1) - Please choose one
The analysis of variance technique is a method for :

- ▶ Comparing F distributions
- ▶ Comparing three or more means
- ▶ Measuring sampling error
- ▶ Comparing variances

Question No: 17 (Marks: 1) - Please choose one
The continuity correction factor is used when:

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Question No: 18 (Marks: 1) - Please choose one
Stem and leaf is more informative when data is :

- ▶ Equal to 100
- ▶ Greater Than 100
- ▶ Less than 100
- ▶ In all situations

Question No: 19 (Marks: 1) - Please choose one
The branch of Statistics that is concerned with the procedures and methodology for obtaining valid conclusions is called:

- ▶ Descriptive Statistics
- ▶ Advance Statistics
- ▶ Inferential Statistics
- ▶ Sampled Statistics

Question No: 20 (Marks: 1) - Please choose one

Which of the following is a systematic arrangement of data into rows and columns?

► Classification



Tabulation

- ▶ Bar chart
- ▶ Component bar chart

Question No: 21 (Marks: 1) - Please choose one
In normal distribution Q.D =

- ▶
- ▶
- ▶
- ▶

Question No: 22 (Marks: 1) - Please choose one
In normal distribution

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ 0

Question No: 23 (Marks: 1) - Please choose one

If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- ▶ Ogive
- ▶ Frequency polygon
- ▶ Frequency curve
- ▶ Histogram

Question No: 24 (Marks: 1) - Please choose one

Which one of the following statements is true regarding a population?

- ▶ It must be a large number of values
- ▶ It must refer to people
- ▶ It is a collection of individuals, objects, or measurements
- ▶ It is small part of whole

Question No: 25 (Marks: 1) - Please choose one

When ,what is the value of Median, if the distribution is symmetrical:

- ▶ 1
- ▶ 2
- ▶ 3
- ▶ 4

Question No: 26 (Marks: 1) - Please choose one

In a simple linear regression model, if it is assumed that the intercept parameter is equal to zero, then:

- ▶ The regression line will pass through the origin
- ▶ The regression line will pass through the point (0,0).

- ▶ The regression line will pass through the point (0,-10).
- ▶ The slope of the line will also be equal to 0.

Question No: 27 (Marks: 1) - Please choose one

The degrees of freedom for a t-test with sample size 10 is:

- ▶ 5
- ▶ 8
- ▶ 9
- ▶ 10

Question No: 28 (Marks: 1) - Please choose one

In testing of hypothesis, we always begin it with assuming that:

- ▶ Null hypothesis is true
- ▶ Alternative hypothesis is true
- ▶ Sample size is large
- ▶ Population is normal

Question No: 29 (Marks: 1) - Please choose one

A failing student is passed by an examiner is an example of:

- ▶ Type I error
- ▶ Type II error
- ▶ Correct decision
- ▶ No information regarding student exams

Question No: 30 (Marks: 1) - Please choose one

How to find ?

- ▶ $f(0, 0) + f(0, 1) + f(1, 2)$
- ▶ $f(2, 0) + f(0, 1) + f(1, 0)$
- ▶ $f(0, 0) + f(1, 1) + f(1, 0)$
- ▶ $f(0, 0) + f(0, 1) + f(1, 0)$

Question No: 1 (Marks: 1) - Please choose one

10!

=.....

- ▶ 362880
- ▶ 3628800
- ▶ 362280
- ▶ 362800

Question No: 2 (Marks: 1) - Please choose one

When E is an impossible event, then $P(E)$ is:

- ▶ 2
- ▶ 0
- ▶ 0.5
- ▶ 1

Question No: 3 (Marks: 1) - Please choose one

χ^2 can never be :

- ▶ Zero
- ▶ Less than 1
- ▶ Greater than 1
- ▶ Negative

The value of

The curve of

Question No: 4 (Marks: 1) - Please choose one

F- distribution depends upon:

- ▶ Degrees of freedom
- ▶ Sample size
- ▶ Mean
- ▶ Variance



If X and Y are

Question No: 5 (Marks: 1) - Please choose one

$$E(X \pm Y)$$

$\pm$

random variables, then

is equal to:

$$E(X) \pm E(Y)$$

▶

$$E(X) \pm E(Y)$$

▶

$$X \pm E(Y)$$

▶

$$E(X) - Y$$

?

Question No: 6 (Marks: 1) - Please choose one

hypothesis, we always begin it with assuming that:

- ▶ Null hypothesis is true
- ▶ Alternative hypothesis is true
- ▶ Sample size is large
- ▶ Population is normal

Question No: 7 (Marks: 1) - Please choose one

In testing

For the

$$\frac{1 \times 0.135}{0.135^1}$$

$$1!$$

Poisson distribution $P(x) =$

the mean value is :

- ▶ 2
- ▶ 5
- ▶ 10
- ▶ 0.135

Question No: 8 (Marks: 1) - Please choose one

When two

coins are tossed simultaneously, P (one head) is:

- 1
- ▶ $\frac{1}{4}$
- ▶ $\frac{1}{2}$
- ▶ $\frac{3}{4}$
- ▶ 1

From point

Question No: 9 (Marks: 1) - Please choose one

estimation, we always get:

- ▶ Single value
- ▶ Two values

The sample

- ▶ Range of values
- ▶ Zero

Question No: 10 (Marks: 1) - Please choose one

variance $S^2 = \frac{\sum (x - \bar{x})^2}{n}$ is:

► Unbiased estimator of σ^2

► Biased estimator of σ^2

► Unbiased estimator of μ

► None of these

Question No: 11 (Marks: 1) - Please choose one

Var(4X + 5)
= _____

► 16 Var (X)

► 16 Var (X) + 5

► 4 Var (X) + 5

► 12 Var (X)

When f (x, y)

Question No: 12 (Marks: 1) - Please choose one

is bivariate probability density function of continuous r.v.'s X and Y, then

$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy$ is equal to:

- 1
- 0
- -1
- ∞

Question No: 13 (Marks: 1) - Please choose one

The area under a normal curve between 0 and -1.75 is

► .0401

► .5500

► .4599

► .9599

When a fair

Question No: 14 (Marks: 1) - Please choose one

is rolled, the sample space consists of:

► 2 outcomes

► 6 outcomes

► 36 outcomes

► 16 outcomes

Question No: 15 (Marks: 1) - Please choose one

When testing

for independence in a contingency table with 3 rows and 4 columns, there are _____ degrees of freedom.

- ▶ 5
- ▶ 6
- ▶ 7
- ▶ 12

Question No: 16 (Marks: 1) - Please choose one

The F- test

statistic in one-way ANOVA is:

- ▶ SSW / SSE
- ▶ MSW / MSE
- ▶ SSE / SSW
- ▶ MSE / MSW

The

Question No: 17 (Marks: 1) - Please choose one

continuity correction factor is used when:

A uniform

- ▶ The sample size is at least 5
- ▶ Both nP and $n(1-P)$ are at least 30
- ▶ A continuous distribution is used to approximate a discrete distribution
- ▶ The standard normal distribution is applied

Which graph

Question No: 18 (Marks: 1) - Please choose one

distribution is defined by:

- ▶ Its largest and smallest value
- ▶ Smallest value
- ▶ Largest value
- ▶ Mid value

Question No: 19 (Marks: 1) - Please choose one

is made by plotting the mid point and frequencies?

- ▶ Frequency polygon
- ▶ Ogive
- ▶ Histogram
- ▶ Frequency curve

Question No: 20 (Marks: 1) - Please choose one

In a set of 20 values all the values are 10, what is the value of median?

- ▶ 2
- ▶ 5
- ▶ 10
- ▶ 20

MIDTERM EXAMINATION

STA301- Statistics and Probability (Session - 2)

Question No: 1 (Marks: 1) - Please choose one

A histogram consists of a set of adjacent rectangles whose bases are marked off by:

- ▶ Class
- ▶ boundaries
- ▶ Class limits
- ▶ Class marks
- ▶ Class frequency

Question No: 2 (Marks: 1) - Please choose one

Stem and leaf was introduced by:

- ▶ Francis Galton
- ▶ Friedman
- ▶ John Tukey
- ▶ Pearson

Question No: 3 (Marks: 1) - Please choose one

**For positively skewed
distribution**

Mean.....Median.....Mode:

- ☐ =
- ☐ <
- ☐ >
- ☐ $\neq$

Question No: 4 (Marks: 1) - Please choose one

For a positively skewed distribution:

► **Mean<Mode<Media**

► **n**

► **Mdian<Mode>Mean**

► **Mode>Mean>Media**

n

Mean>Median>Mod

e

**Question No: 5
) - Please choose one**

(Marks: 1

5C_5 equals :

► **1**

► **5**

► **10**

► **25**

**Question No: 6
) - Please choose one**

(Marks: 1

If a curve has a longer tail to the right, it is called :

► **Positively**

►

►

skewed

negatively

skewedJ-shaped

symmetric

Question No: 7
) - Please choose one

(Marks: 1

In measures of relative dispersion unit of measurement is:

- ▶ **Change**
- ▶ **d**
- Vanish**
- ▶ **Does not**
- ▶ **changed**
- Dependent**

Question No: 8
) - Please choose one

(Marks: 1

Data used by an agency which is originally collected by them is :

- ▶ **Primary data**
- ▶ **Raw data**
- ▶ **Secondary**
- ▶ **dataGrouped**
- data**

Question No: 9
) - Please choose one

(Marks: 1

When we toss a coin , we get only:

- ▶ **1 outcome**

► 2 outcomes

► 3 outcomes

► 4 outcomes

Question No: 10
) - Please choose one

(Marks: 1

Which of the following is the class frequency

- ▶ **The number of observations in each class**
- ▶ **The difference between consecutive lower class limits**
- ▶ **Always contains at least 5 observations**
- ▶ **Usually a multiple of the lower limit of the first class**

Question No: 11
) - Please choose one

(Marks: 1

Standard deviation is affected by the change of_____:

- ▶ **Origin &**
- ▶ **scale**
- ▶ **only Scale**
- ▶ **only**
- ▶ **Not origin & scale**

Question No: 12
) - Please choose one

(Marks: 1

For a moderately skewed distributions, the approximate percentage of case included

between $\bar{X} - 2S$ and $\bar{X} + 2S$ is :

- ▶ **99.7%**
- ▶ **68%**

► 95%

► 50%

Question No: 13
) - Please choose one

(Marks: 1

A die is rolled. What is the probability that the number rolled is greater than 2 and even:

- ▶ $1/2$
- ▶ $1/3$
- ▶ $2/3$
- ▶ $5/6$

Question No: 14
) - Please choose one

(Marks: 1

The probability of drawing any one spade card is:

- ▶ $1/52$
- ▶ $4/52$
- ▶ $13/52$
- ▶ $52/52$

Question No: 15
) - Please choose one

(Marks: 1

Which of the following is not the type of frequency curves?

- ▶ **The symmetrical frequency curve**
- ▶ **The extremely skewed frequency**
- ▶ **curveThe U-shaped frequency curve**
- ▶ **Frequency polygon**

Question No: 16
) - Please choose one

(Marks: 1

$(m$
3

If the third moment about mean is zero

0)

=

, then the distribution is:

► Symmetrical

Negatively

skewed

Positively

skewed

Mesokurti

c

We can apply method of Maximum Likelihood on

Discrete as well as continuous variables

Sample mean is a/an estimate of population mean

Unbiased

When sample size is to be considered large, population standard deviation can be replaced by

Sample Standard Deviation

In testing $H_0: \mu = 100$ against $H_1: \mu \leq 100$ at the 10% level of significance, H_0 is rejected if

The value of the test is static in the acceptance region

If we draw a sample of size “n” from a population then sample is called “large sample” if.

$n > 30$

As sample size goes up, what tends to happen to 95 % confidence interval?

They become more precise and

narrow Confidence interval is also

known as

Non of these

Confidence interval is known as the acceptance region and the region outside the

confidence interval is called the rejection or critical region

Non of these

The precision of an estimates can be increased by increasing the:

size of sample

The precision of an estimator can be increased by decreasing the:

level of significance

ideally, the width of confidence interval should be:

0

The method of maximum likelihood (ML) is used to find out:

Point estimates

A confidence interval has a specified probability___of containing the true value of parameter.

$1-\alpha$

In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to

5

An estimator is said to be___if its expected value is equal to true value of its parameter.

un Biased

Sample mean is a..... estimate of population mean.

Unbiased

In a geometric distribution, maximum likelihood estimator (MLE) for proportion (P) is equal to:

Reciprocal of the mean

Interval estimation and confidence interval are

Same

If $P(6.2 < ? < 12.2) = 0.95$ then

Method of Moments for estimation of parameters is based on the theory of

Non of above

The end points that bound the confidence interval are called:

Lower and upper limits

When sample size is to be considered large, population standard deviation can be replaced by

Sample Standard Deviation

In a random sample of 500 men from Lahore city, 300 are found to be smokers; the proportion of smokers is equal to

0.6

Internal estimation and Confidence interval are:

Same

An estimator is said to be efficient if it has:

Both (a & b)

A statistic whose standard error decreases with an increase in the sample size will be...

Consistent

ideally, the width of confidence interval should be:

0

“A failed student was passed by the examiner”, is an example of:

Type-II error

STA 301 GRAND QUIZ FILE NEEDS TO BE SOLVED WITH REFERENCE

1. A histogram is consists of a set of a set of adjacent rectangle whose bases are marked off by:

- a. Class boundaries (Lecture no 4 Page no 32 Handouts)**
- b. Class limits
- c. Class frequency
- d. Class marks

2. The middle value of an ordered array of numbers is the

- a. Mean
- b. Median (Lecture no 7 Page no 59 Handouts)**
- c. Mode
- d. Midpoint

3. If Mean =25 & S.D is 5 then C.V is:

- a. 100%
- b. 20% (Lecture no 11 Example Page no 88 Handouts)**
- c. 10%
- d. 25%

Solution: $C.V = (S.D/X) \times 100 = (5/25) \times 100 = 20\%$ Answer

4. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, you will get:

- a. Ogive
- b. Frequency polygon
- c. Frequency curve
- d. Histogram

5. The conditional probability $P(A/B)$ is:

- a. $P(A \cap B)/P(B)$ (Lecture no 20 Page no 154 Handouts)**
- b. $P(A \cap B)/P(A)$
- c. $P(A \cup B)/P(B)$
- d. $P(A \cup B)/P(A)$

6. Serious disadvantage of using range as a measure of dispersion is that it is based on only:

- a. Minimum Values
- b. Maximum Values
- c. Both Minimum and Maximum values
- d. None of the above**

7. Frequency of a variable is always in:

- a. Fraction form
- b. Percentage form
- c. Less than form
- d. Integer form**

8. 5C_5 is equal to

- | | |
|-------|-------------|
| a. 5 | b. 1 |
| c. 10 | d. 24 |

9. Chebychev's inequality does not hold for $k = ?$

- a. 3
- b. 2
- c. 1
- d. 0 (Lecture no 12 Page no 94 Read CHEBYCHEV'S THEOREM)**

10. If $A = \{1,2,3,4\}$ and $B = \{3,4,5,6\}$ then $A - B$ will be:

- a. **{1,2}**
- b. {3,4}
- c. {3,2,1}
- d. {1,2,3,4,5,6}

11. If $P(B|A) = 0.25$ and $P(A \text{ and } B) = 0.20$, then $P(A)$ is

- a. 0.05
- b. **0.80**
- c. 0.95
- d. 0.75

Solution: $P(B|A) = [P(A \text{ and } B)] / P(A)$

$$P(A) = [0.20 / P(B|A)]$$

$$P(A) = [0.20 / 0.25]$$

$$\mathbf{P(A) = 0.80}$$

12. According to Empirical rule, approximately 68% of the measurements will fall within:

- a. **(Mean - S.D, Mean + S.D) (Lecture no 12 Page no 90 Handouts)**
- b. (Mean - 2S.D, Mean + 2S.D)
- c. (Mean - 3S.D, Mean + 3S.D)
- d. None of these

13. For Platykurtic distribution, b_2 (moment ratio) will be:

- a. Greater than 3
- b. **Less than 3 (Lecture no 14 Page no 114 Handouts)**
- c. Equal to 3
- d. Equal to zero

14. There are two broad categories of data, which are:

- a. Weighted and Un-weighted
- b. Grouped and Un-grouped
- c. **Qualitative and Quantitative (Lecture no 3 Page no 16 Handouts)**
- d. Primary and Secondary

15. In a linear regression, $Y = a + bX$, the variable "Y" will always:

- a. **A random variable (Lecture no 15 Page no 117 Handouts)**
- b. A non-random variable
- c. Qualitative variable
- d. Quantitative variable

16. In regression line $Y = a + bX$, X is called:

- a. Dependent variable
- b. Independent variable (Lecture no 15 Page no 116 Handouts)**
- c. Explained variable
- d. Regressand

17. Difference between the largest and the smallest data values is called

- a. variance
- b. interquartile range
- c. range**
- d. coefficient of variation

18. The extremely positively skewed curve is also known as:

- a. Frequency curve
- b. U-shaped curve
- c. J-shaped curve
- d. Reverse J-shaped curve (Lecture no 5 Page no 35 Handouts)**

19. A list of 7 pulse rates is: 70, 64, 80, 74, 92, 96, 98. What is the median for this list?

- | | |
|-------|--------------|
| a. 70 | b. 80 |
| c. 92 | d. 98 |

Solution: Median is the middle value. First of all we will arrange them in ascending order 64, 70, 74, 80, 92, 96, 98

20. The value of $10C_9$:

- | | |
|-------|--------------|
| a. 45 | b. 35 |
| c. 35 | d. 10 |

21. The most frequent value in the data is called

- | | |
|----------------|------------------|
| a. Mean | b. Median |
| c. Mode | d. Harmonic mean |

22. An event that contains more than one sample points is called:

- a. Mutually exclusive event
- b. Not mutually exclusive event
- c. Hyper event
- d. Compound event (Lecture no 17 Page no 140 Handouts)**

23. Calculate range for the following data: 22, 22, 30, 32, 37, 48, 60, 88, 90.

- a. 22
- b. 90
- c. 37
- d. 68

Solution: Largest value – Smallest value = Range

$$90 - 22 = \mathbf{68 \text{ Answer.}}$$

24. What is the median of this set of numbers: 4, 6, 7, 9, 2000000?

- a. 9
- b. 6
- c. 7.5
- d. 7

25. The value of the middle term in a ranked (ordered) data set is called the:

- a. Mode
- b. Mean
- c. **Median**
- d. Harmonic mean

26. The median is _____.

- a. The highest number
- b. **The middle point**
- c. The average
- d. Affected by extreme scores

27. The Mode of 8, 5, 7, 10, 15, 21, 5, 7, 2, 5 is

- a. 8
- b. 5
- c. 7
- d. 21

Note: Mode is the most repeated value so in this case 5 is the most repeated value.

28. When all the values falling in a class are equal to the mid-point of the class interval is called?

- a. Random error
- b. Unbiased Error
- c. Biased Error
- d. **Grouping Error (Lecture no 7 Page no 56 handouts)**

29. For a leptokurtic distribution, b_2 (moment ratio) will be:

- a. **Greater than 3 (Lecture no 14 Page no 114 Handouts)**
- b. Less than 3
- c. Equal to 3
- d. Equal to zero

30. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$ Then $A \cap B$:

- a. **$\{3, 4\}$**

- b. {1,4}
- c. {3,5}
- d. {3,6}

31. The range of the scores 29, 3, 143, 27, 99 is:

- a. **140**
- b. 143
- c. 146
- d. 70

Solution: $(143 - 3) = 140$ Answer

32. In how many ways, a team of 11 players can be chosen from a total of 16 players?

- a. **4368**
- b. 2426
- c. 5400
- d. 2680

33. Direct personal investigation is _____ when the area to be covered is vast.

- a. **Costly (Lecture no 2 Page no 6 Handouts)**
- b. **Time-consuming (Lecture no 2 Page no 6 Handouts)**
- c. Localized inquires
- d. Inaccurate

Note: It can be both read lecture no 2 Page no 6 Handouts.

34. What is the mean of this set of numbers: 4, 6, 7, 9, 200000?

- a. 7.5
- b. 7
- c. **400,005.2**
- d. 4

Solution: Mean = $(4+6+7+9+200000)/5 = 400,005.2$ Answer

35. According to this empirical rule, approximately how much values will fall within (Mean – 3S.D, Mean + 3S.D)?

- a. **100% (Lecture no 12 Page no 90 Handouts)**
- b. 95%
- c. 75%
- d. 68%

36. In a linear regression, $Y = a + bX$, the variable "X" will always:

- a. A random variable
- b. **A non-random variable (Lecture no 15 Page no 117 Handouts)**
- c. Qualitative variable
- d. Quantitative variable

37. Variance is expressed in _____ units as the units of data set.

a. **Squared (Lesson no 11 Page no 85 Handouts)**

- b. Cube
- c. Single
- d. Same

38. In a linear regression, best fitted line is obtained through:

- a. Method of moment
- b. Method of likelihood
- c. Method of least square
- d. Method of semi average

39. In case of frequency distribution, the median is given by the formula:

- a. $l + \frac{h}{f} \left(\frac{n}{2} - 2c \right)$
- b. $l + \frac{h}{f} \left(\frac{n}{2} - c \right)$
- c. $l + \frac{f}{h} \left(\frac{n}{2} - c \right)$
- d. $l + \frac{f}{h} \left(\frac{n}{4} - c \right)$

40. The sum of squared deviations from mean is:

- a. Maximum
- b. Minimum
- c. Zero
- d. Undefined

41. In a week the prices of a bag of rice were 350,280,340,290,320,310,300.

- a. 320
- b. 315
- c. 300
- d. 420

42. Which scale will you use to measure the temperature?

- a. Nominal scale
- b. **Interval scale (Lecture no 1 Page no 4 Handouts)**
- c. Ratio scale
- d. Ordinal scale

43. From the given data, which one of the following chart we can draw:

Year	Turnover
1965	350000
1966	42000
1967	43500
1968	48000
1969	48500

- a. Simple bar chart
- b. Multiple bar charts
- c. Component bar chart
- d. Pie chart

44. In a Venn diagram, the overlap between two circles represents:

- a. The union of two sets
- b. The intersection of two sets
- c. The elements that are in either of two sets
- d. The difference between the number of elements in two sets

45. Calculate range for the following data: 10, 32, 33, 34, 37, 42, 55, 58, 70

- a. 50 **b. 60** c. 40 d. 20

Solution: Largest value – Smallest value = Range
 $70 - 10 = 60$ **Answer**

46. When the frequency distribution or curve departs from symmetry, is called

- a. Skewed b. Positively skewed
c. Negatively skewed d. None of these

47. Measure of central tendency is used to measure:

- a. Average b. Variability
c. Location d. Both Average and Location

47. Component bar charts are used when data is divided into:

- a. Parts b. Groups c. Circles d. None of these

48. In a Box and Whisker plot, right end of the box is referred as:

- a. First quartile b. Second quartile
c. Third quartile d. Mode

49. Fourth moment about mean provides information about the _____ of the distribution.

- a. Centre b. Dispersion
c. Symmetry d. Kurtosis

50. Let A and B are two dependent events such that
 $P(A)=1/4$, $P(A/B)=1/2$ and $P(B/A)=2/3$.
Find $P(A \cap B)$.

- a. $1/8$ b. $1/6$ c. $2/3$ d. $1/4$

51. Consider a set $A = \{1, 2, 3\}$. What is the number of subsets of A? **(Doubtful)**

- a. 3** b. 6 c. 8 d. 9

52. The stem for the following data is:

22, 45, 36, 15, 14, 12, 14, 14, 17, 21, 24, 24, 25, 25, 26, 26, 27, 29, 31, 34, 35

- a. 1, 2, 3, 4 b. 1, 2, 3, 4, 5
c. 11, 12, 13, 14, 15 d. 10, 20, 30, 40, 50

53. An event that contains more than one sample points is called: **(Doubtful)**

- a. Mutually exclusive event
- b. Normal event
- c. Simple event**
- d. Compound event

54. Frequency distribution is considered as negatively skewed if all values of distribution moves to

- a. lower tail
- b. median tail
- c. variance tail
- d. upper tail

55. Which of the following is NOT a common measure of central tendency?

- a. Mode
- b. Range
- c. Median
- d. Mean

56. From the table given below, how many students obtained marks between 60 and 69? **(Doubtful)**

Marks	f	Mid-Points
50-59	5	54.5
60-69	7	64.5
70-79	8	74.5
80-89	5	84.5

- a. 64.5
- b. 12
- c. 60
- d. 7**

57. If the first and third quartiles are 22, 16 and 56,36 respectively, then the quartile deviation is:

- a. 17.1
- b. 30.5
- c. 50.5
- d. 51.3

58. Adding all the squared deviations taken from mean and dividing by the number of observations, we get:

- a. Standard Deviation
- b. Variance
- c. Mean Deviation
- d. None of the above

59. Standard deviation divided by mean is known as:

- a. Co-efficient of standard deviation
- b. Co-efficient of variation
- c. Both
- d. None

60. $5C5$ equals to:

- a. 5
- b. 1**
- c. 10
- d. 15

61. Which of the measure of dispersion is used to compare variation between two series?

- a. C.V.
- b. Q.D.
- c. M.D.
- d. S.D.

62. If $Y=3X+5$, then S.D of Y is equal to **(Doubtful)**

- a. 9 s. $d(x)$ b. 3 s. $d(x)$ c. s. $d(x)+5$ **d. 3 s. $d(x) + 5$**

63. Which of the following technique is not used to represent the bivariate qualitative data?

- a. Component Bar Chart
b. Multiple Bar Chart
c. Line Chart
d. Pie Chart

64. When a frequency distribution involves “open-end” classes, then which average is appropriate?

- a. Mean
- b. Mode
- c. **Median (Lecture no 8 Page no 62 Handouts)**
- d. None of these

65. Harmonic mean is extremely useful in averaging _____ types of data.

- Ratios
- Rates
- Both ratios and rates
- None of the above

66. According to Empirical rule, approximately how much values will fall within (Mean-3S.D, Mean+3S.D)?

- a. 100% (Lecture no Page no 90 Handouts)
b. 95%
c. 75%
d. 68%

67. What is probability of drawing two clubs from a well shuffled pack of 52 cards?

- a. $13/51$ **b. $1/17$** c. $1/26$ d. $13/17$

Solution: Total Cards = 52 cards

Number of club cards = 13 Probability of 1st card being club card = $13/52$

Probability of 2nd card being club card = $12/51$

∴ Total Probability

$$= (13/52) \times (12/51)$$

= 1/17 Answer

68. Which pair of measure cannot be calculated when one of numbers in the series is zero?

- a. G.M. and A.M.
c. G.M. and H.M.
- b. H.M. and A.M.
d. None of these

69. Which of the following techniques is used to predict the value of one variable on the basis of other variables?

- a. Correlation analysis
- b. Coefficient of correlation
- c. Covariance
- d. Regression analysis

70. A bag contains 12 red balls and 12 blue balls. A ball is drawn at random. The probability that ball drawn is red is

- a. **1/2**
- b. 5/11
- c. 6/10
- d. 1

Solution: Total Red Balls: 12

Total blue balls: 12

Total Balls: $12 + 12 = 24$

Probability of ball drawn is a red ball:

$P(\text{Red}) = 12/24 = 1/2$ **Answer**

71. If the GM of a set of two observations is 10 and its HM is 8, then the AM of the set of observations is

- a. 100
- b. **12.5**
- c. 64
- d. 7.5

RELATION BETWEEN AM, GM AND HM - FORMULA

Relation between AM, GM and HM is:

$$AM \times HM = GM^2$$

Solution: $GM = 10$, $HM = 8$, $GM^2 = (10)^2 = 100$, $AM \times HM = GM^2$,

$AM \times 8 = 100$, $AM = 100/8 = 12.5$ Answer

72. If $A = \{H, T\}$ then which of the following is power set of A?

- a. $\{\{\}, \{H, T\}\}$
- b. $\{\{H\}, \{T\}, \{H, T\}\}$
- c. $\{\{H\}, \{T\}, \{H, T\}, \{T, H\}\}$
- d. **$\{\{\}, \{H\}, \{T\}, \{H, T\}\}$**

What do understand by the term power set?

In set theory, the **power set** (or **power set**) of a **Set A** is defined as the **set** of all subsets of the **Set A** including the **Set** itself and the null or empty **set**. It is denoted by $P(A)$. Basically, this **set** is the combination of all subsets including null **set**, of a given **set**. Simple formula 2^n . n is the number of subsets. $2^2 = 4$.

$A = \{H, T\} = \{\{\}, \{H\}, \{T\}, \{H, T\}\}$ Answer

73. 2nd Quartile = 5th Decile = 50th Percentile =?

- a. Geometric Mean
- b. **Median**
- c. Mode
- d. Harmonic Mean

74. The total number of observations, which are below a certain value are known as

- A. class boundaries
- B. class marks
- C. **cumulative frequency**
- D. variances

A. Discrete frequency distribution **B Continuous frequency distribution**
C Continuous frequency distribution D Relative frequency distribution

A. one characteristic
B. Two or more characteristic
C. Similar characteristic
D. None of these

a. Mean deviation b. Standard deviation
c. Mean deviation and Standard deviation d. None of these

a. 1 b. 1,2 c. **1,2,5** d. no mode in the data

Important: The mode can be the same value as the mean and/or median, but this is usually not the case.

a. Mean b. Median c. Mode d. Harmonic mean

a. (component part/total)*100
b. (component part/total)*360
c. (component part/total)*180
d. (component part/total)*300

a. $\text{HM} \geq \text{GM} \geq \text{AM}$
c. $\text{GM} < \text{HM} < \text{AM}$

b. $\text{AM} \geq \text{GM} \geq \text{HM}$
d. $\text{AM} > \text{HM} < \text{GM}$

a. nominal b. ordinal c. interval d. ratio

a. Less than zero
b. Equal to zero
c. Greater than zero
d. None of the above

84. Smaller standard error of estimate shows:

- a. Data points are very far to the line
- b. Data points are close to the line
- c. There is no difference between line and points
- d. Difference is additive

85. Data arranged in ascending or descending order of magnitude is called:

- a. Ungrouped data
- b. Grouped data
- c. Discrete frequency distribution
- d. Arrayed data

86. A circle in which sectors represents various quantities is called:

- a. Histogram
- b. Frequency Polygon
- c. Pie Chart
- d. Component Bar Chart

87. Sum of the absolute deviations of the values is least when deviations are taken from:

- a. Mean
- b. Median
- c. Mode
- d. G.M

88. Chebychev's inequality is valid for the data set

- a. Sample
- b. Entire population
- c. Both sample and entire population
- d. None of the above

89. Which of the following terms best describes data that were originally collected at an earlier time by a different person for a different purpose?

- a. Primary data
- b. Secondary data
- c. Experimental data
- d. Field notes

90. Statistics deals with

- a. Individuals
- b. Isolated items
- c. Observations
- d. Aggregates of facts

91. If a box contains six red, three blue and five pink ties then probability of blue ties will be equal to:

- a. $1/14$
- b. $3/14$
- c. $5/14$
- d. $6/14$

92. Which of the rule is applied to any data set, regardless shape of the frequency distribution?

- a. Chebychev's rule
- b. Empirical rule
- c. Combination rule
- d. Permutation rule

93. Which average is used in the situation where the number of floors in the buildings at the center of a city?

- a. Mean
- b. Median
- c. Mode
- d. Variance

94. Rankings of the finishes of competitors in a foot race is an example of a(n)_____.

- a. ratio scale b. ordinal scale c. nominal scaled. interval scale

95. Among 18 articles, six having minor Defects and three have major defects. Determine the probability that an article selected at random has major defect.

- a. $\frac{1}{6}$ b. $\frac{1}{5}$ c. 0.25 d. 0.11

96. A series of data with exclusive classes along with the corresponding frequencies is called:

- a. Discrete frequency distribution b. Continuous frequency distribution
c. Percentage frequency distribution d. Cumulative frequency distribution

97. Using the following table, calculate $P(X < 2)$

X	0	1	2	3
f(x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

- a. $\frac{1}{8}$ b. $\frac{3}{8}$ c. $\frac{4}{8}$ d. $\frac{7}{8}$

98. If a distribution has two modes then the distribution is called:

- a. Uni-Modal b. Bi-Modal c. Tri-Modal d. Multi-Modal

99. A group consists of three persons, A, B, & C. How many ways are there of selecting a group of two persons out of these three?

- a. 1 b. 2 c. 3 d. 6

100. If any of the value in data set is zero then it is not possible to compute

- a. Mean b. Mode
c. Geometric Mean d. Harmonic Mean

1.If A and B are mutually exclusive events then P (A or B) equals:

- a. $P(A)+(B)-P(A \text{ and } B)$
- b. $P(A)*P(B)$
- c. $P(A)+P(B)$ ✓
- d. $P(A|B) + P(B|A)$

2. In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is

- a. $1/52$
- b. $4/52$
- c. $13/52$ ✓
- d. $52/52$

3. The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is

- a. $4/52$
- b. $2/52$ ✓
- c. $13/52$
- d. $26/52$

4. If $P(B|A) = 0.25$ and $P(A \text{ and } B) = 0.20$, then P (A) is

- a. 0.05
- b. 0.80 ✓
- c. 0.95
- d. 0.75

5. When a coin is tossed 3 times, the probability of getting 3 tails is

- a. $1/8$ ✓
- b. $3/8$
- c. $3/6$
- d. $2/8$

6. A set that contains all possible outcomes of a system is known as

- a. Finite Set
- b. Infinite Set
- c. Universal Set
- d. Non of these ✓

7. The probability can never be:

- a. 1
- b. $1/2$
- c. 1
- d. $-1/2$ ✓

8. The probability distribution of a statistic is called the:

- a. Population distribution
- b. Frequency distribution
- c. Sampling distribution
- d. Sample distribution ✓

9. When a coin is tossed 3 times, the probability of getting 3 tails is

- a. $1/8$ ✓
- b. $3/8$
- c. $3/6$
- d. $2/8$

10. The classical definition of probability assumes:

- a. Exhaustive events
- b. Mutually exclusive events

c. Equally likely events✓

11. Which of the following can never be probability of an event?

- a. 10
- b. 0.5

c. -0.5✓

12. The Special Rule of Addition is used to combine:

- a. Independent Events

b. Mutually Exclusive Events✓

- c. Events that total more than 1.00
- d. Events based on subjective probabilities

13. If $P(E)$ is the probability that an event will occur, which of the following must be false:

- a. $P(E) = -1$ ✓
- b. $P(E) = 1$
- c. $P(E) = 1/2$
- d. $P(E) = 1/3$

14. If we roll a die then probability of getting a '6' will be

- a. $2/6$
- b. $1/6$ ✓
- c. $4/6$
- d. 1

15. A standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:

- a. $1/5$
- b. $1/13$
- c. $5/52$ ✓
- d. $1/52$

16. The probability of drawing any one spade card is:

- a. $1/52$
- b. $4/52$
- c. $13/52$ ✓
- d. $52/52$

18. When we toss a fair coin 4 times, the sample space consists of.... Points.

- a. 4
- b. 8
- c. 12
- d. 16✓

19. A die is rolled.

What is the probability that the number rolled is greater than 2 and even:

- a. $1/2$
- b. $1/3$
- c. $2/3$
- d. $5/6$

20. The probability of drawing a king of spade from a pack of 52 cards is:

- a. $1/4$
- b. $1/13$
- c. $1/26$
- d. $1/52$

21. A fair coin is tossed three times, the probability that at least one head appear is:

- a. $1/8$
- b. $7/8$
- c. $3/8$
- d. $5/8$

22. Standard deck of 52 cards is shuffled. What is the probability of choosing the 5 of diamonds:

- a. $1/5$
- b. $1/13$
- c. $5/52$
- d. $1/52$

23. When two coins are tossed simultaneously, P (one head) is:

- a. $1/4$
- b. $1/2$ ✓
- c. $3/4$
- d. 1

24. When a fair die is rolled, the sample space consists of:

- a. 2 outcomes
- b. 6 outcomes
- c. 36 outcomes
- d. 16 outcomes

25. If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards

- a. $1/52$
- b. $13/52$
- c. $4/52$
- d. $26/52$ ✓

- 1) Who introduced stem and leaf display technique?
 - a. James T. McClave
 - b. George Benson
 - c. John Turkey
- 2) When introduced stem and leaf display technique?
 - a. 1977
 - b. 1979
 - c. 1980
- 3) John Turkey is a-----
 - a. Mathematician
 - b. Statistician
 - c. Both
- 4) A stem is a-----of each number.
 - a. Leading digit
 - b. Trailing digit
 - c. Variable
- 5) A stem is leading digit and it is used in-----
 - a. Sorting
 - b. Displaying
 - c. Both
- 6) A leaf is-----and shown in display.
 - a. Trailing digit
 - b. Rest of number
 - c. Both
- 7) ----- separates the leaf from the stem.
 - a. Horizontal line
 - b. Vertical line
 - c. Dot line
- 8) Construct a stem and leaf display from number 243
 - a. Stem 2, leaf 43
 - b. Stem 24, leaf 3
 - c. Both
- 9) Stem and leaf technique offer quick and novel way for simultaneously sorting and displaying data set where each number in data is divided into-----parts.
 - a. Two
 - b. Three
 - c. Four
- 10) Averages unable us to measure the -----
 - a. Variability

b. Central tendency of variable data

c. Frequency

11) -----enable us to measure its variability.

a. Measure of dispersion

b. Frequency

c. Central tendency

12) A single value which is intended to represent a set of data or distribution as a whole is called----

a. Mean

b. Average

c. Mode

13) -----indicates the location or general position of distribution on X-axis.

a. Average

b. Mean

c. Frequency

14) Average indicates the location or general position of the distribution on X-axis. It is known as-----

a. Measure of location or position

b. Measure of dispersion

c. Central of tendency

15) How many types of averages?

a. Four

b. Five

c. Six

16) Name of types of averages.

• Arithmetic mean

• Geometric mean

• Harmonic mean

• Median

• Mode

17) -----are averages that are mathematical in character and give indication of magnitude of observed values.

a. Arithmetic mean

b. Geometric

c. Harmonic mean

d. All of these

18) Indicated the middle position while mode provides information about most frequent value in distribution or set of data is called-----

a. Median

b. Mode

c. Mean

19) -----is the value which occurs most frequently in a set of data.

a. Mean

b. Mode

c. Median

20) What is model value from the data?

2, 7, 9, 5, 8, 9, 10, 9

a. 10

b. 9

c. 7

21) In case of set of value that have not been grouped into frequency distribution the model is obtained by --
-----the number of times each value occurs.

a. Counting

b. Adding

c. None

22) The numerical value of each measurement in the data set is located on the horizontal scale by a-----
-

a. Graph

b. Frequency

c. Dot

23) The horizontal axis of dot plot contains a scale for the -----that we want to represent.

a. Quantitative variable

b. Qualitative variable

c. Both

24) In discrete frequency distribution, identification of mode is immediate, finds value which has-----
frequency

a. Highest

b. Lowest

25) What is formula of mode for group data?

a.
$$X = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)}$$

b. Mode = 3Median- Mean

c. None

26)
$$X = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)}$$
 in this formula f_m represent-----

a. Frequency of model class

b. Lower class boundary

c. None

27) $X = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)}$ in this formula l indicates-----

- a. Frequency of model class
- b. Lower class boundary of model class
- c. Both

28) In most situation mode exists somewhere in the middle of our data values, hence it is thought as a -----

- a. Measure of central tendency
- b. Measure of dispersion

